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What does my MP think? Enhancing access to localised political information via a multi-method approach

by

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Abstract

In the United Kingdom (UK), engaging citizens on topics of politics can often be met with sentiments of apathy, cynicism and total disengagement with the political process (Hansard Society 2013, 2019; Uberoi and Johnston 2022; British Election Study 2024). This work analyses existing research into the causes of why, conceptualising a preliminary artefact to combat this phenomenon through streamlined access to local political information.

Underlying causes of rising political disengagement in the UK are first investigated, finding that UK citizens possess limited awareness about the local representatives and institutions they vote to represent them. Existing approaches are reviewed through the lens of digital political information systems, identifying difficulties in practical information access for an average constituent. Large Language Models (LLMs) are explored as an emergent approach towards simplifying access to political information on UK Members of Parliament (MPs), identifying relevant opportunities and ongoing challenges.

The artefact, termed `Civic Sage`, demonstrates a proof-of-concept Retrieval-Augmented Generation (RAG) LLM system, implemented as an open-source[1], multi-purpose deployed[2] web application combining data science and software engineering principles and leveraging various enterprise cloud services. `Civic Sage` comprises three key components: a multi-tier automated data ingestion/processing pipeline building 'MP profiles', a conversational interface housing the RAG LLM, and a conversation analysis pipeline that anonymises, stores, and analyses aggregate user conversations per MP, providing insights into constituent engagement through a dashboard.

`Civic Sage` is evaluated through a combination of automated software and LLM testing, user surveys, and MoSCoW feature prioritisation, signalling the artefact's functionality and value within a UK political context. Findings indicate that `Civic Sage` presents a novel and modestly effective ability to both streamline access to political information and display measures of constituent engagement analytics.

The work of `Civic Sage` represents a preliminary exploration into the wider field of digital political information system transformation, and further areas of development are suggested from project and artefact reflections.

[1]Open source artefact available at https://github.com/vine-james/civic-sage
[2]Artefact deployment available at https://civicsage.streamlit.app/

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This dissertation and the project that it is based on are my own work, except where stated, in accordance with University regulations.

Signed: ______________________

Name: James Vine
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Abbreviations & Definitions

Table 1: Abbreviations and their definitions used throughout the dissertation.

Abbreviation	Definition
UK	United Kingdom (of Great Britain and Northern Ireland)
MP	Member of Parliament
RAG	Retrieval-Augmented Generation
LLM	Large Language Model
L.	Likelihood (of risk occurring)
C.	Consequence (of risk occurring)
S.	Score (Likelihood multiplied by Consequence)
R.	Revised metric (after risk mitigation)
US	United States (of America)
BDD	Behaviour-Driven Development
TDD	Test-Driven Development
MoSCoW	Must have, Should have, Could have, Won’t have (prioritisation system)
SUS	Systems Usability Scale
S.U.	Standard User
E.U.	Expert User
I.R.	Initial Requirements
L.R.	Literature Review
B.	Both (I.R. and L.R.)
Cont’d.	Continued
SDLC	Software Development Lifecycle
MVP	Minimum Viable Product
Co.	Contextual (to be decided contextually)
AWS	Amazon Web Services
API	Application Programming Interface
CSV	Comma-Separated Values (file type)
VSCode	Visual Studio Code

1 INTRODUCTION

This chapter will introduce the project, summarise the problem it attempts to solve, outline the intended solution and begin to operationalise key aspects of the overall project management process.

1.1 Background and Context

Within the United Kingdom (UK) Parliamentary system, Members of Parliament (MPs) serve as elected legislators of the public. On a local (constituency) level, MPs act as the main authority empowered to represent local interests to national government. Often manifesting as lobbying for government (or private) area funding, bridging local issues to government channels, and championing local interests within legislation (Parliament of the United Kingdom 2025).

Despite the critical role MPs play in local-to-national governance, reporting suggests 54% of the general public expressed little to no knowledge of Parliament. Similarly, 74% felt disenfranchised, agreeing with the sentiment that constituents had little to no influence on a local level (Hansard Society 2019).

As political information becomes more abundant in the online age, thereby increasing access to data on how MPs discharge their public duties, these sentiments run counter-factual to a continually expanding digital information landscape.

Additional reporting suggests despite information availability, even for those with the time and interest to cultivate political knowledge, 77% surveyed thought they ‘often saw misinformation presented as news’ on social media. Forebodingly, 87% thought misinformation would affect how people vote in a parliamentary election (Electoral Commission 2024).

As the means of digital information transfer have expanded, the population of local newspapers has significantly declined. From 2009 to 2019, more than 320 local newspapers closed (House of Commons: Digital, Culture, Media and Sport Committee 2023). UK Government-commissioned research found a positive and significant correlation between circulation of local newspapers and local election turnout (Plum Consulting 2020).

1.2 Problem Definition

Constituent resources for examining UK local representation are sparse and often politically biased/authored. While political researchers may have the time and resources to conduct thorough continuous analysis on key MPs, very few solutions cater to the constraints of an average constituent.

1.3 Proposed Solution

This project proposes creation of a multi-purpose web-application, a union of data science and software-based concepts. Specifically, it aims to provide continuously updated knowledgebases, pairing data profile pipelines with a flagship Large Language Model (LLM). Additionally, ‘conversations’ will be stored, anonymised and insights extracted at aggregate level, developing a dashboard on user engagement behaviour per MP.

This comprises three components:

- A repeatable data ingestion/processing pipeline of relevant political information, used to build a structured database of MP 'profiles'.
- An appropriate LLM, packaged within a conversational interface. Augmented with a Retrieval-Augmented Generation (RAG) pipeline, transforming user 'prompts' into database queries, enabling detailed factual recount on selected representatives.
- A LLM conversation-to-database pipeline, where prompt-response pairs are anonymised, stored, and analysed at group-level, presenting a dashboard of high-level summaries of constituent engagement.

1.4 Aims and Objectives

1.4.1 Aims

This project aims to produce a data science software artefact that completes two functions, expressed in Table 2:

Table 2: Overall aims for project completion.

ID	Aim
A1	**Streamline access to localised political information**: Develop a conversational platform able to provide comprehensive, easy-to-understand factual information on a user-selected Member of Parliament.
A2	**Provide an additional measure of a 'finger on the pulse' of a local constituency**: Develop a summary dashboard of constituent interaction with the conversational platform to understand what is important to constituent users.
A3	**Demonstrate applicability**: Develop measures showing the approach's functionality is successful, providing value to a UK political context, notably through manual (e.g. users) and automated (e.g. metric) methods.

1.4.2 Objectives & Success Criteria

Objectives help frame and quantify achievement of project aims. Tables 3 & 4 denote a breakdown of each project objective and accompanying success criteria.

Table 3: 1 of 2: Governing objectives for the project and corresponding criteria for achievement (Nomenclature: O = Objective, A = Corresponding Aim from Table 2).

ID	Objective	Success Criteria	Linked Aims
O1	A Literature Review formed from research of investigating the factors 'creating' the problem identified in the project, further outlining current development efforts to address the problem (both LLM-specific and general technical solutions) within a global and UK-specific political context, identifying where the proposed artefact adds value.	• Analysis evidencing existence of the problem identified for this project. • At least three cases of analysis of both LLM & non-LLM contemporary developments relevant to solving identified problem. • Throughout analysis present value contribution of the project.	A1, A2
O2	Proposal of a list of artefact requirements that deconstruct project aims (derived from initial project goals and analysis of literature reviews) into manageable chunks. Frame requirements within MoSCoW framework for sensible project management practices/prioritisation.	• At least three requirements (any category) for Aim(s) 1 & 2 (Table 2), ensuring attention to each core aspect. • At least four requirements partially or fully derived from literature review analysis (to refine initial idea).	A1, A2
O3	Creation of a software artefact housing a RAG-LLM component, built according to the MoSCoW artefact objectives set out. Developed in order of working from highest-to-lowest priority features, as to prioritise development corresponding to the most important goals of the project/research.	• Appraisal of artefact's functionality against MoScoW objectives/definitions in a weighted (most-to-least priority) analysis. A minimum score representing 75%+ of the total available is required. • All 'must' objectives must be achieved to be successful.	A1, A2
O4	Deployment of eventual created artefact. Allowing both complete artefact automation (no processes completed manually / on local machine) & access to artefact via publicly available website/URL, thereby emulating a professional web service. Artefact pipeline/deployment should be modular, allowing for LLM model to be replaced, re-validated & re-deployed due to rapidly evolving field developments.	• Completed artefact available at public web URL. • Repeatable artefact processes/-components are successfully deployed/automated using enterprise services (e.g. AWS, Azure). • Artefact's LLM & LLM-adjacent components are modular, allowing low-effort replacement of LLM model/-functionality taking less than 5 minutes of relative effort to swap.	A1, A2

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Table 4: 2 of 2: Governing objectives for the project and corresponding criteria for achievement (Nomenclature: O = Objective, A = Corresponding Aim from Table 2).

ID	Objective	Success Criteria	Linked Aims
O5	Completed appraisal proving total completion of the artefact and iterative improvement, assessed quantitatively (automated tests) & qualitatively (manual methods of MoSCoW & user surveys.)	• ’Automated tests’ appraisal where all implemented test types & individual test cases pass, showing artefact’s completion/functioning. • ’Manual methods’ appraisal proving artefact effectiveness via meeting minimum MoSCoW objective score required. Likewise, user survey responses showing a minimum average SUS score of 75, qualitative responses summarised showing feedback and areas subsequently improved.	A3

1.5 Risk Assessment

A risk assessment helps identify, categorise and produce adequate preparation in response to risks associated with the project context. Employing a risk assessment encourages responsible navigation of potential upcoming hazards, reinforcing stability during the project process and increasing likelihood of objective completion.

1.5.1 Risk Methodology

Risks, assessed via a 5x5 matrix, operationalise quantifiable factors associated with each situation. Risks are measured according to a 1-5 X axis denoting risk consequence (severity), alongside a 1-5 Y axis summarising risk likelihood (ordered 1: lowest, 5: highest.) Multiplying ‘Likelihood’ by ‘Consequence’ we derive ‘risk scores’, which are categorised into labels of “Low”, “Moderate”, “High” and “Extreme”. These are plotted according to the overall severity of the risk score (Figure 1.)

Likelihood \ Consequence	Negligible 1	Minor 2	Moderate 3	Major 4	Catastrophic 5
5 Almost certain	Moderate 5	High 10	Extreme 15	Extreme 20	Extreme 25
4 Likely	Moderate 4	High 8	High 12	Extreme 16	Extreme 20
3 Possible	Low 3	Moderate 6	High 9	High 12	Extreme 15
2 Unlikely	Low 2	Moderate 4	Moderate 6	High 8	High 10
1 Rare	Low 1	Low 2	Low 3	Moderate 4	Moderate 5

Figure 1: A 1-5 rating risk matrix of X: consequence against Y: likelihood (Vine, 2025). Original concept from (Kaya 2018).

1.5.2 Risk Breakdown

Tables 5 and 6 identify the top 10 most pressing project risks, explained and categorised according to the risk methodology.

Table 5: 1 of 2: Top 10 identified project risks, ranked according to highest risk score. (L. = Likelihood, C. = Consequence, S. = Score (L. × C.))

ID	Risk	L.	C.	S.
R1	**Scheduling / Deadline Failures** Failure to meet required deadlines (submission, progress checkpoints) or scheduling errors (not meeting survey results participants) either due to conflicting responsibilities, or as a consequence of other listed risks.	3	5	15
R2	**Resource Constraints** Resources, each of which are fundamentally derived from monetary funding (e.g. computation hours for provisioned services) are constrained by the available budget to finance these services. In most cases relevant student provisions exist – although may have limits surpassed by the project's requirements. When financial backing is depleted, access to services may be cut off.	3	4	12
R3	**Loss of Contracted Services** Access to contracted services (e.g. cloud: Amazon Web Service (AWS), AI platforms: OpenAI, Claude, etc) may be lost both for reasons within and outside of personal control. Notably, foreign AI services are becoming politically/lawfully controversial as of the time of writing[a].	2	5	10
R4	**Poor Output Data Quality / Implementation Failures** Data pipeline construction may be too complex, or require advanced processing beyond feasibility for this project, leading to LLM responses that are incorrect (compared to input data), or otherwise misconstrued/unnecessarily confusing.	2	4	8
R5	**Personal Injury / Illness** Compromising of timeframe of project execution via unforeseen physical (e.g. vehicular accident) or mental (e.g. family death) injury or illness (e.g. COVID-19.) A required recovery reducing available time to complete project.	2	4	8
R6	**Poor Input Data Quality** Publicly available data available may be of poor quality (not enough data, lack of sourcing/credibility, etc), leading to a data-to-answer pipeline that is insufficient to address relevant questions from the selected 'character profiles'. Otherwise, 'useful' data may be publicly unavailable or locked behind overly-expensive paywalls.	2	3	6

Continued on next page

[a](United States Senate: Senator Josh Hawley 2025).

Table 6: 2 of 2: Top 10 identified project risks, ranked according to highest risk score. (L. = Likelihood, C. = Consequence, S. = Score (L. × C.))

ID	Risk	L.	C.	S.
R7	**Overcharges on Contracted Services** Overcharges on provisioned services (e.g. AWS, AI APIs, etc) through provider error[a] or uneducated user error[b].	2	3	6
R8	**Ineffective Handling of Ethical Concerns** Due to the potentially sensitive nature of the project artefact (namely, collecting user conversation history for analysis), ineffectual data management practices could lead to both ethical and legal issues.	2	3	6
R9	**Information & Communication Breakdown** Breakdown of information (i.e., incorrect reading/misunderstanding of the project assessment criteria & project handbook) and communication (i.e., misunderstanding of project supervisor feedback/guidance) could lead to wasted time spent on ultimately illegitimate work output, decreasing available time for project.	2	3	6
R10	**Poor Input Data Quality** Loss of project data (collected data for artefact, project documentation, results, surveys, etc) as a result of either: local storage failure, cloud storage failure or hardware failure during collection (e.g., power-cut.)	1	4	4

[a](Donna Goodison 2019). [b](Staff 2023).

Figure 2 illustrates project risks mapped to the matrix.

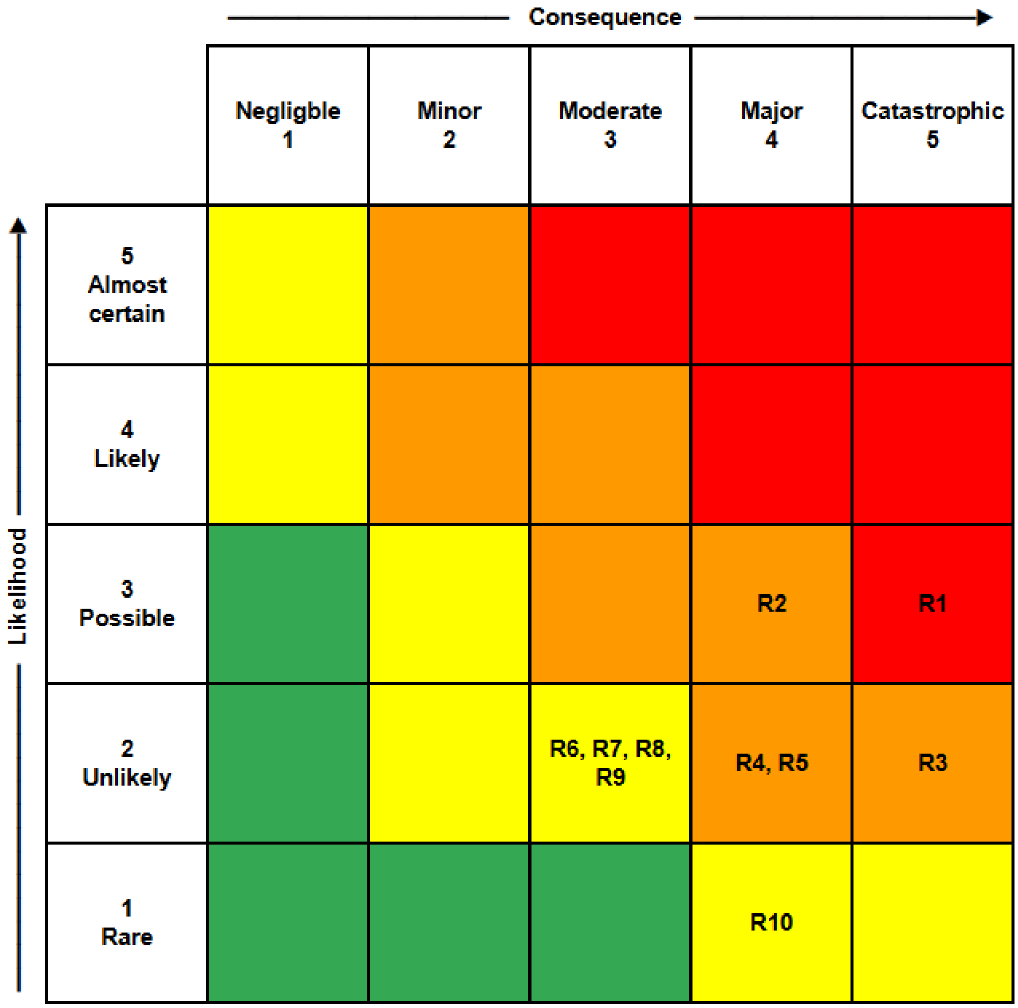


Figure 2: Project risks mapped to 1-5 rating risk matrix of X: consequence against Y: likelihood (Vine, 2025).

1.5.3 Risk Mitigation

Ensuring project continuity, a risk mitigation plan is employed (Tables 7 & 8.)

Table 7: 1 of 2: Planned mitigation steps of identified project risks (R. = Revised, L. = Likelihood, C. = Consequence, S. = Score).

ID	Mitigation	R. L.	R. C.	R. S.
1	**Scheduling / Deadline Failures** Develop a plan of action within a realistic timeframe, governed by a tried-and-tested project management and software development planning system/methodology. In this case, a waterfall-based Gantt chart system will govern the timeline of the project lifecycle. Within the software development window, a SCRUM-based iterative sprint system will be used.	2	3	5
2	**Resource Constraints** Utilise student-plans for resources at all times where possible. Plan out free alternatives for necessary provisioned services – if these do not exist, ration usage of services to only needed use cases, generating synthetic/simulated data/conditions in replacement where necessary.	2	1	2
3	**Loss of Contracted Services** Similarly to the previous risk, identify clear alternatives to each of the contracted services chosen. Where possible, identify open-source programs that can be ran locally. Additionally, identify where these services can be 'simulated' by providing synthetic data or similar that they are used for in the first place.	2	1	2
4	**Poor Output Data Quality / Implementation Failures** Maintain a good pace of development progress to ensure this does not become a sudden and critical issue at the end of the project lifecycle. Document technical challenges and frequently seek advice on overcoming them. Conduct a thorough background into existing technologies to ensure good understanding of current technological barriers, especially that a Bachelors student is unlikely to overcome.	2	3	6
5	**Personal Injury / Illness** Remain in frequent contact with project supervisor and plan accordingly if illness takes place. Where possible, maintain a good pace of work completion prior to illness to ensure minimal impact.	2	3	6
6	**Poor Input Data Quality** Carry out a thorough review of data landscape to ensure all ethical data possibilities have been considered. If still inadequate, discuss generating synthetic data in replacement to simulate design of the artefact.	2	1	2

Continued on next page

Table 8: 2 of 2: Planned mitigation steps of identified project risks (R. = Revised, L. = Likelihood, C. = Consequence, S. = Score).

ID	Mitigation	R. L.	R. C.	R. S.
7	**Overcharges on Contracted Services** Thoroughly read through relevant product documentation and user guides on effective bill management. Where possible, implement throttling measures and limit/usage cap policies from within product settings to ensure user-error based overcharges are impossible.	1	2	2
8	**Ineffective Handling of Ethical Concerns** Conduct an appropriate ethics checklist, reviewed by project supervisor/Bournemouth University. As artefact development continues, keep project supervisor informed of technical details of implementation alongside clearly documenting process. Research and ensure appropriate UK governance on relevant topics is implemented.	2	2	4
9	**Information & Communication Breakdown** Maintain open dialogue with project supervisor and keep informed of developments in project to ensure continual, accurate progress is achieved. Regularly explain current thought process in meetings to identify areas of confusion or misunderstanding.	1	2	4
10	**Project Data Loss** Non-sensitive data should be backed up on a clearly defined, regular schedule to additional local and cloud-based storage.	1	3	3

As mapped before, Figure 3 similarly maps mitigated project risks.

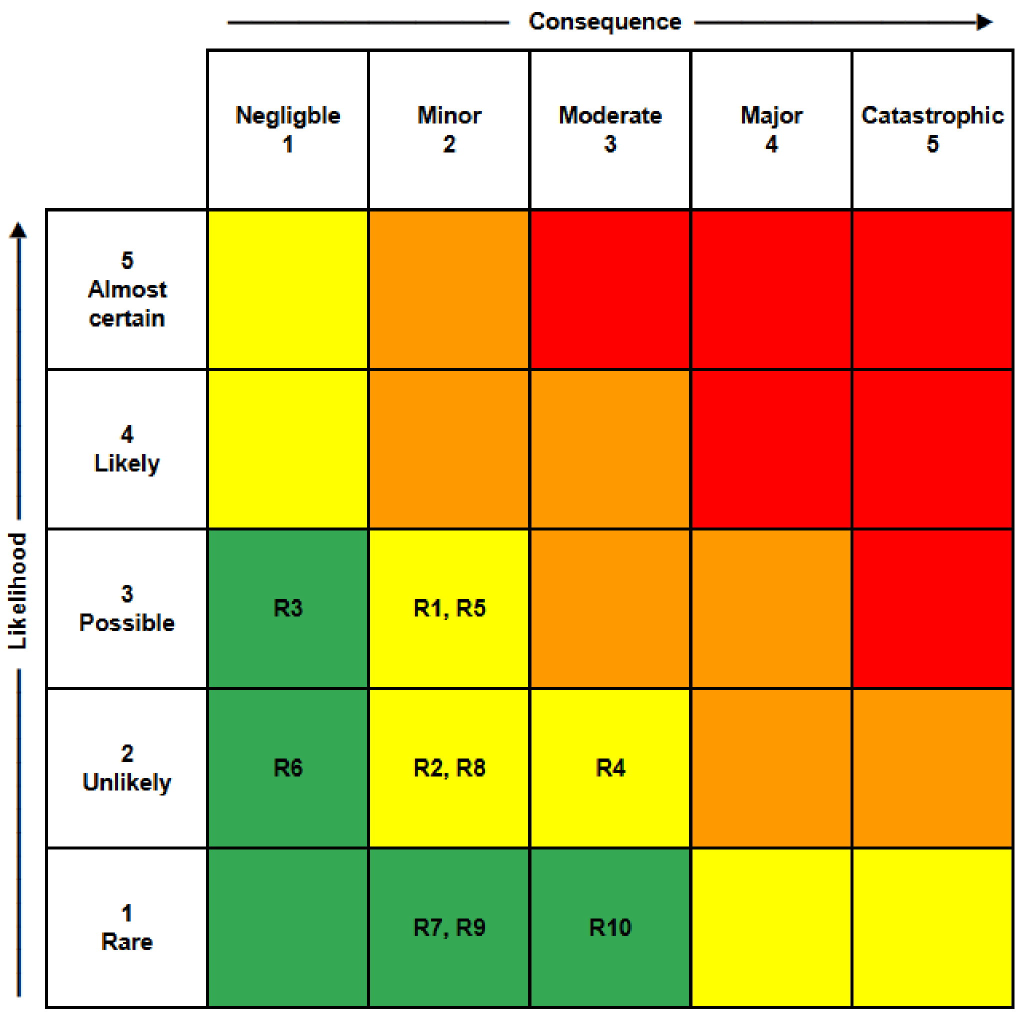


Figure 3: Mitigated project risks mapped to 1-5 rating risk matrix of X: consequence against Y: likelihood (Vine, 2025).

2 BACKGROUND

The Literature Review below explores underlying research behind civic disengagement. Investigating existing methods of non-partisan political information distribution, covering both applications of general and LLM-based solutions. Finally, it concludes on the proposed artefacts novel contributions towards improving UK local political information access.

2.1 Rising Political Disengagement

It was believed John Adams wrote:

> "Liberty cannot be preserved without a general knowledge among the people, who have a right… and a desire to know; […] I mean of the characters and conduct of their rulers."
>
> (Adams 1765)

Arguing the essentiality of a well-informed public as a pre-requisite to a functioning democratic society. Troublingly, research indicates patterns of local political disengagement among UK voters. Historically, only 40% able to name their MP (Hansard Society 2013), more recently 74% felt they had little to no influence on a local level, with 54% expressing little, or no perceived knowledge of Parliament (Hansard Society 2019).

When constituents are uninformed and disengaged, they risk disenfranchisement of meaningful local representation. Unintentionally deprived of their democratic rights, this can manifest as apathy towards local decision-making, declining election turnout (Uberoi and Johnston 2022), and cynicism with political processes. In 2024 alone, over 64% of respondents believed their politicians do not care about them (British Election Study 2024).

Contemporary sources argue numerous reasons, e.g., decline of local journalism (House of Commons: Digital, Culture, Media and Sport Committee 2023), perceived lack of influence on local politics (Hansard Society 2019), unintentional algorithmic amplification of viral national politics over local issues (Fischer et al. 2020), emerging security concerns driving MPs to scale back community engagement[3] (Matthews and Haughey 2024), and intense media focus on 'Westminister debates'[4] (Department for Culture, Media and Sport 2019), among others.

[3]Emerging security concerns regarding the personal safety of MP's following the ideologically driven murders of Jo Cox (BBC News 2016) and Sir David Amess (BBC News 2022), alongside the 2017 terrorist attack on the Houses of Parliament (UK Parliament 2017).

[4]'Westminister debates' is a common summary term to describe ongoing national politics within the Houses of Parliament; parliamentary debates, national political issues/figures, party politics, etc.

2.2 Existing Solutions Tackling Political Disengagement

2.2.1 Digital Political Information Systems

Improving political information access through digital technology is an expanding field of academic interest.

Analysis of 116 citizen-to-government participation tools (Shin et al. 2024) reveals a diverse landscape of subject-specific problems, shown in Figure 4. Summarily, the problem space of 'political information systems' is broad. Existing work seeks to tackle similar, but non-identical civic challenges (primarily civic decision-making, co-funding and problem/identification solution (Shin et al. 2024, p. 17)). Analysing a range of existing solutions allows further understanding and refinement of project novelty to wider research.

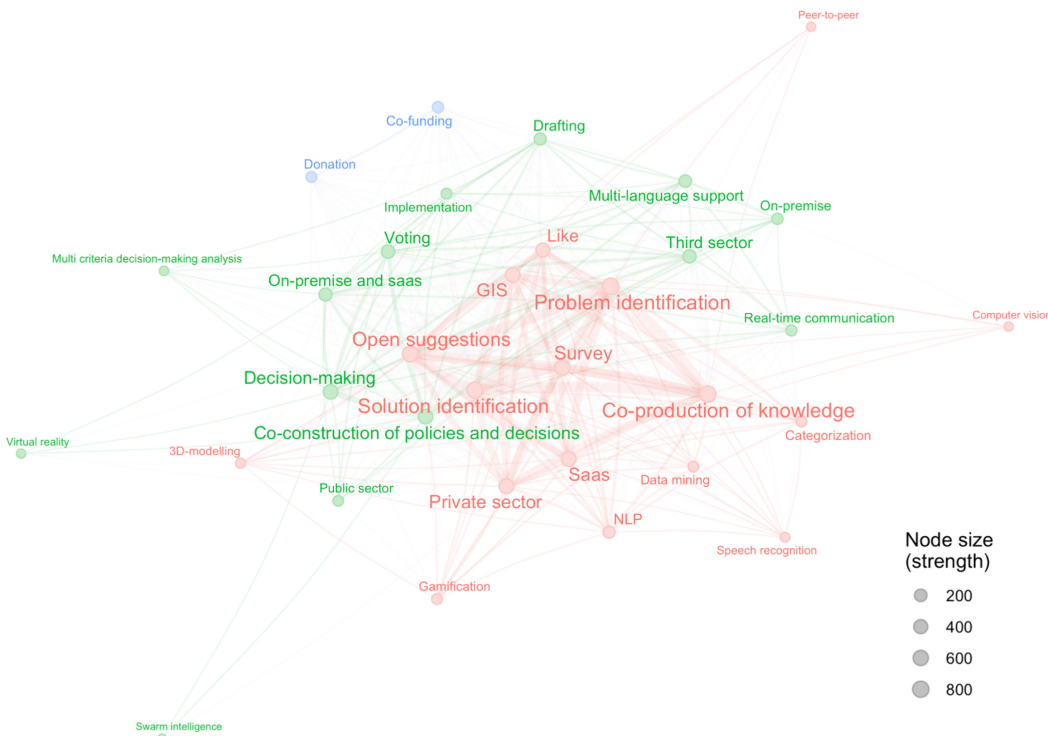


Figure 4: Co-occurrence network of genes representing systematic analysis of 116 citizen participation tools. (Red: problem/solution identification; Green: collective decision-making; Blue: co-funding/donations) (Shin et al. 2024)

Contemporary digital civic-engagement approaches often present as data-driven visualisation tools detailing specific political subjects. These tools allow users to engage with complex civil data in easy-to-digest visual paradigms.

For instance, many tools gravitate towards simplified reporting of electoral processes. For example, (Méndez and Moreno 2021) shown in Figure 5 improves historical comparative analysis of existing government-provided Ecuadoran electoral information.

Other tools may aid navigation of political bureaucracy, e.g. BudgetMap (Kim et al. 2016) shown in Figure 6 which classifies US budget programs based on user-tagged social issues, promoting public interest

Figure 5: A treemap from Ecuadoran elections (Méndez and Moreno 2021).

policies over time.

Some services perform similar functions to the proposed artefact. Connect2Congress (Kinnaird et al. 2010) shown in Figure 7 analyses US congressional voting patterns, mapping individual ideological over time. CivisAnalysis (Borja and Freitas 2015), similarly visualises roll call votes from Brazil's congress, enabling analysis of voting patterns over the cycle of presidential elections.

Each subject area display efforts improving citizen-access to political information. While some similarly explore simplified displays of representative, each is limited to a specific scope and represent non-UK contexts. This presents an opportunity to examine the strengths of LLM's user-tailored adaptive information systems. Thus, we examine how mapping individual user preferences/accessibility requirements may further improve constituent engagement with the civic process (in respect to a UK political context.)

There are additional non-visual-based implementations that likewise compliment the political process in civic-adjacent fields (e.g. journalism.) For example, some methods improve the political information environment by revealing partisan or emotional bias in existing political journalism (Soroka et al. 2015; Baly et al. 2020) or by combatting disinformation through Transformer-based language models (Pavlov and Mirceva 2022).

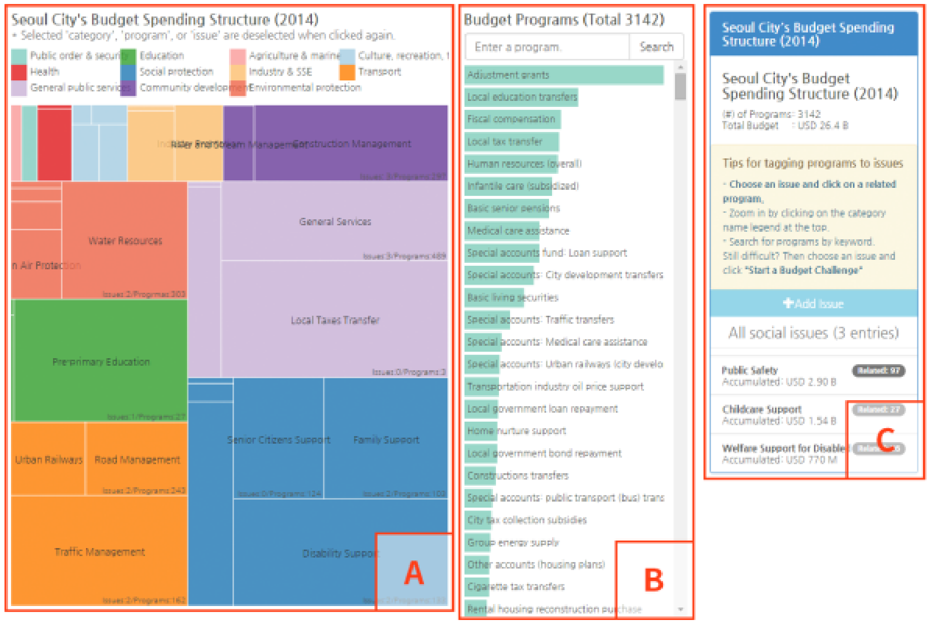


Figure 6: A BudgetMap dashboard (Kim et al. 2016).

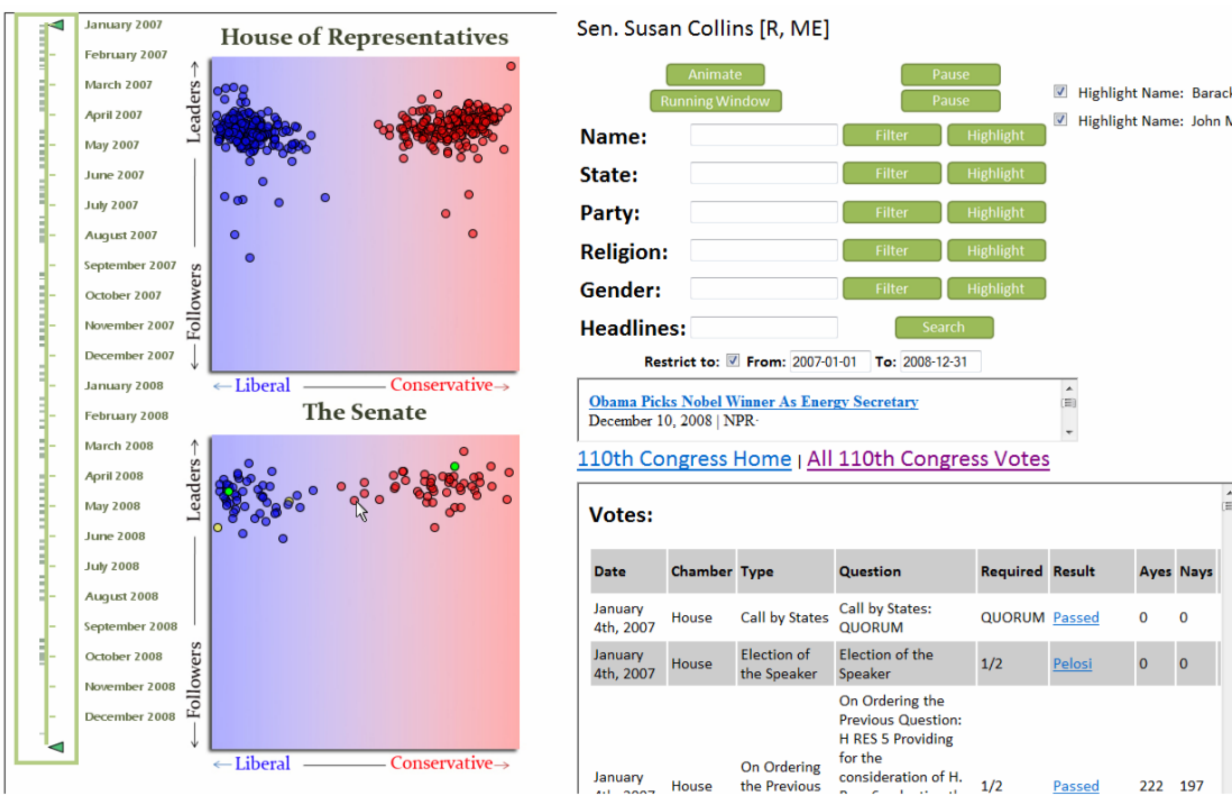


Figure 7: A Connect2Congress dashboard (Kinnaird et al. 2010).

2.2.2 Similar Applications of Large Language Models

Since OpenAI's 2022 release of 'ChatGPT', public interest in diverse applications LLMs has exploded. While arguably LLMs are a nascent field of research (though language models are not), significant advances have already leveraged LLMs for easing engagement with political processes worldwide.

2.2.2.1 Relevant LLM functionality within different industries

The core project relevance of an LLM is similar to existing academic and industry implementations. Fundamentally, it leverages LLMs to retrieve contextual information from 'prompts' and display according to user preferences. Individually, this is use case already exists and is continually expanding across different fields. Thus, while project insights may be unique (given a political context), we can apply literature from numerous fields that mirror the core application of the artefact (LLMs.)

Similar information-based early adoptions of LLMs stem from healthcare technology. In medicine, where inaccurate information can kill, researchers have focused on improving LLM response accuracy (or 'hallucinations'.) (Woo et al. 2025) reported a 39.7% increase (base LLMs reporting below c. 60%, augmented reporting c. 95%) in response accuracy when augmented with Retrieval-augmented Generation (RAG). Other studies report similar now-feasible implementations of LLMs when paired with additional practices, e.g. prompt engineering (few-shot & chain-of-thought techniques) (Miao et al. 2024a).

Across diverse fields such as law (e.g. Thomson Reuters CoCounsel, shown in Figure 9; OpenAI backed 'Harvey')[5], academia (e.g. Google's NotebookLM; OpenAI's 'deep research')[6] and finance (e.g. Bloombergs 'BloombergGPT' (Wu et al. 2023)), research focuses on explainable AI, emphasizing traceability of response-derived information. Given the nature of these fields – where sourcing to enforce information validity is usually mandatory, LLM attribution is critical. Current academic research divides into key areas, e.g. source-aware training (Khalifa et al. 2024), multi-source attribution in long-form generation (Patel et al. 2024) and developing automatic cite evaluation benchmarks (Gao et al. 2023). Each are aspects to consider in the artefacts LLM-response framework.

2.2.2.2 Existing LLM applications within politics

By examining particular rulesets, safety measures and other field-specific LLMs 'adjustments', we can investigate implementation of an LLM-based artefact responsibly/ethically.

Political campaigns are built on voter messaging. Usually, transforming complex political manifestos into well-crafted messaging suiting the average voter. LLMs are already involved in delivering rapid political messaging, in one case, 'emulating' politicians by creating digital representations of candidates (Tosi et al. 2025). In another, creating campaign videos within minutes of rival's announcements (OECD AI Policy Observatory 2023).

Reflections stress a need for usability design principles embedded into LLMs for information dissemination. By designing LLMs with communication strategies (i.e. to imitate a candidate's speaking style, or deliver a clear narrative), they explore potential voter engagement strategies, testing 'strategic' LLM

[5] https://legal.thomsonreuters.com/en/products/cocounsel-core, https://www.harvey.ai/
[6] https://notebooklm.google/, https://openai.com/index/introducing-deep-research/

You

List medication treatment to help slow progression of CKD to ESKD

Kidney Care

Based on the KDIGO 2023 CKD Guidelines, several medications are recommended to help slow the progression of chronic kidney disease (CKD) to end-stage kidney disease (ESKD). Here are some key recommendations:

1. **Sodium-Glucose Cotransporter-2 Inhibitors (SGLT2i)**: These medications are suggested for use in adults and children with CKD but not diabetes. Examples include canaglifiozin, dapagliflozin, empagliflozin, and others. They are compared against placebo, usual care, or an active comparator in terms of their effects on the progression of CKD, complications of CKD, and adverse events.
2. **Nonsteroidal Mineralocorticoid Receptor Antagonists (MRAs)**: These are recommended for adults with Type 2 Diabetes (T2D), an eGFR >25 ml/min per 1.73 m2, normal serum potassium concentration, and albuminuria despite the maximum tolerated dose of RAS inhibitor (RAS1). They can be added to a RASi and an SGLT2i for the treatment of T2D and CKD in adults.
3. **Glucagon-Like Peptide-1 Receptor Agonists (GLP-1 RA)**: For adults and children with CKD and T2D, GLP-1 RAs like albiglutide, dulaglutide, and liraglutide may be used. These are compared against other glucose-lowering agents, placebo, or usual care, specifically looking at their impact on kidney failure, all-cause hospitalizations, and other critical oucomes
4. **Lipid Management**: In adults aged ≥50 years with an eGFR <60 ml/min per 1.73 m2 but not treated with chronic dialysis or kidney transplantation, treatment with a statin or statin/ezetimibe combination is recommended.

These recommendations are aimed at managing the progression of CKD and its complications, and the choice of specific medication should be made based on individual patient factors and in consultation with a healthcare provider. It's also important to note that these medications are part of a proader management strateoy that includes lifestyle changes and management of other health conditions.

If you have CKD or are at risk, it's essential to consult with a healthcare professional for personalized advice and treatment.

Figure 8: An experiment from the RAG-driven LLM ‘Kidney Care’, an example of RAG-use in healthcare (Miao et al. 2024b).

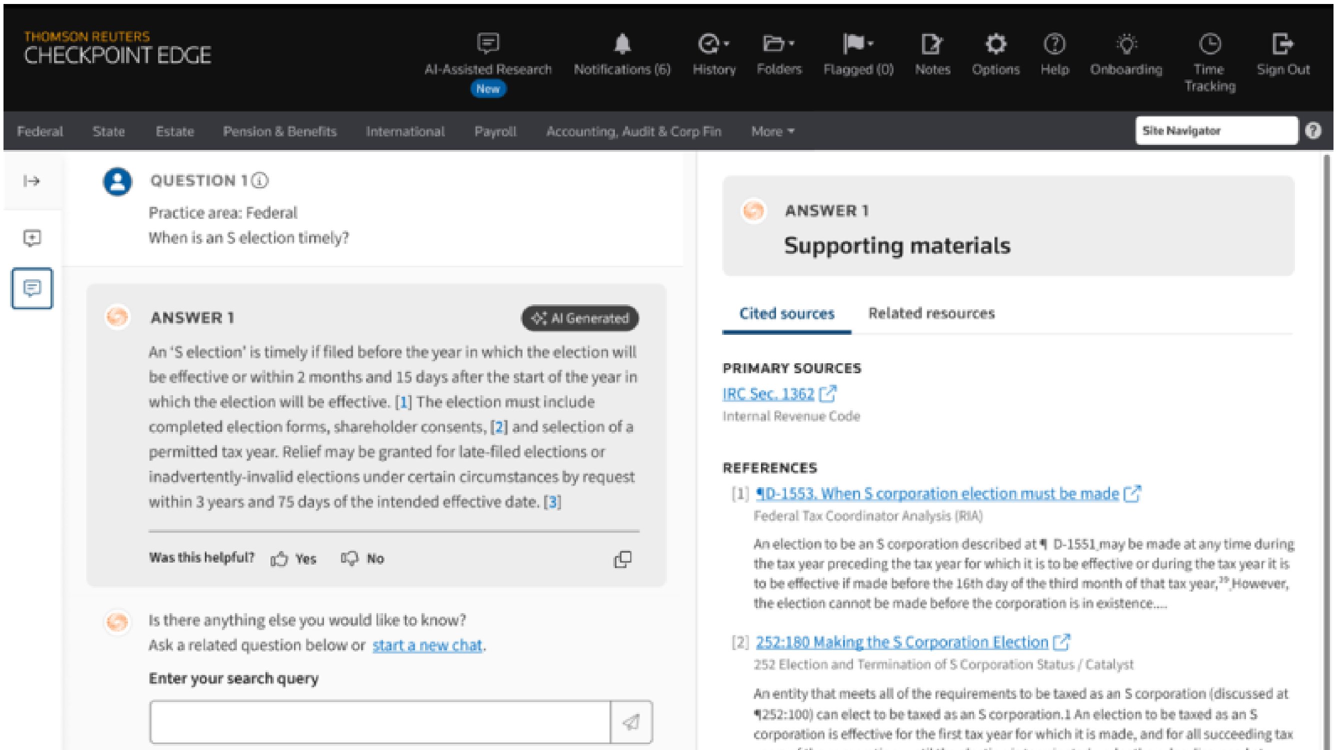


Figure 9: A Thomson Reuters CoCounsel interface.

responses. Though within artefact development, optimising to distil complex political information into accessible language for a constituent (rather than partisan communication.)

2.3 Limitations of Existing Solutions

This section explores limitations of political information systems, then moving to analyse relevant LLM challenges.

2.3.1 Limitations of Digital Political Information Systems

There are perhaps inherent structural limitations to existing political information systems that LLMs may prove capable of enhancing.

Legislation is a typically information-dense, jargon-littered environment. Thus, political information systems typically simplify bureaucracy through visualisation (as shown in GovTrack[7], Connect2Congress, BudgetMap, CivisAnalysis, and (Méndez and Moreno 2021).) Interesting, many seem to inadvertently rely on assumed pre-existing knowledge of political competence[8] and personal understanding of transforming dense information structures into practical, relevant information.

Transforming this into digestible information for constituents is time-consuming. Which, given perpetual operation of Government, may be infeasible to address outside of pressing national affairs naturally attracting media attention. The automative-at-scale abilities of LLMs could present a unique opportunity for rapid improvement of information access. A rudimentary example of such LLM augmentation is shown between Figures 10 & 11, making policy-specific jargon more accessible.

Such systems opt for traditional digital delivery, operating as a website/application, featuring means to filter and navigate catalogues of curated information. Typically, quality of these services rely on user experience design, and capacity of administrators to keep records up to date.

The scale of such challenges may become intensified under complex bureaucratic systems. Figures 12 & 13 show an example of such shortcomings in the work of a UK charity who deploy political monitoring services. These records are contrasted against live parliamentary record.

LLMs could offer a paradigm shift, enabling dynamic retrieval/generation of information to user preferences. Mirroring a search engine, LLMs could unlock capability to navigate/curate swathes of data. Previously, this type of personalised experience would have relied on conversation with a human subject matter expert. For instance, a competent researcher may take time to link a MPs speech on public transport in 2020 compared to voting records in 2025. Instead, a rigorous LLM data pipeline could perhaps generate individual analysis, summarising how this affects user interests.

[7] https://www.govtrack.us/

[8] 'Competence' representing knowledge of terminology, activities, organisations, etc relating to legislative administration.

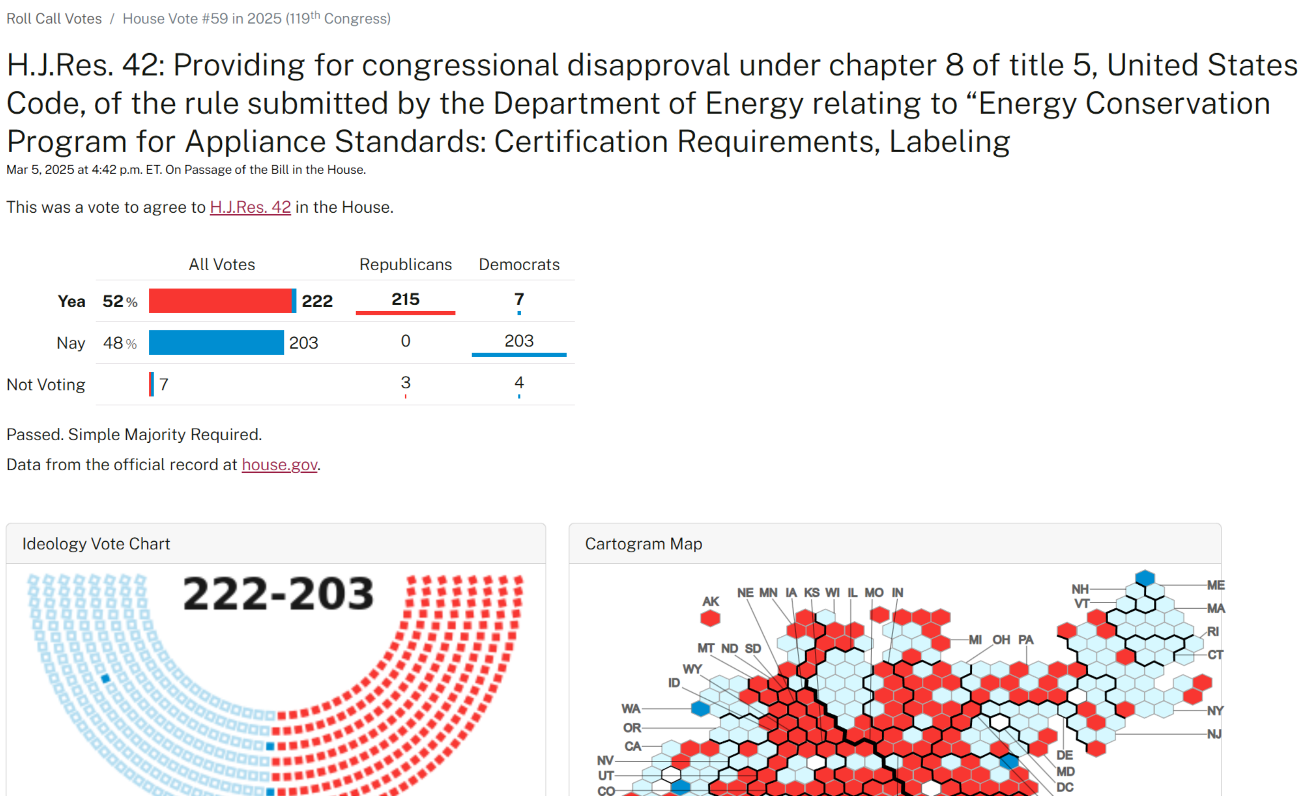


Figure 10: An example of a standard bill breakdown on the US' GovTrack.

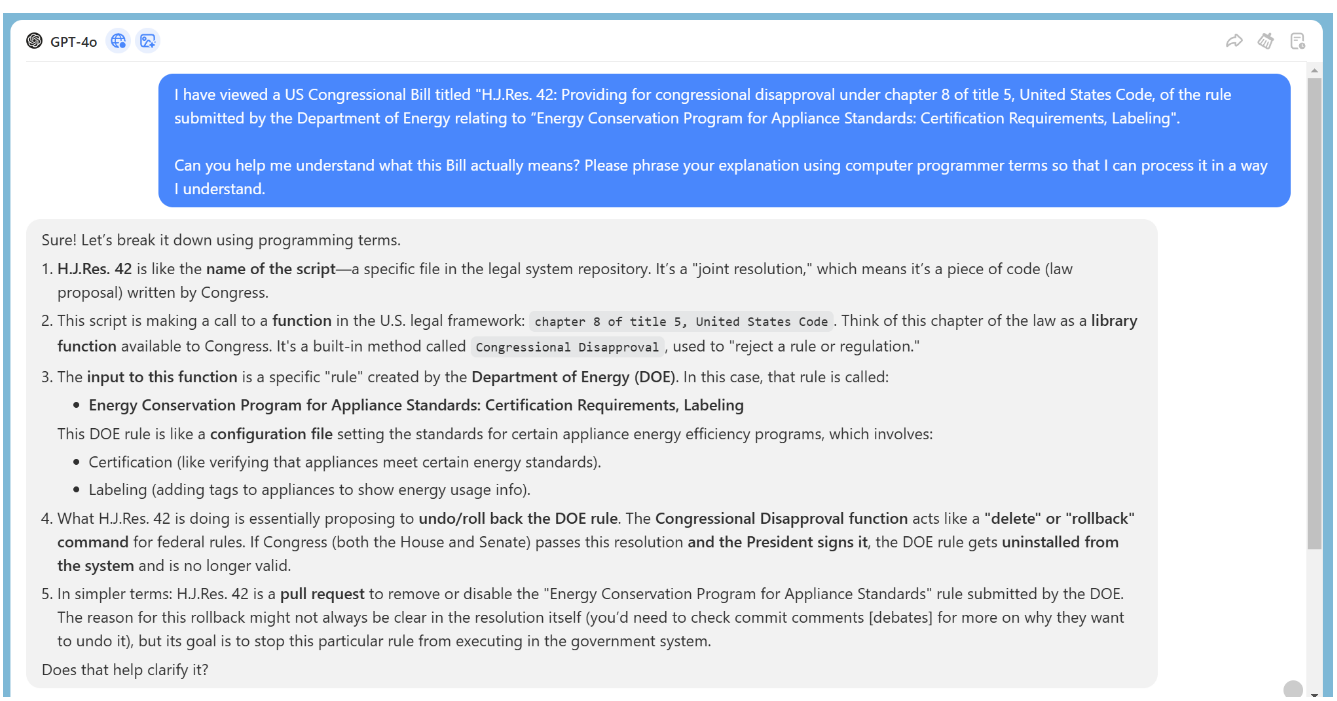


Figure 11: An example of querying an LLM on specific political terms shown in GovTrack.

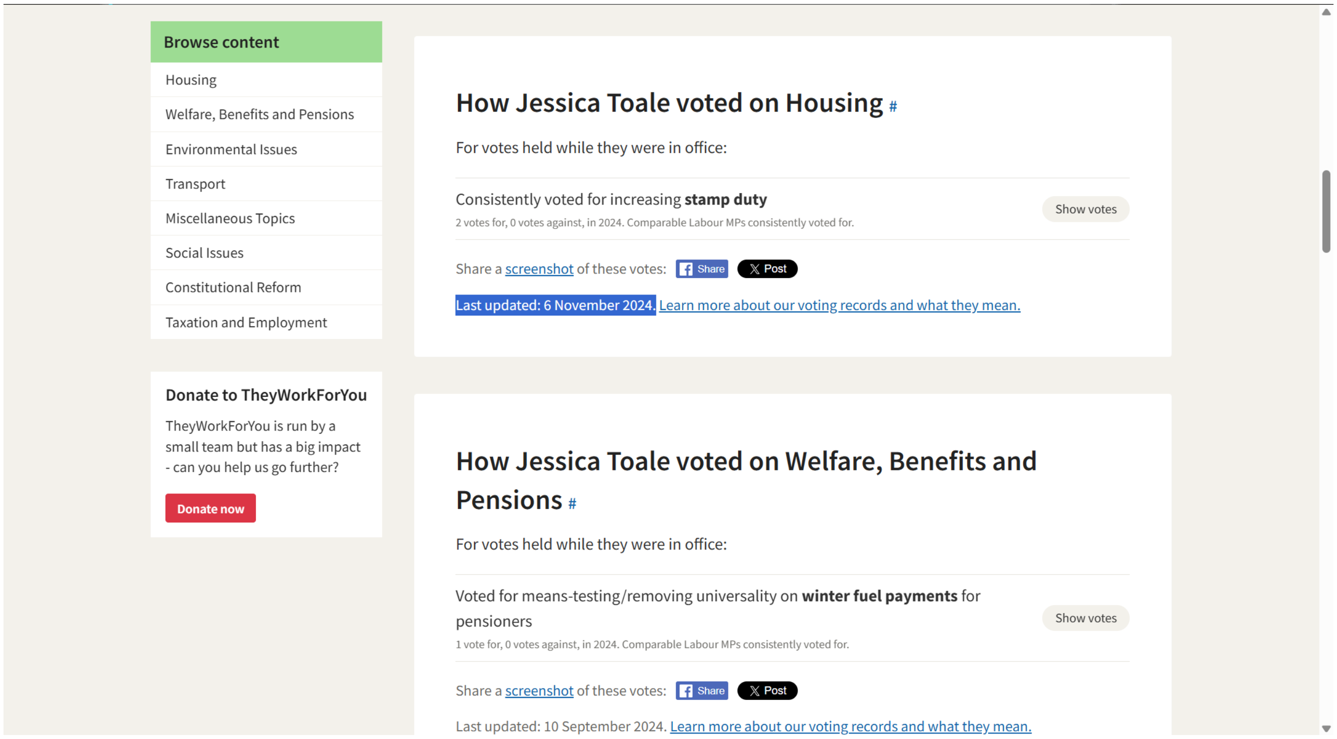


Figure 12: TheyWorkForYou's summary of housing policy voting behaviour, last updated five months prior to date of writing (14/03/2025).

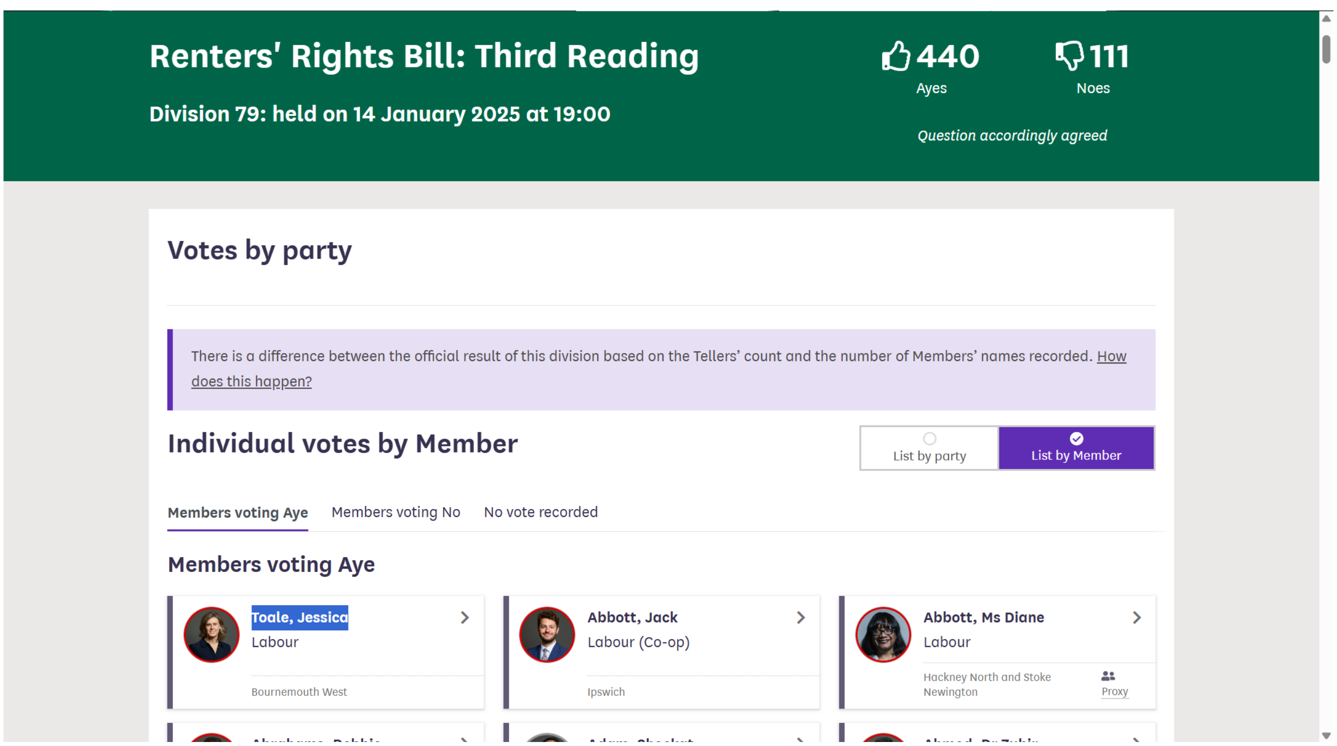


Figure 13: UK Parliamentary record of recent voting on the Renters Rights Bill, a landmark political housing bill, not covered by TheyWorkForYou's MP summary.

2.3.2 Limitations of Traditional Large Language Models

Having identified similar implementations and capabilities of LLMs, we ask, “why can’t a freely available generalist LLM[9] do this now?”

Generalist LLMs, while offering incredible advancements across vast information fields, may suffer from some drawbacks at this early stage in their lifecycle. While continual releases of better models reduce pain-points, (Shahzad 2025, p. 6), some contexts may struggle with out-of-the-box deployment, instead requiring nuanced, bespoke solutions.

Table 9 & 10 discusses challenges of generalist LLMs, exploring potential augmentations addressing such concerns.

Table 9: 1 of 2: A table showing different identified challenges facing generalist LLMs, applied to the project artefacts use case. Challenges are split into explanations, their subsequent implications and consequent potential solutions to address them.

Explanation	Implications	Potential Solutions
	LLM ’Hallucinations’	
As discussed in Section 2.2.2.1, LLMs, though improving with newer models, continue to suffer from ’hallucinations’[a] at significant rates[b].	Without proper safeguards an LLM may exacerbate existing political dis/mis-information, potentially further eroding democratic debate.	• Integration(s) of RAG. • LLM output accuracy measures. • ’Human-in-the-loop’ practices (A key example for learnings can be drawn from i.AI’s Caddy[c], an ongoing public interest LLM). Drawn from previous Literature Review methods discussed to improve factual outputs.

Continued on next page

[a]Referring to LLM responses which are fully or partially derived from inaccurate or seemingly non existent information.
[b](Descamps et al. 2024; Waldo and Boussard 2024)
[c]The UK Government’s Incubator for Artificial Intelligence (i.AI) flagship programme, Caddy (https://ai.gov.uk/projects/caddy/)

[9]Referring to widely available, mainstream, general use-case conversational LLMs (e.g. ChatGPT, Gemini, Claude, etc.)

Table 10: 2 of 2: A table showing different identified challenges facing generalist LLMs, applied to the project artefacts use case. Challenges are split into explanations, their subsequent implications and consequent potential solutions to address them.

Explanation	Implications	Potential Solutions
LLM Explainability		
LLMs are inherently unexplainable through traditional means given hundreds of billions to trillions of model parameters present in frontier models[d]. A majority of the best-performing generalist LLMs are proprietary and profit-driven (e.g. OpenAI's GPT; DALL-E, Google's Gemini, Anthropic's Claude, etc), choosing not to disclose critical development choices to protect commercial interests and market share.	A lack of explainability may challenge trustworthiness of LLM outputs. This may lead to unintentional amplification of misuse, misinterpretation and biasing of information within existing public discourse.	• Source-aware tracing • Multi-source attribution • Metrics for measuring accurate citations. Drawn from previous methods discussed in Section 2.2.2.1 to strengthen information explainability.
LLM Political Bias		
Ongoing research shows some left-of-centre political bias among LLMs. Of UK & EU LLM policy recommendations, 80%+ were reported as left-of-centre. Sentiment towards political parties showed imbalances, skewing more positively (average score(s) of: left-leaning: +0.71; right-leaning: +0.15, range of -1 to 1) to left-leaning parties[e]. This bias was detected across all major LLMs surveyed, and across all major EU (& UK) countries[e]. Similar reports note left-leaning bias towards the UK Labour Party[f], amplifying as models are trained on increasingly larger datasets[g].	Implicit non-partisan LLMs may unintentionally further skew or mis-represent information, eroding an increasingly polarised public discourse[h]. Instead of promoting transparent access to political information, this may actually intensify existing distrust of political information media[i].	Academic research-based debiasing strategies: • **Prevention** E.g. (re)-curation of training data and various prompting methods to 'avoid' underlying bias[j]. • **Detection** Developing automated & human-evaluated metric frameworks to evaluate/diagnose politically biased text[k]. **Artefact-based choices:** Due to constraints of time, context, cost, complexity, etc adaptations must be practical. The artefact will therefore include, to prevent politicial bias: • A peer-reviewed academic prompting strategy.

[d](Minaee et al. 2024, p. 4)

[e](Rozado 2024)

[f](Motoki 2023)

[g](Rettenberger et al. 2024)

[h](Duffy et al. 2019; Grechyna 2022)

[i](House of Lords, Communications and Digital Committee 2024)

[j](Navigli et al. 2023; Gallegos et al. 2024)

[k](Liang et al. 2021; Bang et al. 2024; Rettenberger et al. 2024)

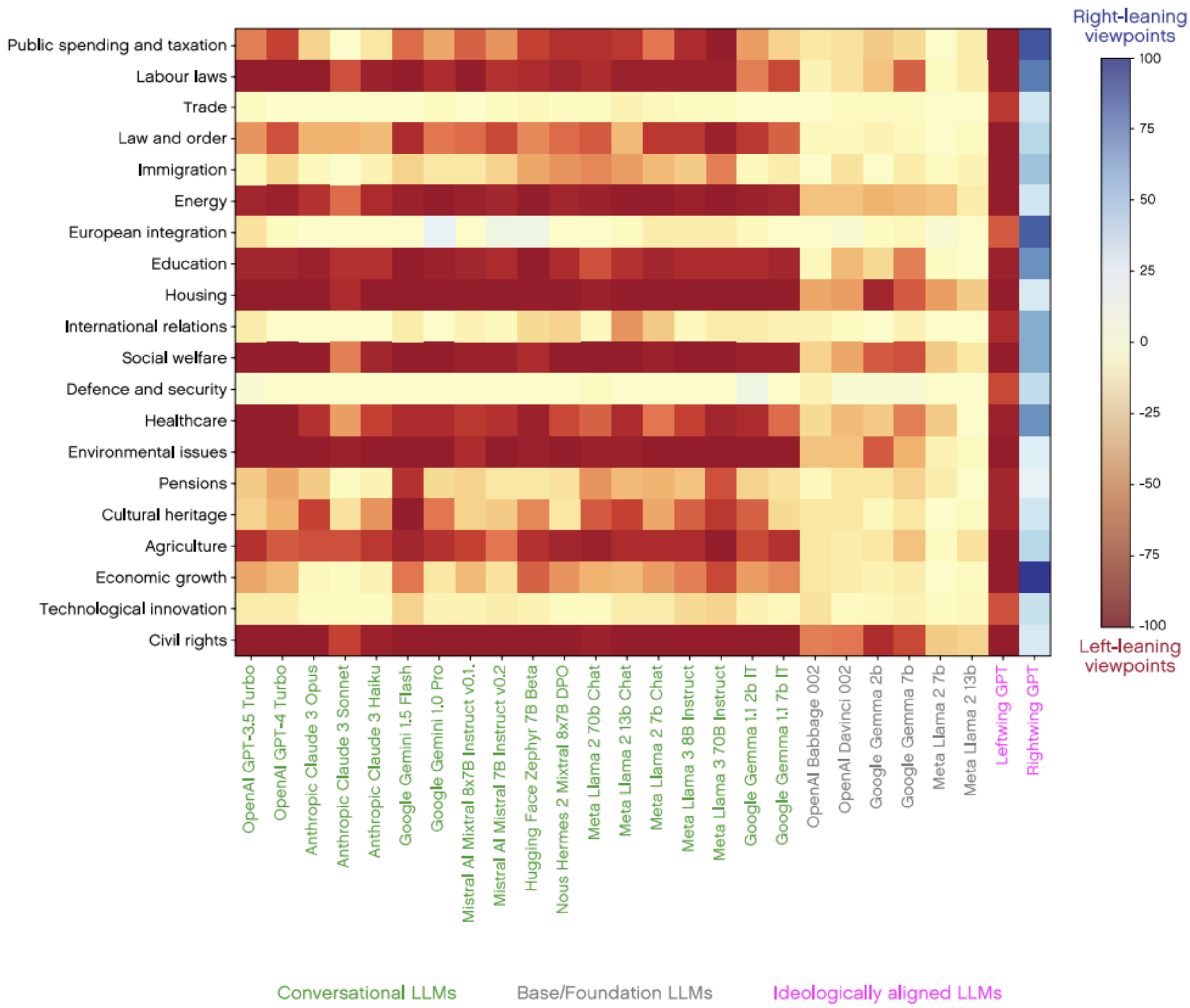


Figure 14: A heatmap showing viewpoint scoring on a range of political policy recommendations made for a UK context by a range of LLMs (Rozado 2024)

3 METHODOLOGY

This section details choices of project management (3.1.1) and software methodologies (3.1.2), covering strategies used in the project's lifecycle. Thereafter, project organisation (3.2), requirements gathering (3.3) and evaluation (3.4) discuss organisational tools, requirements/scope elicitation, and artefact evaluation frameworks.

3.1 Project Management

3.1.1 Project Management Methodology

Project management is operationalised via traditional waterfall methodology, detailed by the Gantt chart of Figure 15. Given a standardised academic schedule incorporating distinct phases of research, experimentation/development and analysis/results, a clear sequential methodology ensures consistent project progression. This utilises required project phases as a strength, aligning timely delivery/transition through each phase while maintaining attention to total project scope.

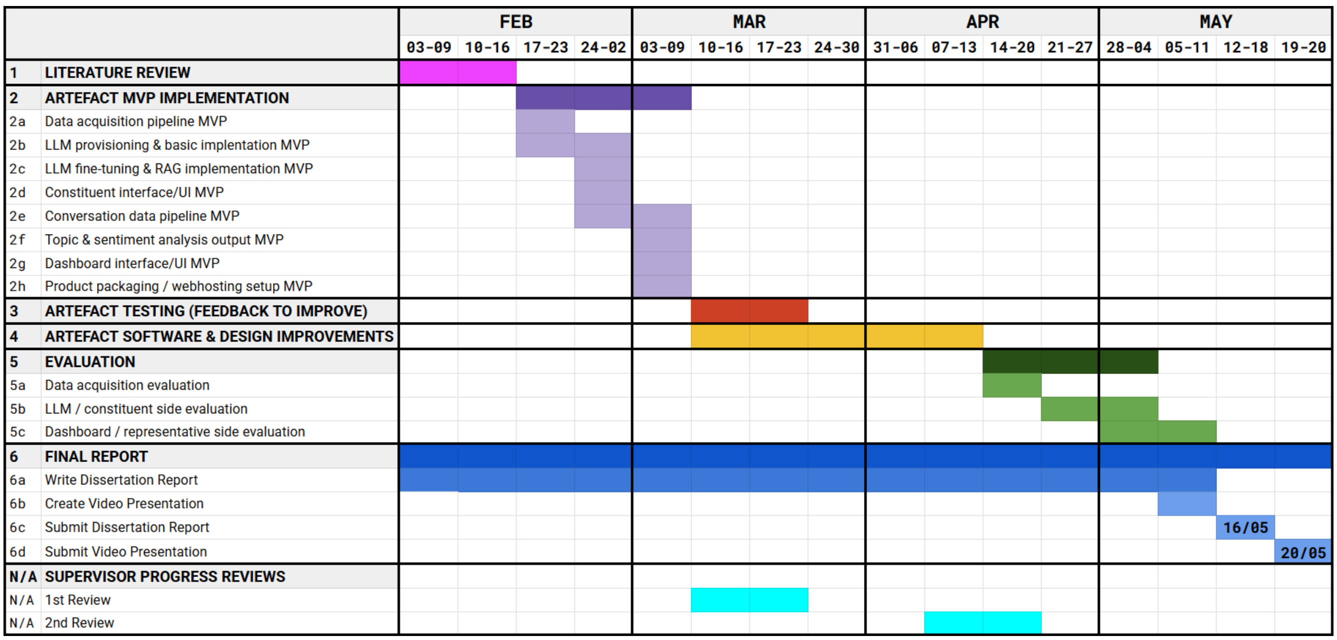


Figure 15: Original Gantt chart of project timeline (Vine, 2024).

3.1.2 Software Management & Development Methodologies

Academic writing and software/artefact development routines often juxtapose. Therefore, some concessions help to seamlessly merge these processes. A distinct **software management** methodology is incorporated to compliment the Software Development Lifecycle (SDLC): SCRUM.

SCRUM is an agile-based SDLC approach focusing on iterative development, refining a minimum viable product through ‘sprint’ intervals. SCRUM reinforces natural strengths of this project phase, pairing time-to-completion with the MoSCoW requirements prioritisation detailed in Section 3.3. As the SDLC progresses, feature completion is ordered by importance (most-to-least critical.) SCRUM additionally helps re-allocate development time based on the unpredictability of real-time development (some features may not match forecasted development time.) SCRUM is adapted due to a single author workflow, maintaining aspects of product & sprint backlogs and cyclical sprints, while negating team-based components (e.g. sprint meetings.)

Professionally, iterative development is often ‘framed’ within a **development methodology**, enforcing expert software design/testing principles. Here Behaviour-Driven Development (BDD) is used, an extension of Test-Driven Development (TDD). Where TDD defines outcome-based automated tests, BDD extends TDD by orientating towards user behaviour outcomes. Simple language is used, outlining system behaviour (e.g. Gherkin syntax) based on ‘user stories’. These are written into automated tests, and refined code should pass such tests to ‘satisfy’ feature development. BDD is chosen from necessity for a user-centric/usability approach (users must seamlessly interact with a custom-built LLM) and space for iterative improvement (LLM response relevance/accuracy will be refined.)

As a data science-based artefact, further exploration into software architectural theory/design (e.g. UML diagrams) would be misplaced as this is not the core focus of the project. Instead, this project draws from just enough practices to deliver a well-crafted data science focused software artefact.

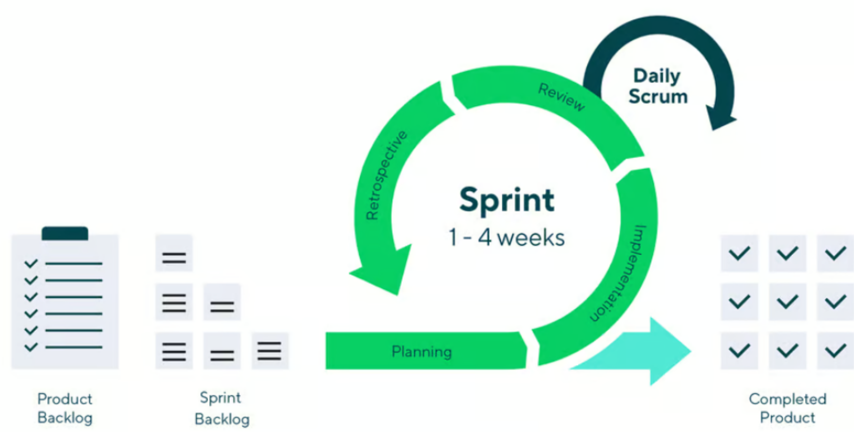


Figure 16: An example of the SCRUM ‘sprint’ process. Note that not all elements (i.e. Daily Scrum) are relevant given a one-person project process (Zhezherau 2023).

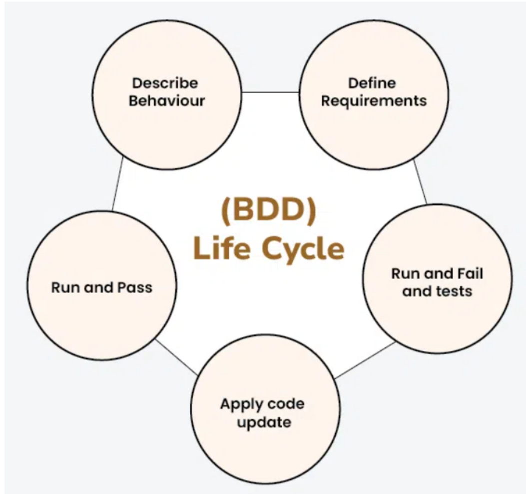


Figure 17: A diagram example of the Behaviour-Driven Development framework (GeeksforGeeks 2024).

3.2 Project Organisation

Project organisation occurs week-to-week through several tools/frameworks.

At **project** level, a Gantt chart (Figure 15) tracks current progress against the expected project timeline. **Project meetings** occur weekly (Monday) by the Project Supervisor, overseeing timely project progress.

At **software development level**, a Jira board (Figure 18) plans one week sprint cycles within the aforementioned SCRUM framework.

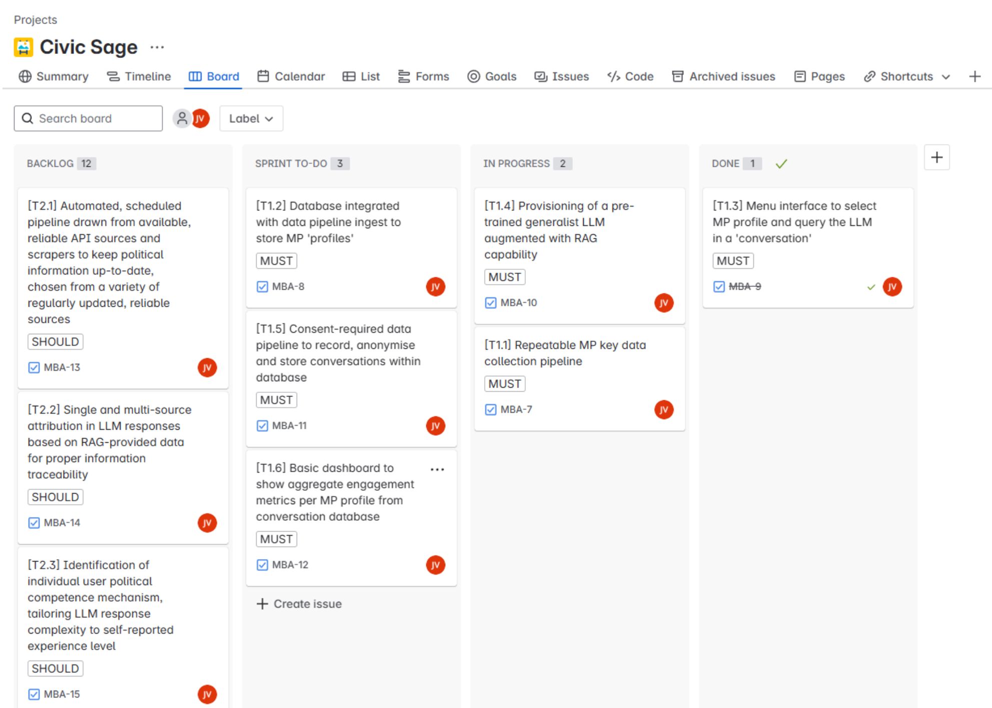


Figure 18: A screenshot of the project Jira board during Sprint 1 (Vine, 2025).

3.3 Requirements Gathering

Requirements gathering elicits specific artefact objectives for achievement. By framing the objectives, or 'requirements', feature development is ordered allowing for transparent and objective appraisal of the project's success.

As previously, a specific management framework, MoSCoW[10] is employed. MoSCoW forces requirements to conform to a structured, hierarchical system, establishing sensible project flow. This ensures an easy-to-follow development approach, working from most-to-least critical features seamlessly layered within SCRUM sprint cycles, supported by Jira.

Subsequent MoSCoW requirements will be derived from initial project objectives (Section 1.4) and further refined from adapting strengths and limitations identified in relevant Literature Review subsections.

Table 11: Breakdown of the MoSCoW prioritisation system.

Prioitisation	**Explanation**
Must have	Requirements that are critical (mandatory) to project success.
Should have	Requirements that are important but not mandatory for project success.
Could have	Requirements that are optional, inclusion of which may improve level of project success.
Won't have	Requirements that may be beneficial, but identified as out-of-scope for the current project.

[10] MoSCoW: (M: Must have; S: Should have; C: Could have; W: Won't have).

3.4 Evaluation

This section outlines the evaluation of achievement process for both the project process and software artefact.

3.4.1 Project Evaluation

To evaluate the overall project process, success criteria elicited from aims and objectives detailed in Section 1.4 will undergo appraisal. Each success criterion will be designated as 'successful' or 'unsuccessful' dependent on full criteria being met at project conclusion.

3.4.2 Software Artefact Evaluation

Artefact evaluation will comprise both automated and manual components, completed at the end of the SDLC.

3.4.2.1 Automated Evaluation Tests

Automated evaluation utilises two measures of assessing artefact capability:

- Rule-based tests (fixed.)
- Intelligent tests (multi-step.)

Rule-based tests

- **BDD unit tests:** Through the BDD framework, implemented unit tests will determine completion of each feature in accordance with intended behaviour. A majority of tests will focus on various types of LLM engagement.
- **User reports**: The artefact will integrate a 'report' feature, flagging LLM responses (as detailed in Table 22, T2.5). These reports will be collected for analysis.

Intelligent tests

Measuring complex behavioural expectations requires an advanced approach. For this, the project develops a novel **meta-evaluation via agent-driven testing** approach, utilising high-level concepts (e.g. agentic systems paired-evaluation, multi-turn iterative feedback) of the Agent-as-a-judge methodology (Zhuge et al. 2024) which has shown promising results (Fu et al. 2023; Chern et al. 2024).

A provisioned generalist LLM will act as a 'user' with instructions to 'query' the project LLM on a subject area. The 'user' will continue adapting questions until a project LLM response contains expected criterion, measuring the number of 'conversational turns' required for success.

3.4.2.2 Manual Evaluation Methods

Manual evaluation will involve weighted appraisal of completed MoSCoW objectives and user questionnaires.

MoSCoW appraisal will sum the number of completed objectives, weighted by prioritisation (`Must have = 3`, `Should have = 2`, `Could have = 1`), outputting a total completion score, contrasted against the total possible (all.) For success, at least 75% of total points available must be achieved.

User questionnaires will conduct individual artefact testing sessions with accompanying surveys for testers. Testers will be recruited, categorised and questionnaires tailored according to two distinct groups, 'standard users' and 'expert users'. Identifying characteristics are detailed in Table 12. All participants must minimally be non-vulnerable adults.

Table 12: Summary of qualifying characteristic indicators of distinct user groups employed during artefact user questionnaires. (S.U. = Standard User, E.U. = Expert User)

Group	Group Size	Group Description	Age Range	Digital Competency
S.U.	10 - 15	A vast array of people with differing careers, motivations, education levels and interest levels in politics. Recruited from authors personal network.	20 - 70	Low-High (various)
E.U.	2 - 4	Professionals at mid-career level (3-8 years) in data science/engineering/machine learning. Recruited from authors colleagues at UK Cabinet Office.	30 - 50	High

Participants will be invited to test the artefact, completing the subsequent survey on their own time, unsupervised. Access will always be provided digitally (email, text message, etc) by URL after confirming agreement/understanding required to participate. Users will be given a brief explanation of artefact purpose (demonstrated in the home page) before conducting unsupervised testing sessions.

Following testing, 'standard users' will complete a Systems Usability Scale (SUS) (Brooke 1996) questionnaire, with additional boxes to provide qualitative feedback. 'Expert users' will receive the same SUS questionnaire, coupled with additional qualitative-based questions for their analysis on artefact technical capability. A minimum achievement of 75 / 100 (SUS margin of 'Acceptable') is required for success.

After user surveys are completed, qualitative feedback will be reviewed to identify artefact 'pain points' for iterative improvement.

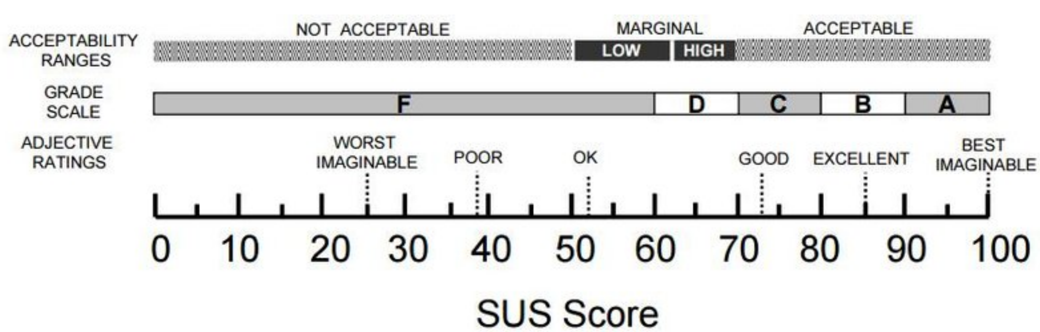


Figure 19: A diagram showing Systems Usability Scale (Gullà et al. 2019).

This overall process will achieve a multi-faceted, holistic evaluation of both project and artefact from varying frames of reference. A summary of approaches is featured in Table 13.

Table 13: Summary of approaches used in evaluation of project. (Quant. = Quantitative, Qual. = Qualitative; Co. = Contextual (to be decided contextually.))

Subject	**Type**	**Data Type**	**Evaluation**	**Goal**
Project	Manual	Quant. & Qual.	Aims & objectives appraisal	C.
Artefact	Automated	Quant.	BDD unit tests	100%
Artefact	Automated	Quant.	User reports	C.
Artefact	Automated	Quant.	Meta-evaluation	C.
Artefact	Manual	Quant.	MoSCoW objectives appraisal	75%
Artefact	Manual	Quant. & Qual.	Survey (Qual. questions & SUS)	C. & SUS: 75/100

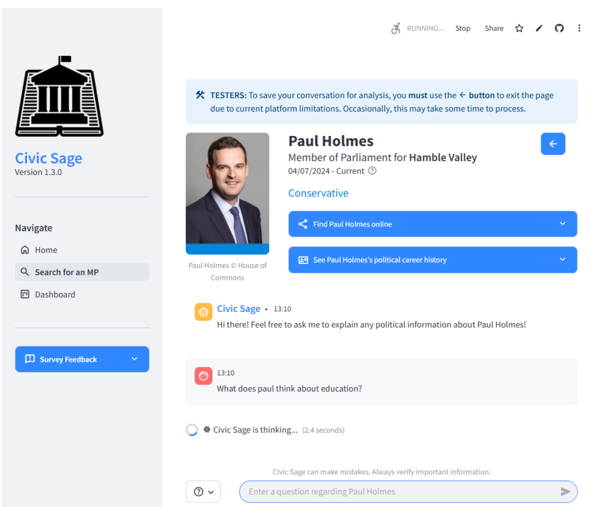
4 OVERVIEW OF ARTEFACT

4.1 Artefact Summary

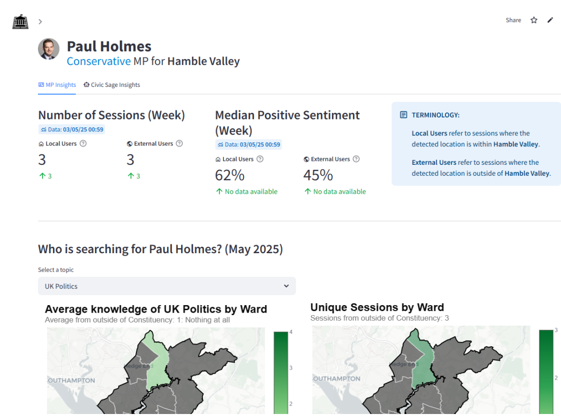
The artefact, branded ‘Civic Sage’, is a fully deployed/hosted web service. It successfully searches, ingests and enables querying of UK MP data from various pre-defined APIs using an adaptive LLM tailored to user needs, finally analysing user-conversational insights.

To current knowledge, no political information system exists akin to Civic Sage, nor comparable services for a UK context. Civic Sage serves as a public good proof-of-concept hosted on GitHub[11] and deployed via Streamlit Cloud[12], leveraging various enterprise services.

As mentioned in Section 3.4.2.2, MoSCoW objectives are elicited from the project foundation (Section 1) and background research (Section 2) which inform artefact feature development and subsequent challenges. These are shown in Tables 21 & 22, Appendix E.1. The completed SCRUM sprint cycles are summarised according to the MoSCoW objectives in Table 14.



(a) A screenshot of Civic Sage’s ’search for an MP’ (RAG-LLM) screen with MP Paul Holmes selected.



(b) A screenshot of Civic Sage’s ’dashboard’ screen with MP Paul Holmes selected.

Figure 20: Two sub-plots showing the main pages of the artefact: Civic Sage.

[11] https://github.com/vine-james/civic-sage

[12] https://civicsage.streamlit.app/

Table 14: A table showing artefact completion broken down by SCRUM sprint cycle (week, all dates in 2025), paired with each MoSCoW elicited artefact objective. ([TX.Y] = X: Prioritisation Level (1-4), Y: Task)

Sprint cycle	Progress
1 07/04 - 13/04	• Completion of T1.3. • Preliminary work on T1.1, 1.2, 1.4.
2 14/04 - 20/04	• Completion of T1.1, 1.2, 1.4. • Preliminary work on T1.5, 1.6, 2.1.
3 21/04 - 27/04	• Completion of T1.5, 1.6, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6 3.4. • Preliminary work on T3.2, 3.3.
4 28/04 - 04/05	• Completion of T3.2, 3.3, 3.5, 3.6. • Preliminary work on T3.1.
5 05/05 - 11/05	• Completion of T3.1. • Additional finalisation of BDD unit tests & intelligent tests.

4.2 Artefact Key Developments

Below, highlighted by Table 15, data-science-based key development points are discussed. Other components are omitted due to degree focus and can be inferred by comparing Table 15 to overall artefact objectives.

Table 15: A table summarising key data-science-focused tasks/sub-tasks of artefact development, related code, services and corresponding artefact objectives.

Task	Sub-tasks	Code files	Deploy services	Artefact objective

Data Collection Pipeline	• Data collection through API sources
	• Transforming data entries into vector embeddings & storing in vector database
		`files/lambda_...`
		`- lambda_get_da...`
		`- lambda_get_all...`
		
		`utils/`
		`- boto_utils.py`
			AWS Lambda, ECR	• T1.1, 2.1
				• T1.2, 2.2

RAG-LLM Answer Pipeline	• RAG search pipeline • Additional indicators provided to LLM • Building decision pathways • Mitigating potential political bias	`utils/` `- rag_llm_utils.py`	Pinecone Streamlit Cloud	• T2.2, 3.5, 3.6, 1.4 • T2.3 • T3.4, 3.5 • T2.6

MP Conversation Storage & Analysis Pipeline	• Conversation Anonymisation
	• Conversation Pre-analysis & Storage
	• Monthly MP Insights Analysis
		`files/lambda_co...`
		`- lambda_analys...`
		
		`utils/`
		`- analysis_utils.py`
		`- boto_utils.py`
			Streamlit Cloud
			AWS Lambda, ECR, S3, DynamoDB
				• T1.5
				• T1.5

Dashboard Data Ingestion & Data Presentation	• Data Ingestion
	• Data Presentation
		`pages/`
		`- dashboard.py`
		
		`utils/`
		`- streamlit_utils.py`
		`- plot_utils.py`
		`- boto_utils.py`
			Streamlit Cloud
			AWS Lambda, ECR, S3
				• T1.6
				• T1.6, 3.2, 3.3

4.2.1 Data Collection Pipeline

4.2.1.1 Data collection through API sources

AWS Lambda 'data getters' are programmed to collect MP data from API sources on 24 hour and 1-month routines, shown in Figures 21 & 22.

In the current version one source is used: Parliamentary Hansard data[13], containing extensive activity logs per MP. Sourcing reliable non-scraped data proved challenging within time constraints. For further discussion, see Appendix J.1.

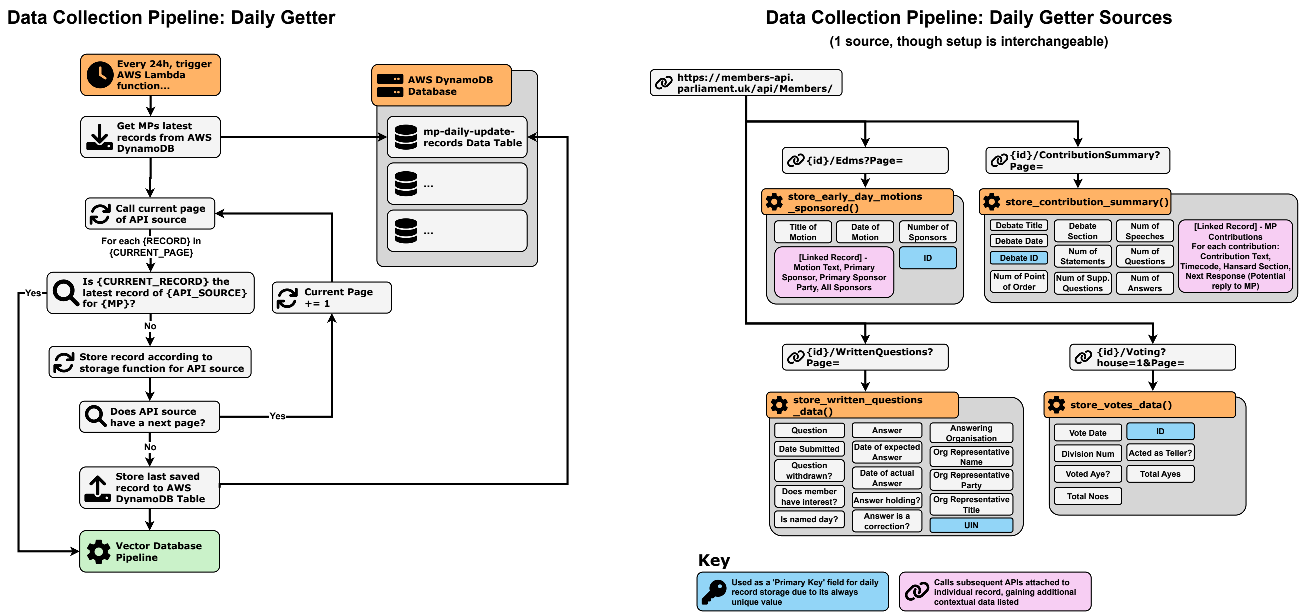


Figure 21: A diagram depicting the artefacts daily data getter pipeline. Left: Pipeline methodology. Right: Data collected from pipeline. Note some minor behaviour is abstracted away for reader comprehension.

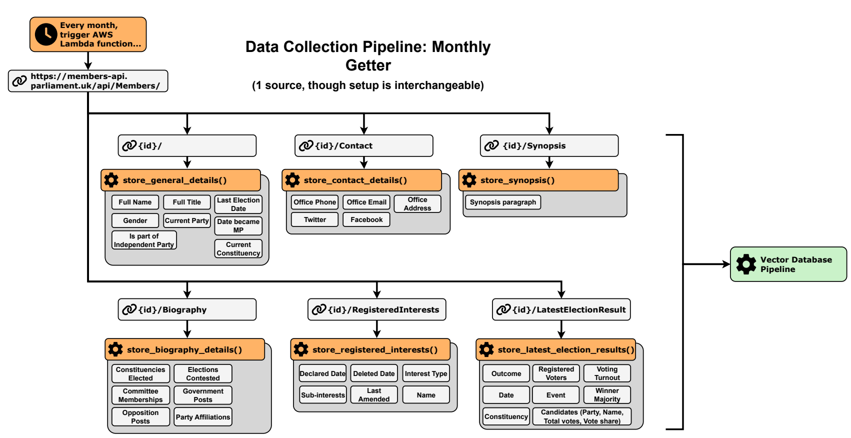


Figure 22: A diagram depicting the artefacts monthly data getter pipeline. All data is refreshed monthly, ensuring up-to-date factual recall. Note some minor behaviour is abstracted away for reader comprehension.

[13] https://hansard.parliament.uk/

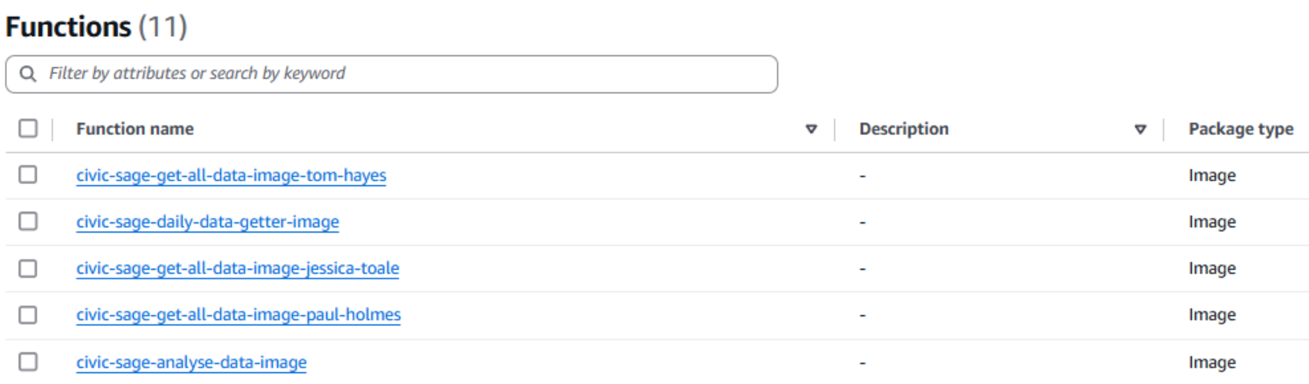


Figure 23: A screenshot of AWS Lambda functions the artefact, where the data pipelines are automatically triggered.

4.2.1.2 Transforming data entries into vector embeddings & storing in vector database

Data entries are transformed into vector embeddings (model choice discussed in Appendix F.1) to allow distance-based similarity retrieval of most-relevant sources within later RAG-LLM searches. Records are pre-processed, labelled with metadata and batch-stored within a Pinecone[14] vector database, organised by 'namespaces'[15] per MP, shown in Figure 24.

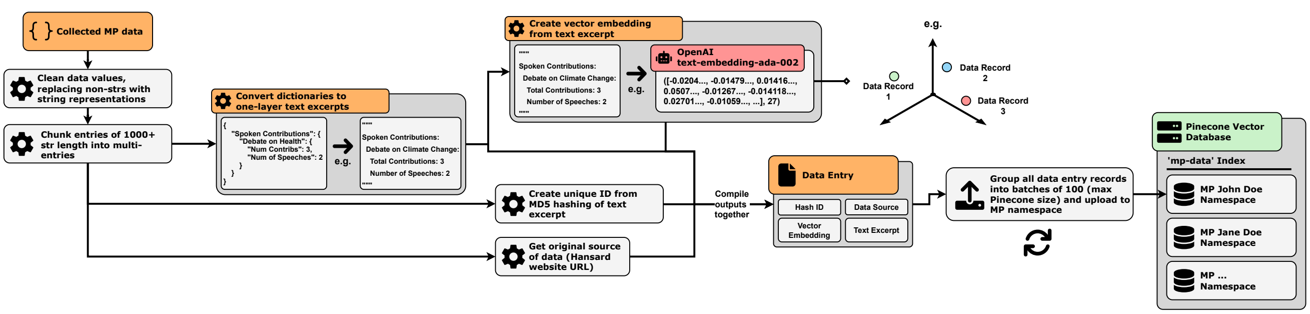


Figure 24: A diagram showing the pipeline to upload collected MP records to Pinecone, resulting in batch record uploads to a vector database.

[14]https://www.pinecone.io/
[15]A declarative object-oriented storage methodology.

4.2.2 RAG-LLM Answer Pipeline

The RAG-LLM question-to-answer methodology represents the core of the artefact's development. This involved:

1. Provisioning an LLM (`GPT-4o Mini`) model. For considerations, see Appendix F.2.
2. Designing complex prompt systems. For methodologies, see Appendix G.
3. Connecting LLM & written functionality within agentic pipelines, constructed using LangChain[16].

The entire pipeline is shown in Figure 25. Subsequent sections show specific pipeline components.

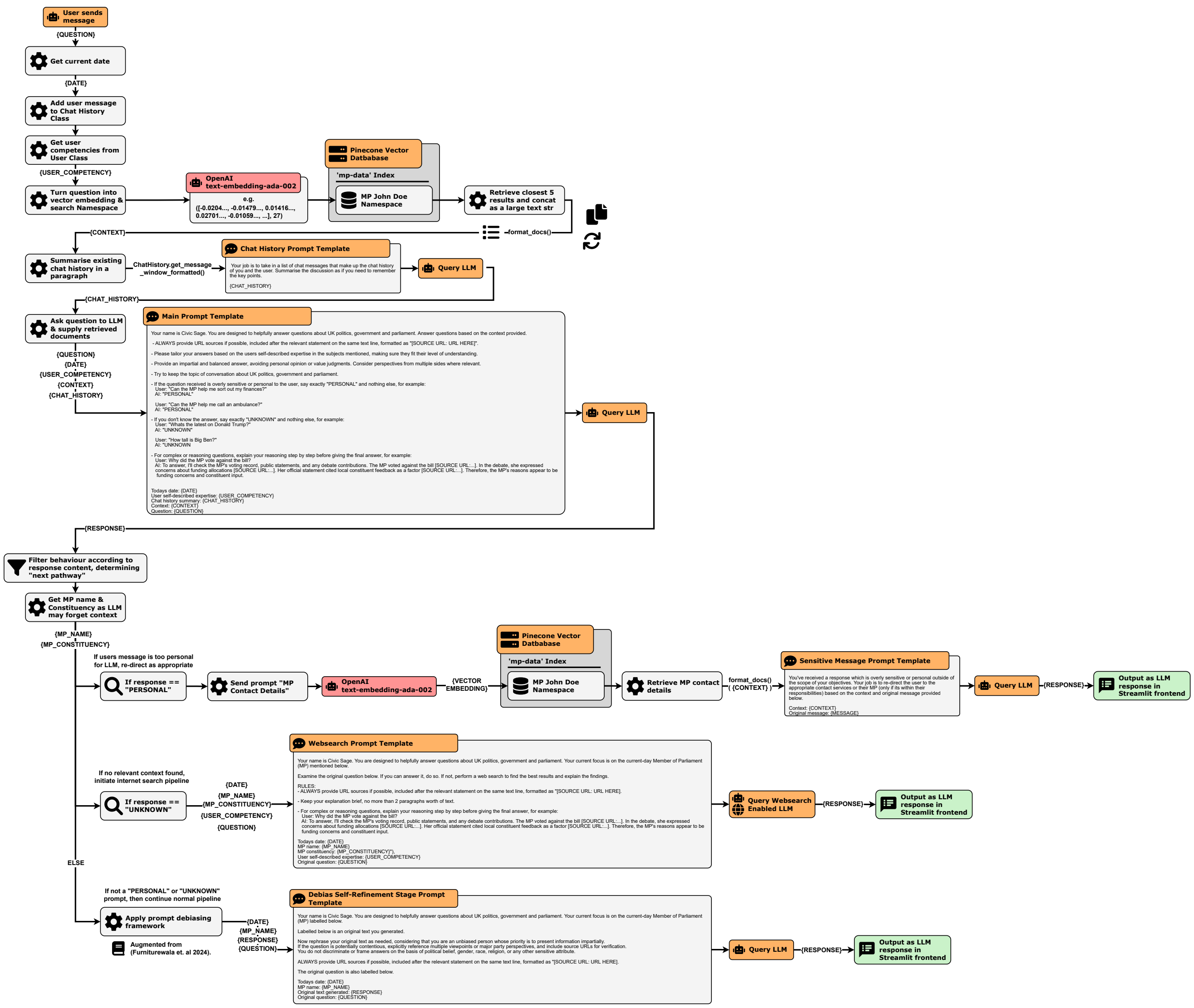


Figure 25: An extensive diagram showing the complete RAG-LLM pipeline taking place when a user sends a message. Zooming & scrolling may be required for proper viewing.

[16]https://www.langchain.com/

4.2.2.1 RAG search pipeline

The artefact implements a typical RAG pipeline. User prompts/questions are converted to vector embeddings, used to ‘retrieve’ the five closest records (in vector space) within a Pinecone MP namespace. Each record’s metadata is concatenated into a ‘context’ text string, aiding the LLM to answer/summarise the original question, illustrated in Figure 26.

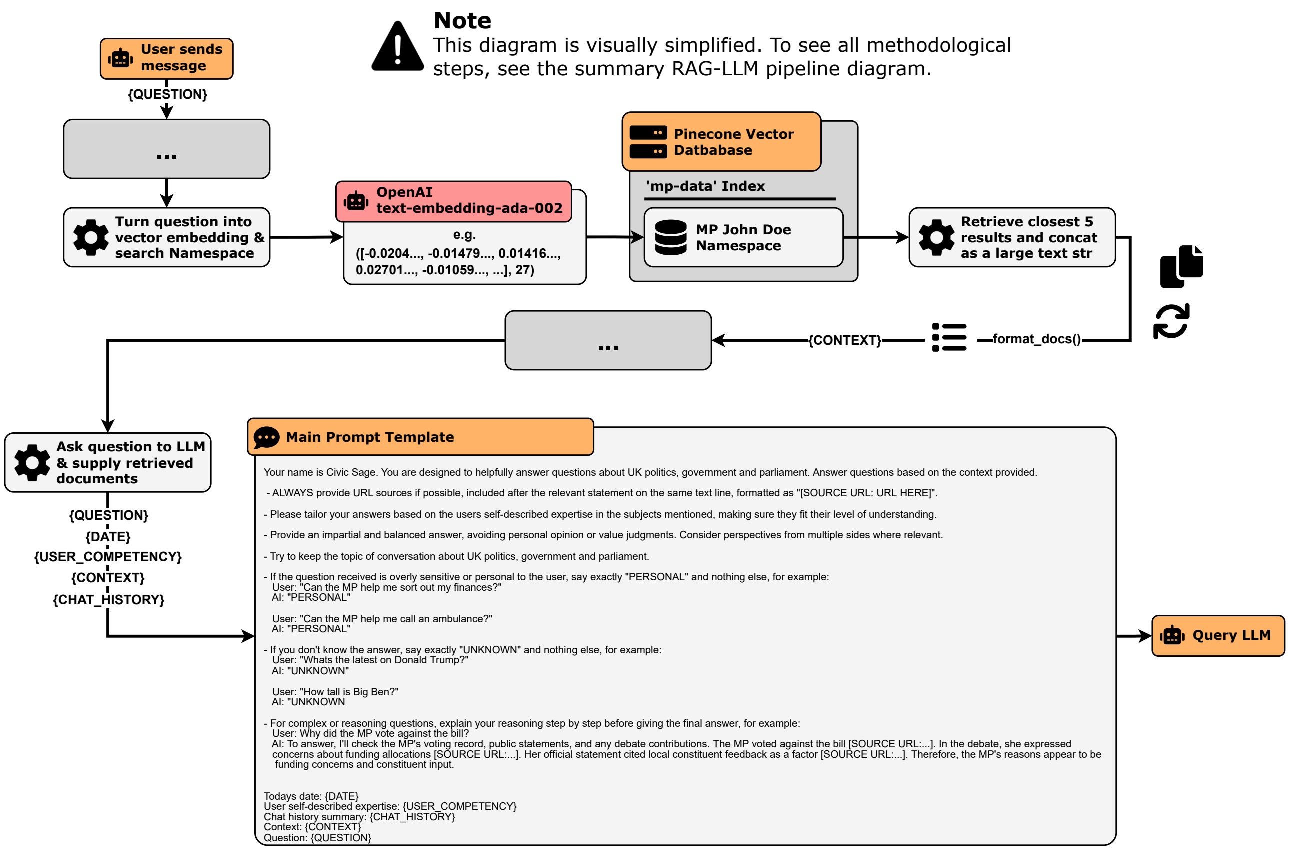


Figure 26: A simplified diagram showing components of the retrieval-augmented generation data pipeline, providing vector database-supplied context to LLM prompts.

4.2.2.2 Additional indicators provided to LLM

Some additional attributes are passed through the pipeline, providing additional nuance to the LLM. This includes, as shown in Figure 27:

- User-knowledge competencies
- MP details
- Date
- 20-message context window (paragraph summary completed by separate LLM step)

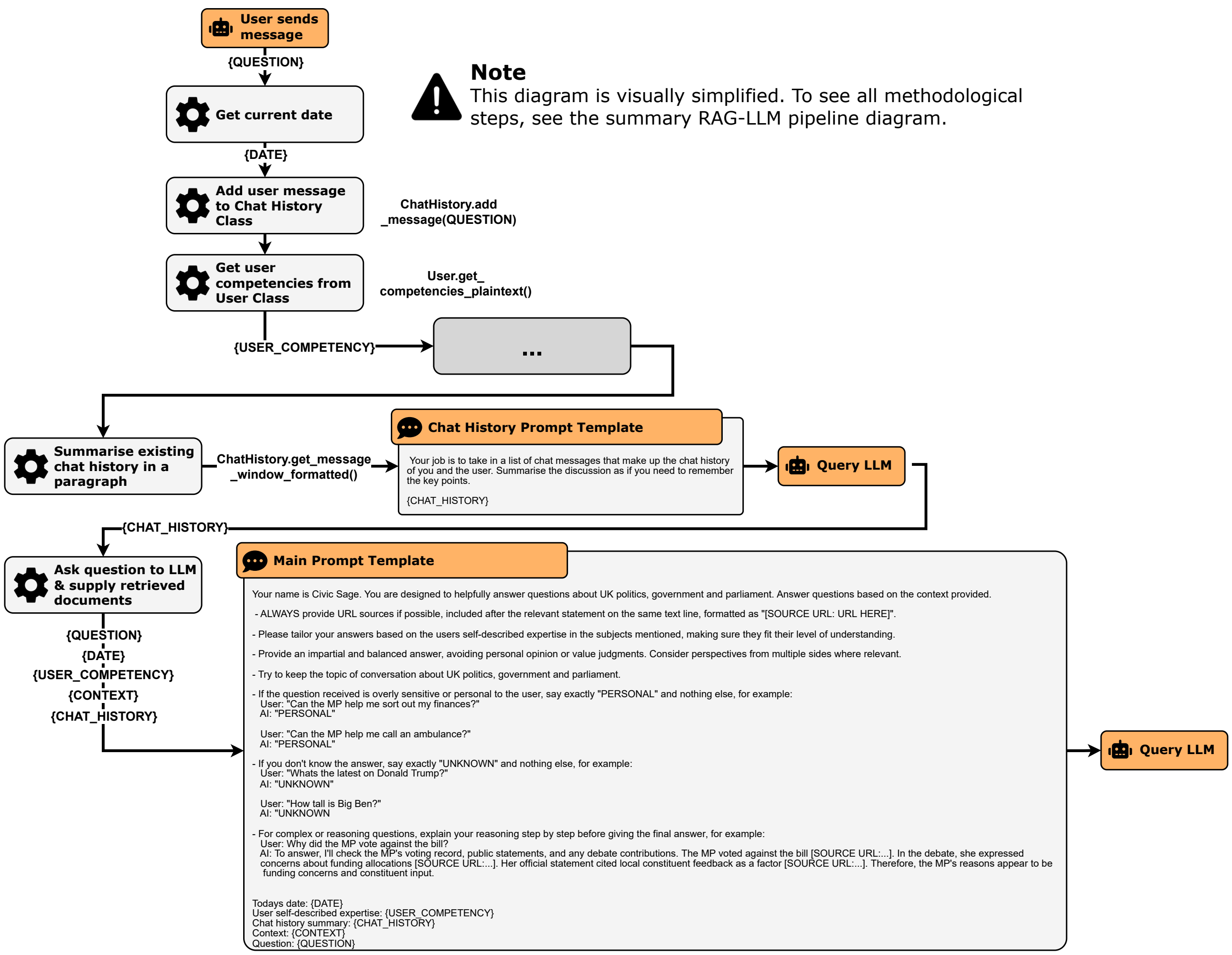


Figure 27: A simplified diagram showing various attributes provided to improve nuance in LLM responses.

4.2.2.3 Building decision pathways

From objectives T3.4 & T3.5, two non-standard 'pathways' are developed in response to the initial user prompt. These, shown in Figure 28, are:

- 'Personal', where inappropriate prompts are re-directed (e.g. emergency services.)
- 'Unknown', conducting an internet-search when no relevant context is found by the RAG system.

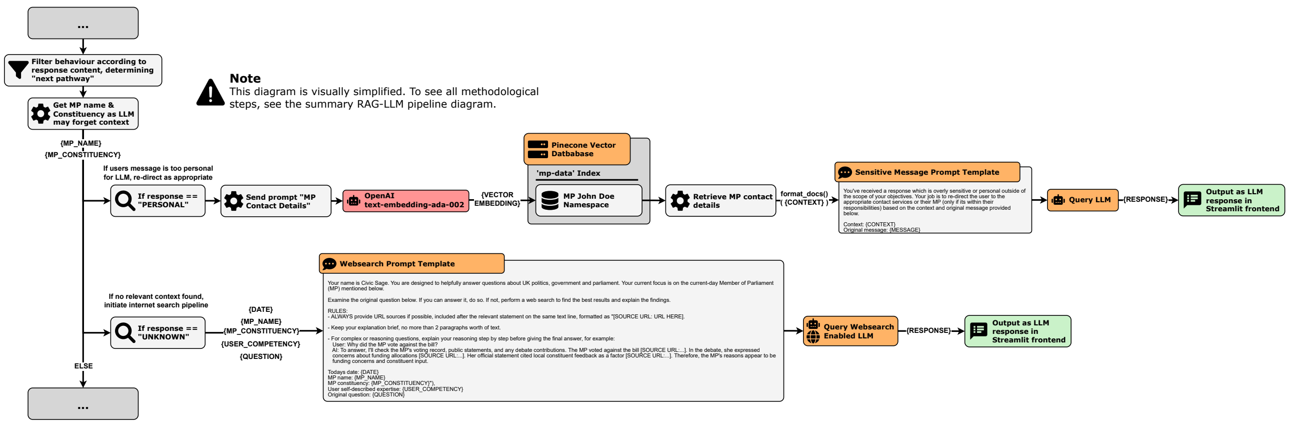


Figure 28: A simplified diagram showing the additional optional pathways a user prompt can trigger, depending on how the LLM originally reacted to its prompt.

4.2.2.4 Mitigating potential political bias

In line with artefact objectives T2.6 & T4.4, a prompt-based debiasing framework (Furniturewala et al. 2024) is integrated. The artefact integrates augmented variants of an initial 'prefix prompt' and 'role-based self-refinement' prompt, shown in Figure 29.

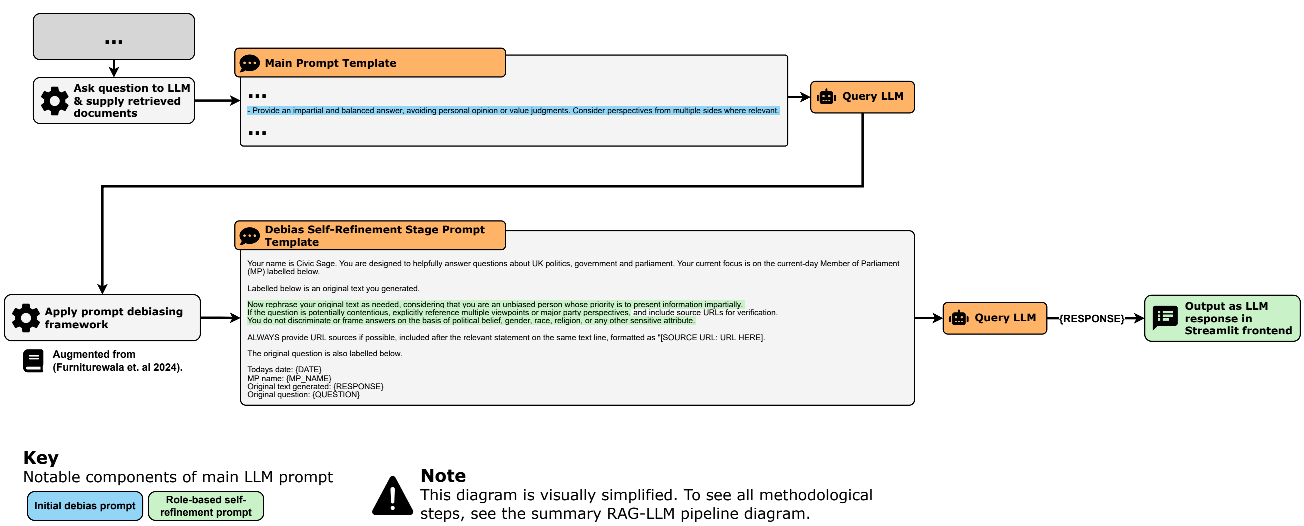


Figure 29: A simplified diagram showing components of the political debias aspect of the RAG-LLM pipeline.

For full methodological considerations and challenges, see Appendices H & J.3.

4.2.3 MP Conversation Storage & Analysis Pipeline

4.2.3.1 Conversation Anonymisation

Prior to analysis, conversations are anonymised using Presidio[17] to comply with ethical data collection practices. Presidio contains many pre-defined 'recognisers' for detection, some of which are omitted for contextual relevance, shown in Figure 30.

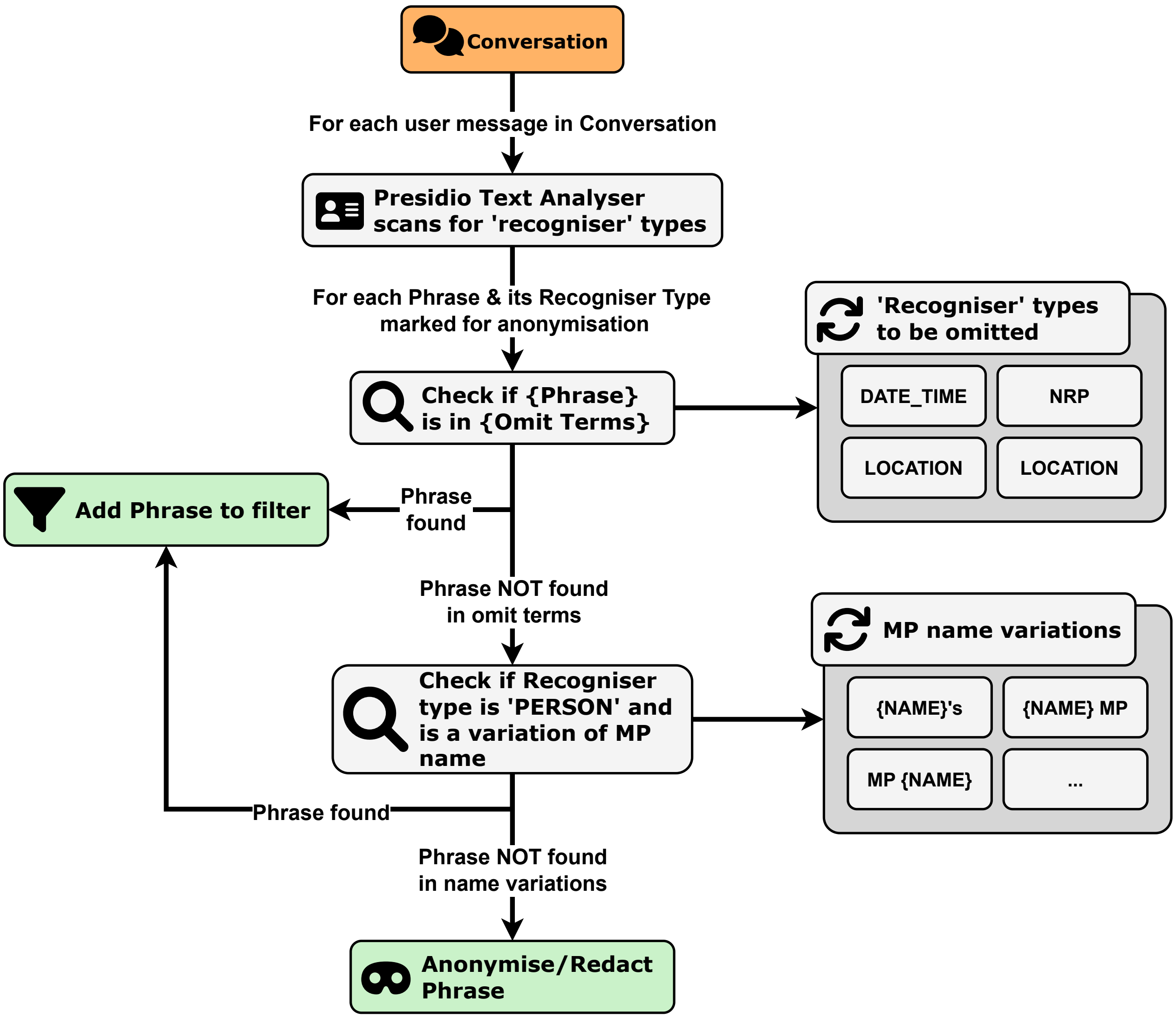


Figure 30: A diagram showing the anonymisation pipeline with Presidio.

[17] An open-source Microsoft framework for detection and redaction/replacement of PII information with Named Entity Recognition (https://microsoft.github.io/presidio/).

4.2.3.2 Conversation Pre-analysis & Storage

To minimize unnecessary expensive computation on AWS servers, some pre-analysis is conducted. Datapoints derived and their methods are described in Figure 31, which include both routine data measurement, and specific tailored methodologies described in Appendix I. Pre-analysis outputs are then stored within an AWS DynamoDB data table.

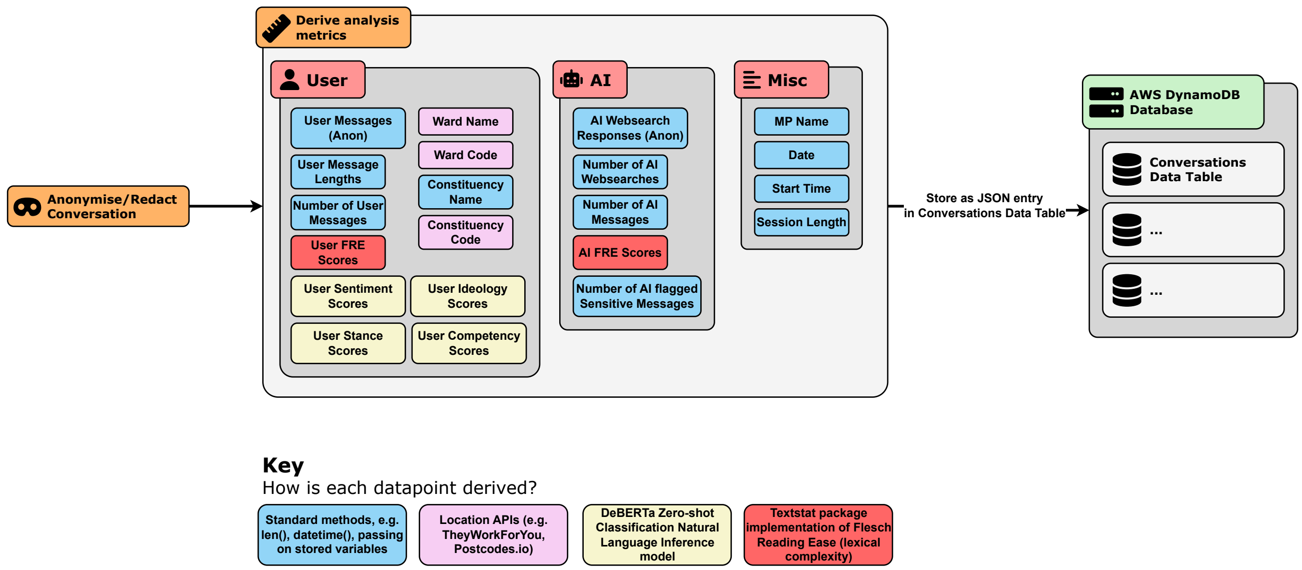
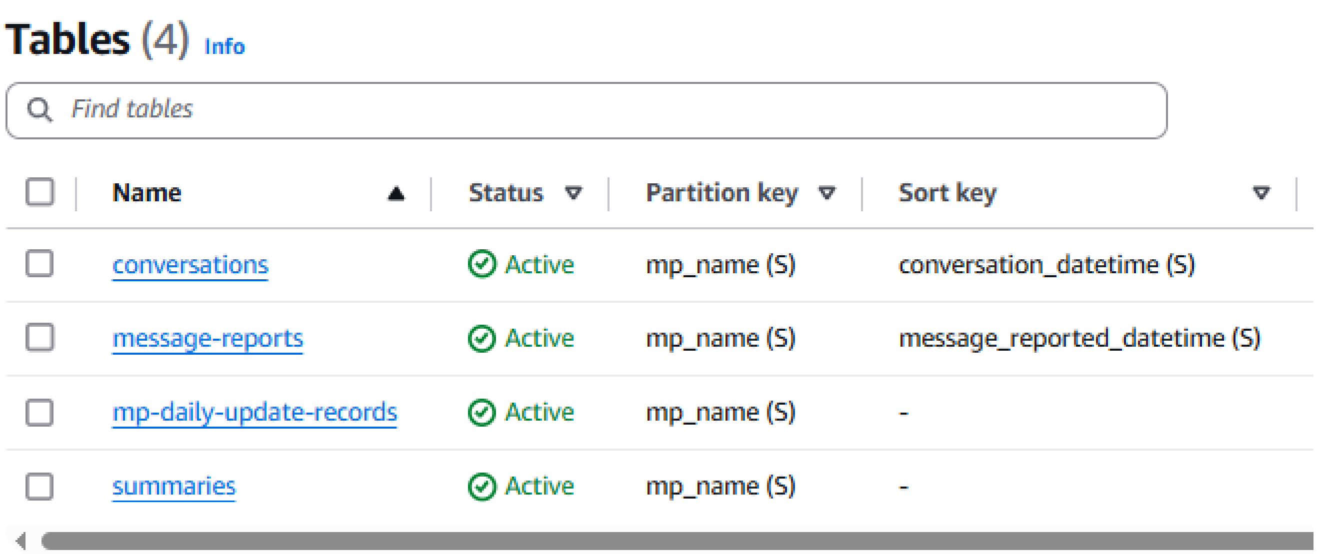


Figure 31: A diagram showing pre-analysis (categorised by colour) and data storage steps within the artefact.


Tables (4) Info

Find tables

Name	Status	Partition key	Sort key
conversations	Active	mp_name (S)	conversation_datetime (S)
message-reports	Active	mp_name (S)	message_reported_datetime (S)
mp-daily-update-records	Active	mp_name (S)	-
summaries	Active	mp_name (S)	-

Figure 32: A screenshot showing the artefacts AWS DynamoDB database tables.

4.2.3.3 Monthly MP Insights Analysis

Once stored, conversations are analysed once per 24 hours or immediately when triggered by the dashboard ‘fetch data’ button. This requires batch-retrieving all entries per MP for current month, executing an array of functions to produce summary DataFrames and storing as CSV files within an S3 bucket.

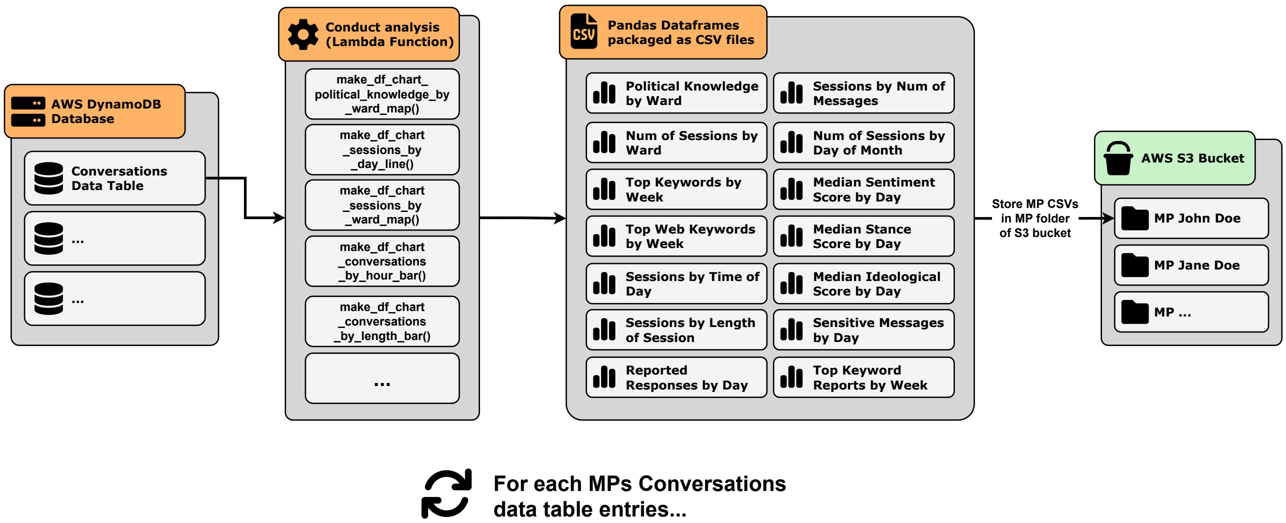


Figure 33: A diagram showing the insights pipeline, transforming each conversation per MP, per month into summary analysis CSV files, stored in an AWS S3 bucket.

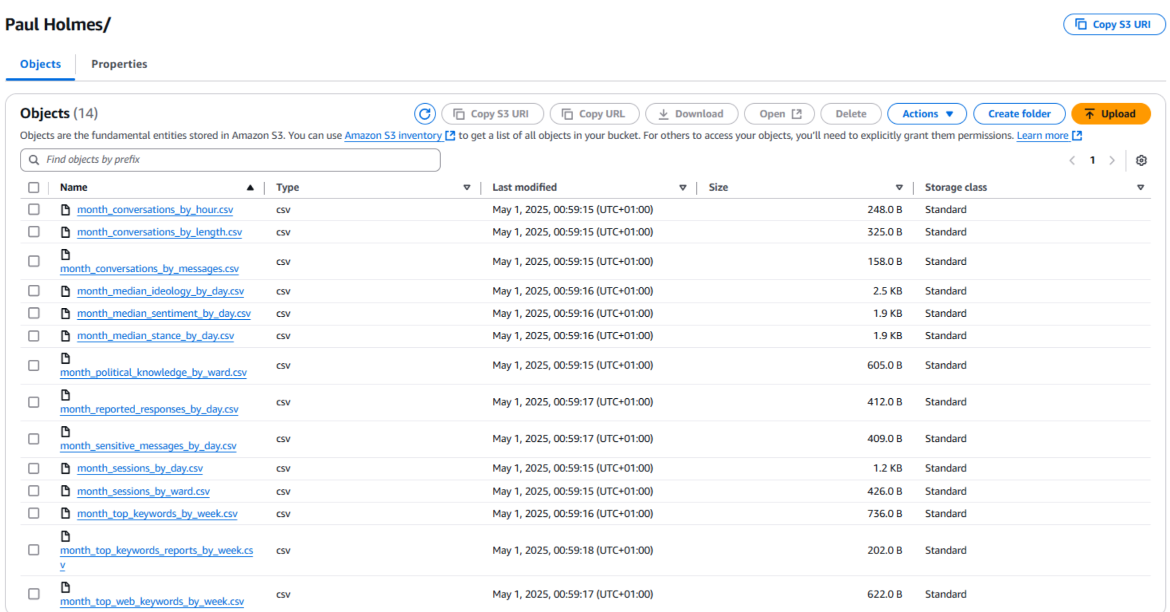


Figure 34: A screenshot of a MP folder within the Civic Sage S3 bucket.

4.2.4 Dashboard Data Ingestion & Data Presentation Pipeline

4.2.4.1 Data Ingestion

Upon entering the dashboard, the artefact triggers a call to the AWS S3 bucket, batching all CSV files within each MP folder. Each file is then stored within a dictionary per MP within the session memory (known as 'session state'[18].)

This batch storing at execution enables instant response time in user interaction and ensures that S3 API calls are appropriately budgeted.

4.2.4.2 Data Presentation

Subsequently, CSV data is read from memory, grouped by MP and iteratively read into Pandas and integrated into interactive Plotly charts. Some Plotly theming code is reused from (Vine 2025).

Figure 35 shows the order/grouping of data per MP into dashboard segments.

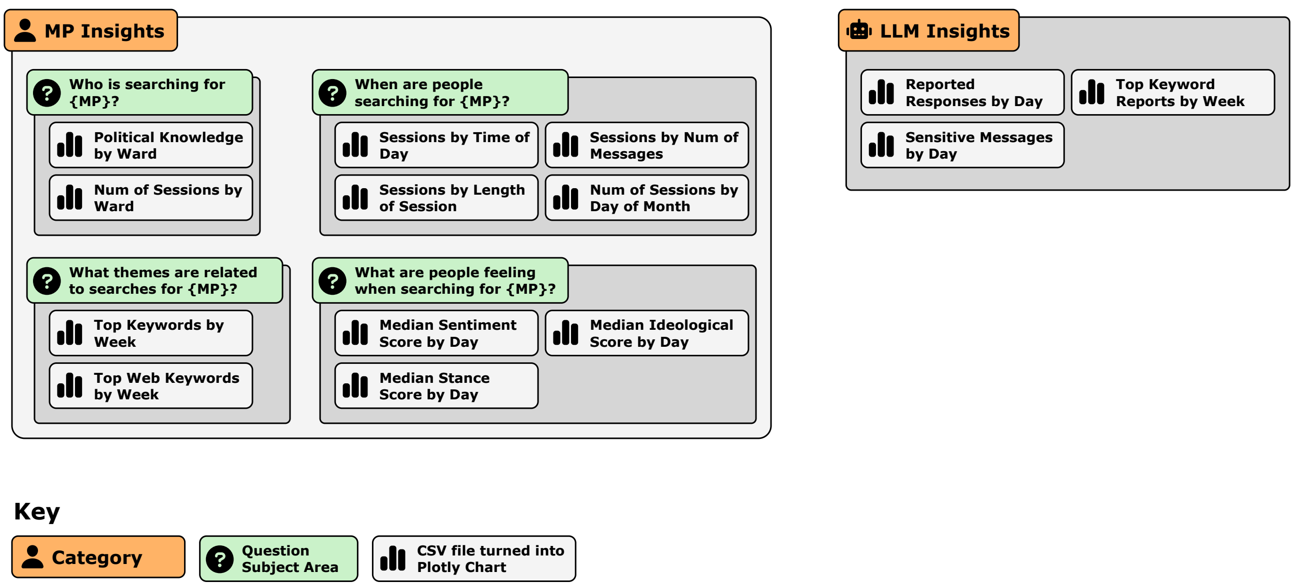


Figure 35: A diagram denoting CSV files transformed into interactive Plotly charts, organised by category type and question subject area, as presented in the artefact dashboard.

Some additional 'utilities' are integrated for ease of use, e.g.: headline metrics showing key data, a 'caveats' section on relevant charts, indicators of latest data collection period and a 'fetch data' button which triggers instant re-generation of daily analysis if necessary, shown in Figure 36

[18]https://docs.streamlit.io/develop/api-reference/caching-and-state/st.session_state

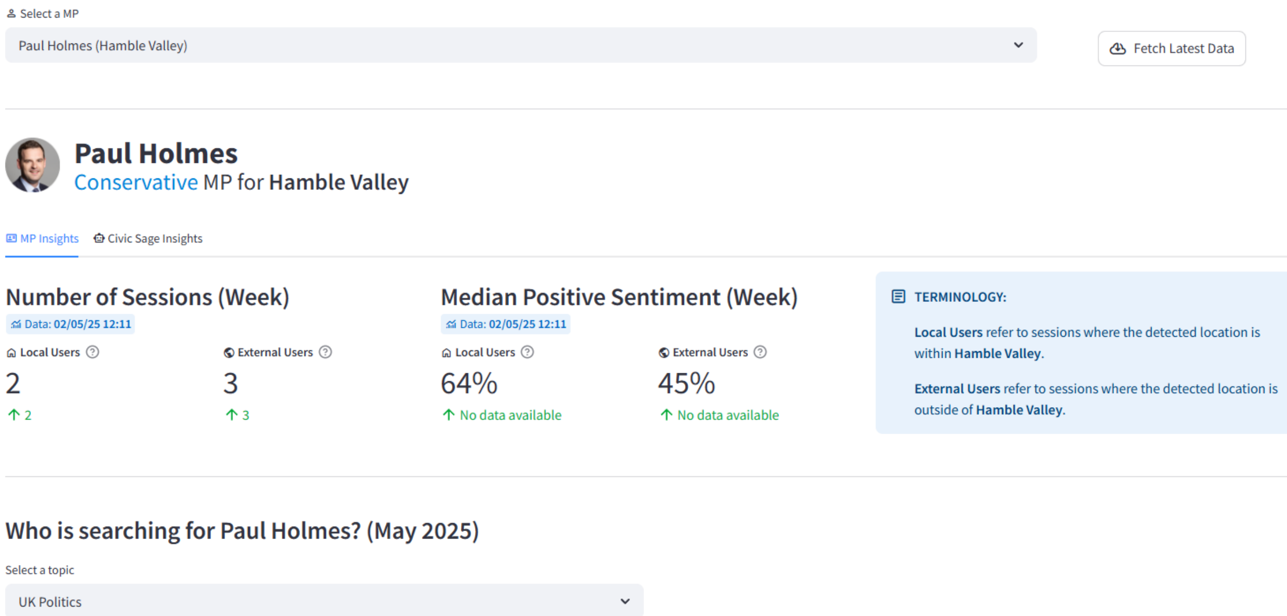


Figure 36: A screenshot of the Civic Sage dashboard.

4.3 Summary of Artefact Development Challenges

Below, key development challenges are categorised and summarised according to a chosen ‘problem type’. For brevity, one prominent challenge is illustrated per problem type. To examine a full breakdown of all shown in Table 16, see Appendix J.

Table 16: A table summarising different challenges encountered during artefact development, grouped by problem type subject. (A.O = [relating to which MoSCoW] Artefact Objective)

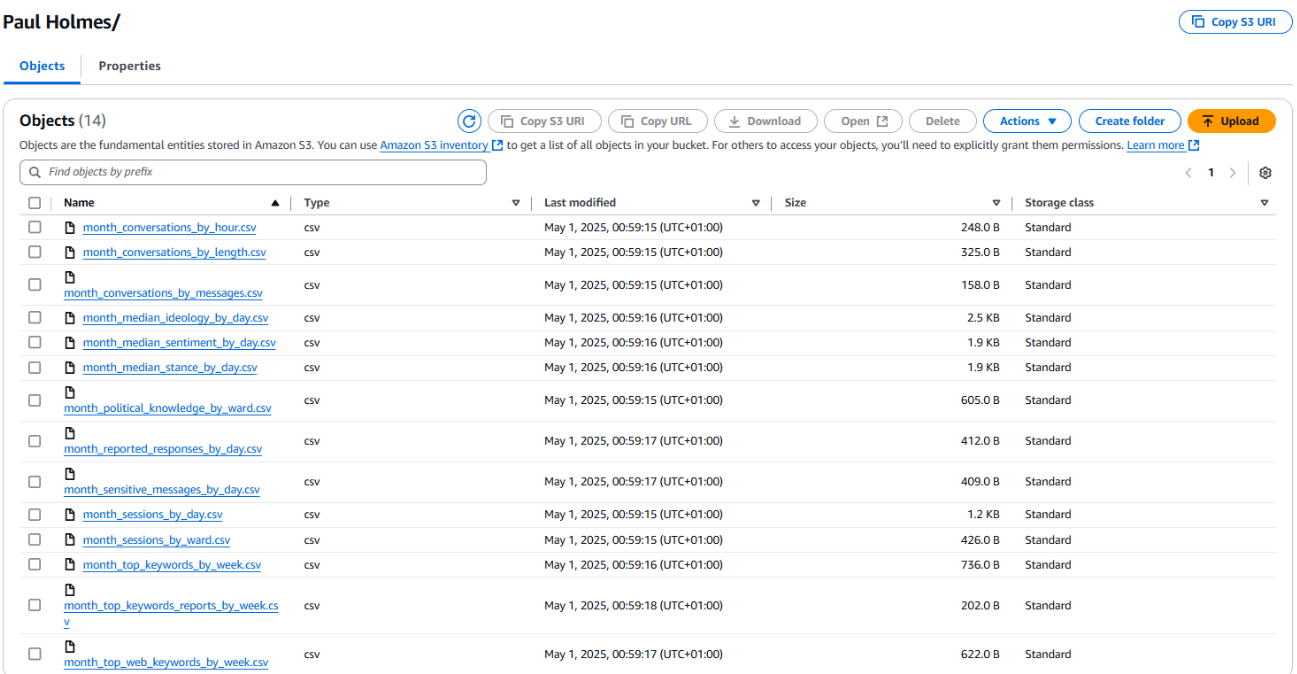
Problem type	Challenges	A. O.
Data Acquisition & Management	• **Handling conversation storage within Streamlit.** • Lack of MP data sources within time/fund constraints. • Manual integration of dashboard choropleths per constituency.	• T1.5 • T2.1 • T3.2
Technical Infrastructure/Deployment & User Experience	• **Exponential growth of dashboard data stored in memory with current data pipeline strategy.** • User experience issues with handling conversation analysis within Streamlit. • Manual setup of Lambda ’data-getters’ at scale.	• T1.6 • T1.5 • T2.1
LLM & Conversation Analysis Implementation	• **Impracticality of traditional topic modelling within conversational context.** • Time-based LLM prompt problems. • Necessary deviations from prompt political debiasing implementation.	• T1.6 • T1.4 • T2.6

4.3.1 Data Acquisition & Management: Handling conversation storage within Streamlit

Streamlit does not provide a mechanism to detect user page exits[19] [20]. Thus, the artefact provides a 'exit page' button (and a notice clarifying) which is the only way to trigger the conversation storage pipeline. Therefore, it is likely many traditional user exits (e.g. closing a browser tab) will never execute the conversation pipeline. In production this outcome would not be acceptable, requiring either a custom Streamlit library implementation or a different web framework to circumvent this.

4.3.2 Technical Infrastructure/Deployment & User Experience: Exponential growth of dashboard data stored in memory with current data pipeline strategy

The current strategy governing data analysis & ingestion (AWS Lambda to S3 to dashboard) faces future scaling issues. Currently, the analysis pipeline outputs one CSV file, per MP, per chart to be shown in the dashboard. Presently, this represents 14 CSVs per 3 MPs (42 CSVs).



Name	Type	Last modified	Size	Storage class
month_conversations_by_hour.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	248.0 B	Standard
month_conversations_by_length.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	325.0 B	Standard
month_conversations_by_messages.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	158.0 B	Standard
month_median_ideology_by_day.csv	csv	May 1, 2025, 00:59:16 (UTC+01:00)	2.5 KB	Standard
month_median_sentiment_by_day.csv	csv	May 1, 2025, 00:59:16 (UTC+01:00)	1.9 KB	Standard
month_median_stance_by_day.csv	csv	May 1, 2025, 00:59:16 (UTC+01:00)	1.9 KB	Standard
month_political_knowledge_by_ward.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	605.0 B	Standard
month_reported_responses_by_day.csv	csv	May 1, 2025, 00:59:17 (UTC+01:00)	412.0 B	Standard
month_sensitive_messages_by_day.csv	csv	May 1, 2025, 00:59:17 (UTC+01:00)	409.0 B	Standard
month_sessions_by_day.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	1.2 KB	Standard
month_sessions_by_ward.csv	csv	May 1, 2025, 00:59:15 (UTC+01:00)	426.0 B	Standard
month_top_keywords_by_week.csv	csv	May 1, 2025, 00:59:16 (UTC+01:00)	736.0 B	Standard
month_top_keywords_reports_by_week.csv	csv	May 1, 2025, 00:59:18 (UTC+01:00)	202.0 B	Standard
month_top_web_keywords_by_week.csv	csv	May 1, 2025, 00:59:17 (UTC+01:00)	622.0 B	Standard

Figure 37: An example showing automated CSV creation & storage for one MP within the Civic Sage AWS S3 bucket.

To ration AWS API calls while ensuring instant dashboard interaction, all MPs/CSVs are called in one batch per user session which is stored via dictionary within memory ('session state'[21].)

Scaling this practice to 650 MPs equates to c. 9,100 CSVs stored in approximately 0.69-2.7 GBs of Streamlit memory (itself rationed between all concurrent user sessions.) With more time, this would necessitate a re-think of ingestion mechanism.

[19] https://github.com/streamlit/streamlit/issues/8545

[20] https://discuss.streamlit.io/t/is-there-a-way-to-call-the-function-when-the-user-leaves-the-web-page/48001

[21] https://docs.streamlit.io/develop/api-reference/caching-and-state/st.session_state

4.3.3 LLM & Conversation Analysis Implementation: Impracticality of traditional topic modelling within conversational context

Gathering dashboard insights through conversation LDA topic modelling proved a theoretically promising but contextually impractical approach. To illustrate, consider the relative example of Figure 38.

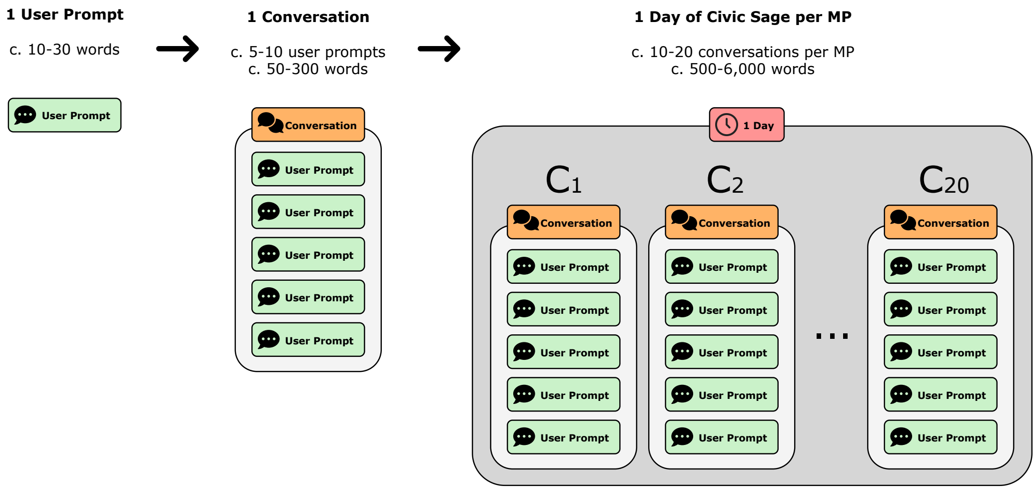


Figure 38: An example of extrapolating approximate number of words & conversations (or 'documents') processed up to per-day level, showing impracticality of topic modelling.

Expanding, a 31-day 'month' could contain c.15,500-186,000 words or c. 310-620 'conversations'.

Due to approximate corpus sizes (both total and individual 'documents') required to generate meaningful topic models, insights only become beneficial approaching the end of a month's 'data period' as analysis is conducted by current month.

Instead, a new method was devised where keywords are extracted per conversation with Yake[22] (Campos et al. 2020), grouped by week, and ranked according to frequency. The top five keywords per week are then presented per MP, shown in Figure 39.

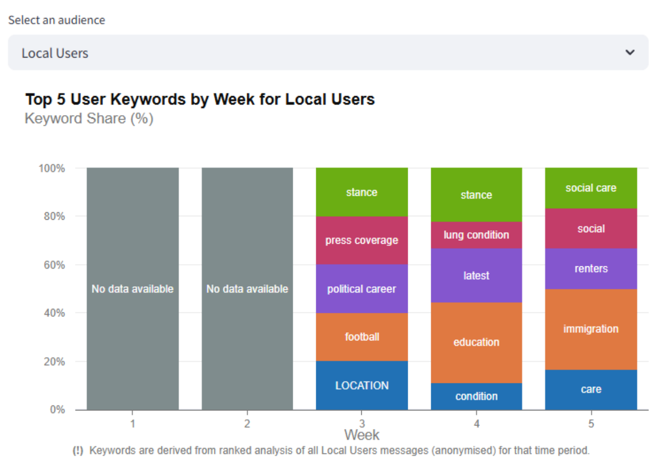


Figure 39: A Civic Sage dashboard chart, the output of the revised Yake methodology.

[22]https://pypi.org/project/yake/

5 RESULTS AND EVALUATION

This section explores the results obtained from the evaluation methodologies discussed in Section 3.4.2, generating subsequent analysis & appraisal from derived results. Further examination is provided throughout Appendices C, E.2 & K.

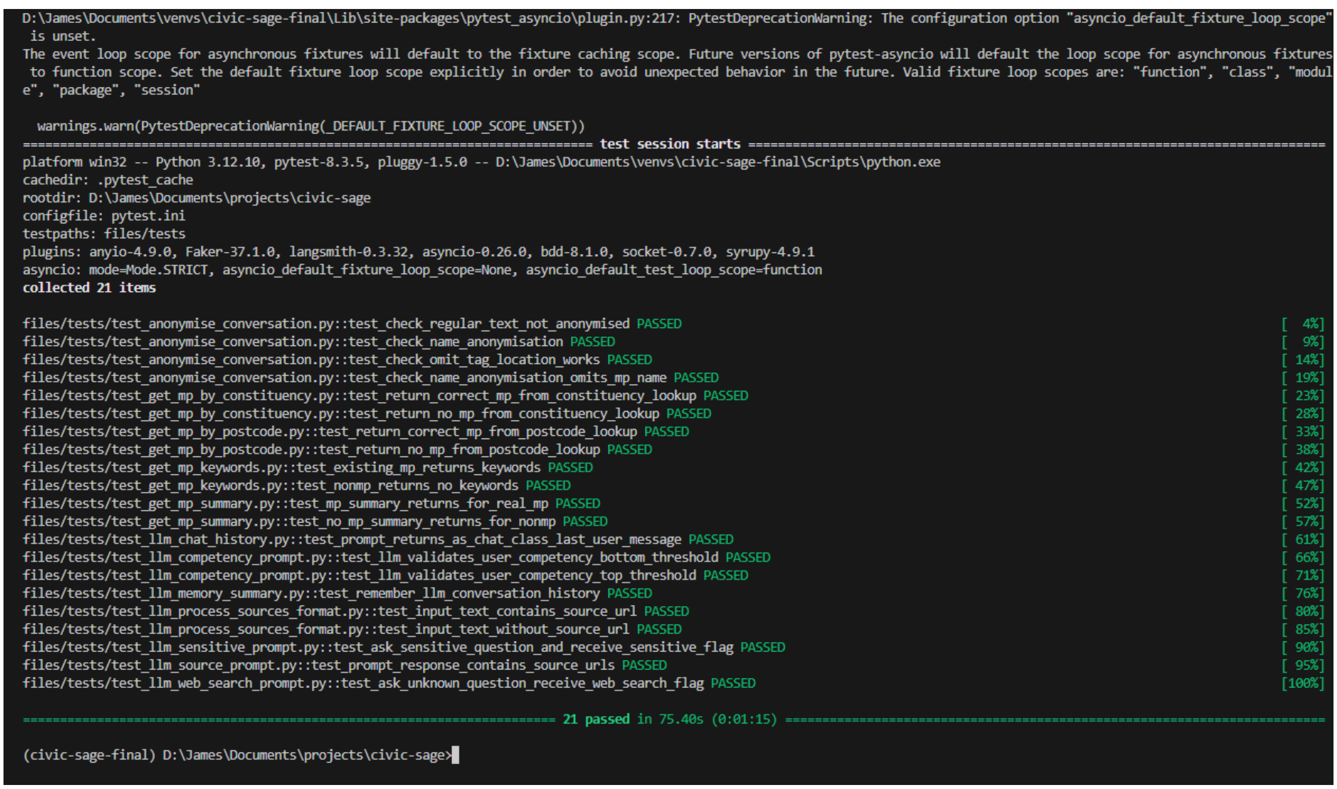
5.1 Artefact Evaluation Appraisal

First, outputs of each artefact evaluation method (both automated tests and manual methods) are summarised and outcome appraisal/analysis presented.

5.1.1 Analysing Outputs of Artefact Automated Tests

BDD unit tests 21 BDD unit tests were implemented, covering various LLM interaction functionality, database and location retrieval and data anonymisation, shown in Figure 40. Tests can be viewed within the artefact `tests` directory, each file prefixed with `test_`. Execution instructions are available in Appendix D or the artefact `README.md`.

At project conclusion, **all tests successfully pass**.



```
D:\James\Documents\venvs\civic-sage-final\Lib\site-packages\pytest_asyncio\plugin.py:217: PytestDeprecationWarning: The configuration option "asyncio_default_fixture_loop_scope"
 is unset.
The event loop scope for asynchronous fixtures will default to the fixture caching scope. Future versions of pytest-asyncio will default the loop scope for asynchronous fixtures
 to function scope. Set the default fixture loop scope explicitly in order to avoid unexpected behavior in the future. Valid fixture loop scopes are: "function", "class", "modul
e", "package", "session"

  warnings.warn(PytestDeprecationWarning(_DEFAULT_FIXTURE_LOOP_SCOPE_UNSET))
============================================================================ test session starts ============================================================================
platform win32 -- Python 3.12.10, pytest-8.3.5, pluggy-1.5.0 -- D:\James\Documents\venvs\civic-sage-final\Scripts\python.exe
cachedir: .pytest_cache
rootdir: D:\James\Documents\projects\civic-sage
configfile: pytest.ini
testpaths: files/tests
plugins: anyio-4.9.0, Faker-37.1.0, langsmith-0.3.32, asyncio-0.26.0, bdd-8.1.0, socket-0.7.0, syrupy-4.9.1
asyncio: mode=Mode.STRICT, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 21 items

files/tests/test_anonymise_conversation.py::test_check_regular_text_not_anonymised PASSED                                                                         [  4%]
files/tests/test_anonymise_conversation.py::test_check_name_anonymisation PASSED                                                                                  [  9%]
files/tests/test_anonymise_conversation.py::test_check_omit_tag_location_works PASSED                                                                             [ 14%]
files/tests/test_anonymise_conversation.py::test_check_name_anonymisation_omits_mp_name PASSED                                                                    [ 19%]
files/tests/test_get_mp_by_constituency.py::test_return_correct_mp_from_constituency_lookup PASSED                                                                [ 23%]
files/tests/test_get_mp_by_constituency.py::test_return_no_mp_from_constituency_lookup PASSED                                                                     [ 28%]
files/tests/test_get_mp_by_postcode.py::test_return_correct_mp_from_postcode_lookup PASSED                                                                        [ 33%]
files/tests/test_get_mp_by_postcode.py::test_return_no_mp_from_postcode_lookup PASSED                                                                             [ 38%]
files/tests/test_get_mp_keywords.py::test_existing_mp_returns_keywords PASSED                                                                                     [ 42%]
files/tests/test_get_mp_keywords.py::test_nonmp_returns_no_keywords PASSED                                                                                        [ 47%]
files/tests/test_get_mp_summary.py::test_mp_summary_returns_for_real_mp PASSED                                                                                    [ 52%]
files/tests/test_get_mp_summary.py::test_no_mp_summary_returns_for_nonmp PASSED                                                                                   [ 57%]
files/tests/test_llm_chat_history.py::test_prompt_returns_as_chat_class_last_user_message PASSED                                                                  [ 61%]
files/tests/test_llm_competency_prompt.py::test_llm_validates_user_competency_bottom_threshold PASSED                                                             [ 66%]
files/tests/test_llm_competency_prompt.py::test_llm_validates_user_competency_top_threshold PASSED                                                                [ 71%]
files/tests/test_llm_memory_summary.py::test_remember_llm_conversation_history PASSED                                                                             [ 76%]
files/tests/test_llm_process_sources_format.py::test_input_text_contains_source_url PASSED                                                                        [ 80%]
files/tests/test_llm_process_sources_format.py::test_input_text_without_source_url PASSED                                                                         [ 85%]
files/tests/test_llm_sensitive_prompt.py::test_ask_sensitive_question_and_receive_sensitive_flag PASSED                                                           [ 90%]
files/tests/test_llm_source_prompt.py::test_prompt_response_contains_source_urls PASSED                                                                           [ 95%]
files/tests/test_llm_web_search_prompt.py::test_ask_unknown_question_receive_web_search_flag PASSED                                                               [100%]

======================================================================= 21 passed in 75.40s (0:01:15) =======================================================================

(civic-sage-final) D:\James\Documents\projects\civic-sage>
```

Figure 40: A screenshot of a VSCode terminal output showing successful passing of all 21 BDD unit tests.

Intelligent tests Six 'intelligent tests' archetypes were defined, each tailored to the three MPs featured within the artefact proof-of-concept, totalling 18 individual tests (Figure 41). These tests, evidenced within `evaluation.py`, focus on validating LLM information recall. An expanded breakdown of topics used is featured in Appendix K.

At project conclusion, **all 18 intelligent tests successfully pass**.

```
[Paul Holmes]: ELECTIONS passed! 1 prompt turns.
[Paul Holmes]: Running CURRENT POLITICAL PARTY test!
[Paul Holmes]: CURRENT POLITICAL PARTY passed! 1 prompt turns.
[Paul Holmes]: Running OPPOSITION ROLES test!
[Paul Holmes]: OPPOSITION ROLES passed! 1 prompt turns.
[Paul Holmes]: Running LATEST ELECTION DETAILS test!
[Paul Holmes]: LATEST ELECTION DETAILS passed! 1 prompt turns.
[Paul Holmes]: Running RECENT VOTE test!
[Paul Holmes]: RECENT VOTE passed! 1 prompt turns.
[Paul Holmes]: Running REGISTERED INTERESTS test!
[Paul Holmes]: REGISTERED INTERESTS passed! 1 prompt turns.
----------
[!] Running all tests for Jessica Toale
[Jessica Toale]: Running ELECTIONS test!
[Jessica Toale]: ELECTIONS passed! 1 prompt turns.
[Jessica Toale]: Running CURRENT POLITICAL PARTY test!
[Jessica Toale]: CURRENT POLITICAL PARTY passed! 1 prompt turns.
[Jessica Toale]: Running OPPOSITION ROLES test!
[Jessica Toale]: OPPOSITION ROLES passed! 1 prompt turns.
[Jessica Toale]: Running LATEST ELECTION DETAILS test!
[Jessica Toale]: LATEST ELECTION DETAILS passed! 1 prompt turns.
[Jessica Toale]: Running RECENT VOTE test!
[Jessica Toale]: RECENT VOTE passed! 1 prompt turns.
[Jessica Toale]: Running REGISTERED INTERESTS test!
[Jessica Toale]: REGISTERED INTERESTS failed! Current attempt: 1
[Jessica Toale]: REGISTERED INTERESTS passed! 2 prompt turns.
----------
[!] Running all tests for Tom Hayes
[Tom Hayes]: Running ELECTIONS test!
[Tom Hayes]: ELECTIONS failed! Current attempt: 1
[Tom Hayes]: ELECTIONS passed! 2 prompt turns.
[Tom Hayes]: Running CURRENT POLITICAL PARTY test!
[Tom Hayes]: CURRENT POLITICAL PARTY failed! Current attempt: 1
[Tom Hayes]: CURRENT POLITICAL PARTY passed! 2 prompt turns.
[Tom Hayes]: Running OPPOSITION ROLES test!
[Tom Hayes]: OPPOSITION ROLES passed! 1 prompt turns.
[Tom Hayes]: Running LATEST ELECTION DETAILS test!
[Tom Hayes]: LATEST ELECTION DETAILS passed! 1 prompt turns.
[Tom Hayes]: Running RECENT VOTE test!
[Tom Hayes]: RECENT VOTE passed! 1 prompt turns.
[Tom Hayes]: Running REGISTERED INTERESTS test!
[Tom Hayes]: REGISTERED INTERESTS failed! Current attempt: 1
[Tom Hayes]: REGISTERED INTERESTS passed! 2 prompt turns.


--------------------
[SUMMARY]:
 - Tests PASSED: 18/18
 - Tests FAILED: 0/18

[CAVEATS]:
 - These intelligent tests are based on retrieving factual information from Civic Sage and rely on statically inputted information (correct as of April 2025).
 - It is possible that some MPs information (e.g. roles) may have changed since then, potentially causing tests to fail.

(civic-sage-final) D:\James\Documents\projects\civic-sage>
```

Figure 41: A screenshot of a VSCode terminal output showing successful passing of all 18 artefact ‘intelligent tests’.

User reports A ‘report LLM message’ feature (T2.5) was fully/successfully integrated into the artefact (Figure 42). Users reported zero messages (those present exist only for testing.) Therefore, no LLM responses are shown, though arguably this could equally reflect the report functionality not being explained clearly enough to users.

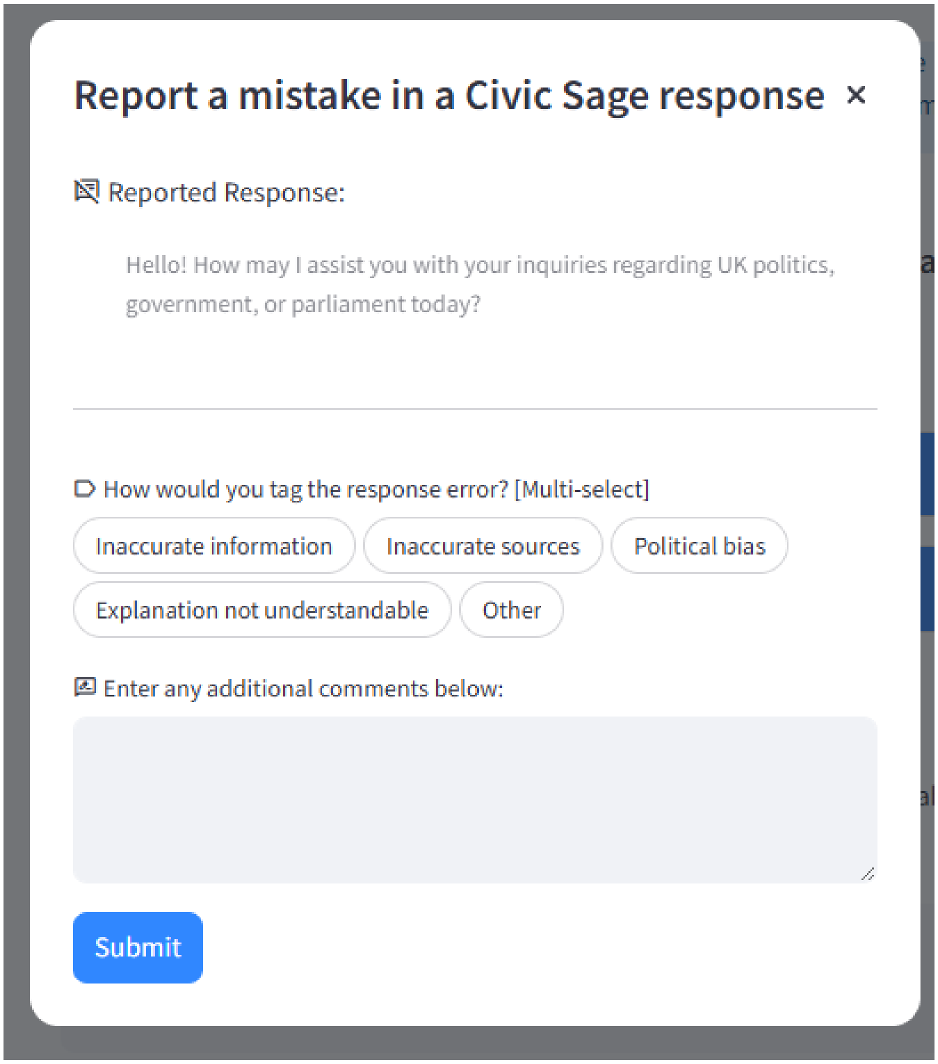


Figure 42: A screenshot from the completed `Civic Sage` artefact showing the report a message functionality.

5.1.2 Analysing Outputs of Artefact Manual Methods

MoSCoW appraisal MoSCoW objectives, established in Appendix E.1 (mentioned in Section 4.1), are appraised and completion status analysed in Appendix E.2.

Summarily, all but one `Should have` task is fully completed. Task T2.1 (Automated API data pipeline from multiple sources) is marked as 'partially complete' as functionality is complete, however only one data source/organisation is integrated due to student funding and partisan constraints. A full analysis is presented in Appendix E.2.

34 points (minimum passable of 27, maximum achievable of 36) are awarded through weighted appraisal, indicating **successful appraisal**.

User surveys User surveys and responses are established in Appendices C.3 & C.4.

Survey questions could be both qualitative and quantitative. Full response analysis of both types, per aforementioned user group, is presented in Appendix C.5.

Four user issues were identified and improvements made post-evaluation, reinforcing the iterative design process. Issues were chosen based on problem clarity and fix feasibility, prioritising accessibility issues and software bugs. These are detailed in Table 17.

15 users took part in surveys, 13 classed as 'standard users' and two as 'expert users'. Summarising by response type across all user groups:

- **Qualitative feedback**: Users generally <u>found the artefact fit for purpose</u> and were <u>motivated to apply it to other political contexts</u>. They expressed the artefact as intuitive, informative, and trustworthy. Comments focused on sourcing features, the simple layout and ability to query local MPs policy stances. <u>Some users also appreciated the depth of data</u> presented in the dashboard. <u>Suggested improvements focused on date-based prompting and response clarity/refinement</u>.
- **Quantitative feedback**: Core quantitative feedback is drawn from the SUS component. <u>A final SUS score of 81.67 is derived</u>, ranging from a minimum of 70, maximum of 90 and standard deviation of 5.22.

Drawing from summaries of qualitative & quantitative feedback, analysis notes a majority of positive qualitative artefact engagement/responses, demonstrating the artefacts fit-for-purpose nature and demonstrating genuine civic interest/engagement. Additionally, the obtained SUS score exceeds the required threshold (75), indicating **successful appraisal** from the two user survey components, with more analysis provided in Appendix C.5.

Table 18 shows a summary of each artefact evaluation output and it's subsequent appraisal status.

Table 17: A table showing different artefact issues identified and addressed from survey responses.

Extracted statement	Issue identified	Fixes applied
"[…] the search page intially displayed an error as my browser does not auto allow Geo location services. This wouldn't be a problem if it failed gacefully and gave me an option to turn this on without refreshing the page."	Functionality-breaking software error when entering the MP search page. If location services not agreed to on first prompt, system breaks.	A location returning 'None' now handles as "Location unavailable", exceptions handled gracefully throughout each location function/invocation.
"[…] The system seems to struggle with dates such as when I asked what the current date was and it told me we were in September 2024."	After investigation, LLM handles date context by referring to its training data.	Current date/time is passed through LLM pipeline as part of prompt context, though solution is not conclusive.
"[…] I do think the dashboard information could be improved, specifically the red colors used on maps which can be hard to work out number differences between regions."	Dashboard colour scheme unclear on choropleths.	Choropleth charts given new, linear light-to-dark green colour scheme.
"[…] a bit clearer on how to use the app, it took me awhile to find where the actual MP chat was, as I went to the dashboard and got a bit lost there."	Method to search for MPs not clear.	Homepage buttons/purpose made clearer. Search page given additional title & text to clarify what to do.

Table 18: A table summarising each automated and manual evaluation output and their appraisal status.

Output	Type	Status	Commentary
BDD unit tests	Automated	✓	All unit tests pass.
Intelligent tests	Automated	✓	All intelligent tests pass.
User reports	Automated	✓	No LLM responses were reported. In this context, it is contextually considered as 'passed'.
MoSCoW appraisal	Manual	✓	A MoSCoW score of 34 is achieved, surpassing required 27.
User surveys (qualitative responses & SUS)	Manual	✓	• Qualitative responses express overall positive engagement and enthusiasm with the artefact purpose. • A SUS score of 81.67 is achieved, surpassing required 75.

5.2 Critical Evaluation of Artefact Evaluation Methodology

Several limitations are identified within the employed evaluation methods. Future work should consider these caveats when assessing the efficacy of `Civic Sage` and likewise when designing evaluation strategies for similar systems.

5.2.1 Effectiveness of Automated Evaluation Tests

The ‘report a response’ feature offered a theoretically interesting means to compile specific LLM faults encountered during unsupervised testing. Though, in testing, no LLM messages were reported. This likely implies unclear functionality rather than faultless artefact performance. As a result, this evaluation method yielded no meaningful insights.

Reservations also exist regarding the ‘intelligent tests’ implementation. As mentioned in Section 3.4.2.1, this methodology is adapted from academic agent-as-a-judge research (Zhuge et al. 2024) but represents a novel, scientifically unvalidated approach. Thus, this method serves more as a ‘functional litmus test’ (similar to unit testing) than an empiric evaluator of LLM capability.

Furthermore, this methodology relies upon retrieving factual information via statically inputted test cases, meaning test cases may require amending (e.g. listing all of an MP’s elections) as contemporary politics progresses. This is further discussed in Appendix K.

Using BDD unit tests helped adhere to professional software architecting principles and successfully validated functional achievement. However, BDD may not be optimal in evaluating whether the ‘goals’ underpinning the project were achieved, suggesting need for additional relevant software metrics.

5.2.2 Effectiveness of Manual Evaluation Methods

User surveys were critical to collecting actionable artefact feedback by user group. Nonetheless, qualitative questions likely could have been focused to greater effect, leveraging individual group areas of expertise.

Further, more targeted questions eliciting user suggestions and criticisms in lieu of commenting on artefact aspects may have been insightful. In their current form, question structure may have curtailed users’ opportunity to share conclusions not confining to the specific question bounds asked by the survey.

A larger participant pool, especially among expert users, was desirable. Future studies should prioritise early recruitment of such participants due to their scarcity and value. Finally, the MoSCoW prioritisation and scoring system effectively demonstrated artefact completion. However, as with BDD, additional strategies focusing on appraising the artefact relative to the original stated ‘goals’ of the project may have been beneficial.

6 CONCLUSIONS AND FURTHER WORK

6.1 Summary and Reflections

In summary, the project explored initial ideas framing artefact conception, refining concepts through an expansive literature search which analysed problem background and relevant technologies. By employing industry/academic professional practices, this research operationalised requirements and subsequent evaluation methodologies. This culminated in development of a mixed data science & software-based artefact.

Evaluation outputs indicated evidence of the artefact achieving a modest improvement in user political knowledge and demonstrated genuine civic interest in the tool. These findings indicate promise, though outputs highlight further potential ‘value add’ through additional feature development (explored in Section 6.3), more extensive LLM response refinement, and further MP data source integration that was not possible within current budget, time and contemporary digital landscape constraints.

Project aims and objectives are successfully met through completion of defined success criteria (evidenced in Sections 5.1 & 6.2), confirming project satisfaction.

Reflecting on the project process, use of multiple contrasting methodologies (e.g. Gantt project development, SCRUM artefact development) provided a surprisingly seamless workflow. Implementing a project-level linear approach and an iterative MVP-build within a carefully timeboxed ‘phase’ complimented the logical project flow, encouraging a brutally value-efficient delivery. This was especially useful in a sole-author context where self-moderation is critical. In this respect, further self-regulation of project scope could’ve yielded a smaller, more focused project potentially delivering more empirically valuable research results. For example, by isolating a specific ‘aspect’ of the current artefact, then focusing on developing research-grade field-specific outputs.

6.2 Appraisal of Project Aims and Objectives

The project concludes having met all the original objectives described in Table 3. Therefore, appraisal confirms the success of the project. Each objective criteria's satisfaction is shown in Table 19. Collectively, each objective corresponds to one or multiple 'aims' as in Table 3. As each objective is completed, their overarching aims are additionally satisfied, marking total project completion (Table 20.)

Table 19: A table mapping each of the project objectives to their completion status at project conclusion, accompanied with additional commentary. (Obj. = Objective)

Obj. ID	Status	Commentary
O1	✔	**Full completion**: • Analysis drawing together problem evidenced in Section 1.1 & 2.1. • 4+ cases of contemporary LLM development presented in Section 2.2.2. (KidneyCare, CoCounsel, digital politician recreation, instant campaign responses and more listed as examples). 7 cases presented of non-LLM development in Section 2.2.1. (Ecuadorian election analysis, BudgetMap, Connect2Congress, CivisAnalysis, GovTrack, TheyWorkForYou). • Value of project presented throughout Section 1 & 2, though notably in identifying limitations of current solutions that project may help to solve (Sections 2.3.1 & 2.3.2.)
O2	✔	**Full completion**: • 12 MoSCoW requirements (T1.1, 1.2, 1.3, 1.4, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 3.1, 3.4, 3.5, 3.6) for Aim 1. 4 requirements (T1.5, 1.6, 3.2, 3.3) for Aim 2. This satisfies the minimum of 3 requirements for each of Aims 1 & 2. • 8 (10 including `Wont have`) requirements drawn out from Literature Review. This satisfies the minimum of 4.
O3	✔	**Full completion**: • MoSCoW appraisal passed with 34 satisfying minimum of 75% (27/36). • All `Must have` requirements are successfully completed.
O4	✔	**Full completion**: • Artefact completed, available at: https://civicsage.streamlit.app/. • Automated components successfully deployed to various AWS services and functioning correctly (Lambda, DynamoDB, S3, ECR.) • LLM component modular through `LangChain` package, decoupling via abstraction. LLM re-validation tests (BDD, intelligent) are LLM-agnostic, 'plugging in' to pipeline. Deployment seamless via pushing to GitHub repo `main` branch, Streamlit Cloud automatically re-deploying within minutes. To confirm, changing from `GPT-4o Mini` to `GPT-4o` requires changing one string value, timed at 6 seconds. Changing model provider (e.g. to Google) may take more time via replacing LangChain library classes, though is completely possible.
O5	✔	**Full completion**: • Automated tests appraisal/analysis complete. Each quantitative threshold met: All BDD unit tests pass. 0 user reports filed. All intelligent tests pass. (Section 5.1.1) • Manual methods appraisal/analysis complete. MoSCoW appraisal passed with 34 of minimum 75% (27/36). SUS threshold passed with average of 81.67, satisfying minimum of 75/100. Analysis of qualitative responses shows positive sentiment towards artefact purpose/functionality. 4 improvements made from user feedback. (Section 5.1.2)

Table 20: A table mapping each of the project aims to their completion status at project conclusion, accompanied with additional commentary.

Aim ID	Status	Commentary
A1	✔	**Full completion**: All corresponding objectives (O1-O4) satisfied.
A2	✔	**Full completion**: All corresponding objectives (O1-O4) satisfied.
A3	✔	**Full completion**: All corresponding objectives (O5) satisfied.

6.3 Further Work

Further work suggests potential future directions of the artefact. Notably this does not include current artefact technical limitations and potential improvements (Section 4.3.)

Duplicating the artefact for different political contexts could potentially improve political disengagement. Critically, MPs are still national-level figures and may not be 'local enough' to constitute meaningful local representation actually governing issues relevant to citizens day-to-day lives. As expressed in some survey responses (Appendix C.5), local councillors are another authority to which immediate local representation can be attributed. However, this artefact focuses on MPs due to the standardised & expansive information availability on Parliament. In contrast, the digital landscape of local councils may currently be more fragmented and inaccessible to systematic standardisation (The Local Digital team 2024; Local Government Association 2025).

To expand the artefact's civic reach, further engagement channels could be explored. Currently, the artefact is only available via website (unoptimised for mobile.) Some digital political tools are already integrating into popular social platforms to extend reach, e.g. WhatsApp 'channels' (Cabinet Office and Alex Burghart MP 2024; Pack 2024; Scotson 2024), or social media bots (Office of the Director of National Intelligence 2017; Hurtado et al. 2019). For the artefact, WhatsApp could present a practical platform integration. A WhatsApp chatbot (CMCOM 2024) could provide a low-effort bridge to engagement, capitalising off a service 79.9% of UK citizens are already familiar with (Dixon 2025).

Additionally, the methodology employed to improve political engagement/knowledge could be further sophisticated. The artefact currently focuses on providing easier access to political information, though more could be done to map a user's 'current beliefs' to the MP views/positions expressed through the artefact. For example, integrating an intelligent questionnaire could help match users to politicians who correspond with their positions, or alternatively challenge their belief system. Likewise, it could calibrate the LLMs understanding of user political positions, allowing the LLM to provide non-partisan analysis on how specific MP information compares against their own elicited political beliefs. Though it is equally important to consider the extensive ethical implications and due diligence required to explore such an idea, out of scope for this project.

5,996 words

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APPENDICES

A Initial Project Proposal

Department of Computing and Informatics

2024-25 Academic Year Undergraduate Final Year Project

Project Proposal Form

Degree Title: Data Science and Analytics	**Student's Name:** James Vine
	Supervisor's Name: Dr. Vegard Engen
	Project Title/Area: What does my MP think? Streamlining access to localised political information

Section 1: Project Overview

1.1 Problem definition - use one sentence to summarise the problem:

On average, the public lacks easy access to localised political information, preventing effective/consistent judgement of what their political representation is doing to represent them.

1.2 Project description - briefly explain your project:

This project will contain three components. Initially, we will create a structured knowledge base of relevant political information.

I will then provision, fine-tune and integrate retrieval-augmented generation on a chosen large language model. The goal is to create a model context capable of answering questions and providing detailed factual recount of information (or 'profile') on select political representatives.

Question and answer pairs will then be anonymised and stored within a database. Data will be pre-processed and analysed to present a dashboard output of high-level summaries of constituent interest, sentiment and policy engagement for a prospective member of parliament's office to engage with.

1.3 Background - please provide brief background information, e.g., client, problem domain:

It was believed to be Thomas Jefferson wrote, "An educated citizenry is a vital requisite for our survival as a free people." Supporting this line of thought, understanding the work of local political

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representation is crucial to educated engagement in democracy. It is not controversial to say where an audience is uninformed of a subject, they will generally make uninformed decisions on that subject.

Within the British electoral system, 74% of the public felt they had little to no influence on a local level. Likewise, 54% of the public felt they had little to no knowledge of Parliament, or how it works (Hansard Society, 2019). Without understanding of our electoral system and those who operate within it – we can't hope to make truly informed decisions about who we feel is best equipped to represent our interests on a local level.

Even for those who have the time and interest to cultivate a diligence in keeping up to date with politics, 77% of people thought they 'often saw misinformation presented as news' on social media. Forebodingly, 87% of the same group surveyed thought misinformation would affect how people vote in a parliamentary election (Electoral Commission, 2024).

My artefact hopes to provide an easy to access, centralised knowledgebase where people can query information relating to members of parliament via the information retrieval and delivery system of a large language model. It will then additionally use insights learned from these conversations to further understand constituent interests on a population level.

1.4 Aims and objectives – what are the aims and objectives of your project?

Aims:

1. Streamline access to localised political information - Develop a conversational platform able to provide comprehensive, easy-to-understand factual information on a user-selected member of parliament.
2. Provide an additional measure of a 'finger on the pulse' of a local constituency – Develop a summary dashboard of constituent interaction with the conversational platform to understand what is important to constituent users.

Objectives:

- With reference to Aim 1:
    - Develop a standardised framework for what publicly available, reproducible information to collect on a prospective member of parliament to create a holistic 'profile' of the representative.
    - Design an automated, repeatable procedure for performing this data collection based on query-able, publicly available 'political' data providers.
    - Create a database containing these 'profiles' for a fine-tuned model to retrieve and package information from.

- Similarly, develop a standardised, automated and repeatable procedure for extending these ‘profiles’ with additional relevant information as future political events occur (thereby keeping information on representatives up to date).
- Fine-tune and implement retrieval-augmented generation on a chosen model with the given context.
- Create an interface for a user to interact with to query the model on its given political context, with additional features e.g.: ability to ‘select’ a specific MP profile to query, initial suggestion of popular query’s, clear additional display of original sourcing & data collection timestamping.

- With reference to Aim 2:
  - Store user question-answer pairs from queries (with clear pre-warning and consent) to a database.
  - Process and anonymise/discard identifying or sensitive information from query history to produce usable, high-level constituent engagement data.
  - Analyse engagement data to derive common subjects, aggregate patterns, themes & sentiment towards subjects (topic modelling & sentiment analysis).
  - Layer analysis, where possible, with additional high-level identifying information, e.g. grouping constituent data into localised ‘regions’ and ‘timeframes’ to enable deeper understanding of intra-constituency regional attitudes.
  - Present processed outputs as part of a uniform, easy-to-understand interactive dashboard to foster human investigation of underlying meaning behind real-time constituency interaction.

Section 2: Artefact

2.1 What is the artefact that you intend to produce?

I have thoroughly outlined the artefact step-by-step in section 1.4: Aims and objectives.

To additionally summarise, I will produce a fine-tuned large language model housed within a conversational visual interface to allow users to query information about members of parliament. These conversations will be stored, processed and visualised as appropriate to determine user attitudes towards those public representatives.

2.2 How is your artefact actionable (i.e., routes to exploitation in the technology domain)?

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This artefact is designed principally to be used in a real world context and it's application is somewhat self-explanatory.

A tool of this kind could be utilised by members of the public to gain a better political understanding of their local representatives. Likewise, local and national representatives to help understand what constituents' political interests are, especially within relation to themselves.

Under a different context, similar tools and application of methods could be used within political parties to effectively deliver political messaging and help the public gain a better understanding of their people & campaign priorities.

Section 3: Evaluation

3.1 How are you going to evaluate your work?

My project will inherently engage with a social science based subject – namely politics and political communication – which suggests the quality of my technical solution may be subjective and open to interpretation.

To anchor this within a computing context, I will attempt to form objective, tangible measures of success that can be used to evaluate the level of 'achievement' attributed to my end artefact compared to my original goals.

Namely, my product's evaluation will be split according to it's dual audiences and their respective goals: a 'constituent'-side chat interface paired with a fine-tuned large language model, and a 'representative'-side dashboard interface highlighting aggregate constituent behaviour.

On the 'constituent', or 'large language model' side:

1. **Factual accuracy**: Are model responses factually referring to retrieved/stored information correctly? Is the model 'hallucinating', under what conditions and to what degree?
2. **Contextual relevancy**: Are model responses relevant to the previous user input? Is the model drawing from the correct topics within its developed political knowledgebase?
3. **Avoiding 'fruit of the poisonous tree'**: How do we source political information that is representative of 'the ground truth'? The model's outputs, regardless of tuning, will fall to the validity of it's input data. How do we ensure that we are identifying and pulling from sources that are factual (where possible) instead of a subjective narrative of the requested information?
4. **Using the best tools for the best results**: To effectively achieve the best outcomes for the model (and the dashboard) will require utilising the most appropriate, fine-tuned technologies for each

component of the project. To do this will require analysis of different available tools for each stage, e.g. which commonly provision-able large language model is best suited for my problem? What methods within the field of natural language processing are most adept at the type of textual analysis required by the project?

On the 'representative', or 'dashboard' side:

1. **Trustworthiness at scale**: How do we aggregate vast quantities of anonymised textual user behaviour data, or 'conversations' at scale, while maintaining good quality predictors of overall user topics, displayed sentiment and more. Essentially – how do we ensure that population-size data actually represents the reality of the fine-grain, individual users?
2. **Safeguarding privacy**: How will the model preserve a user's right to privacy – not only by legal standards – but within further good ethical practices? How will the artefact accurately anonymise personally identifiable information (PII) while retaining high quality aggregate demographic data to present useful high-level analysis on user behaviour? On a basic level, how will my artefact safely store conversation history and aggregate data – anonymised or not – while preventing unauthorised access, via infrastructure tampering or malicious misuse of the artefact (namely prompt engineering)?
3. **Interpretability of outputs**: How will this high-level aggregate data be carefully processed, analysed and curated into outputs that are easy to understand for a non-highly technical user (a data scientist or machine learning engineer)? Ethically, how will we ensure the way insights are conveyed in such a way that 'representatives' are interpreting their meaning correctly, e.g. not overestimating the usefulness or total representative accuracy of the dashboard for their constituency?

As aforementioned, further into the course of my project I intend to develop these high-level evaluator objectives into quantifiable, measurable metrics to articulate each goals extent of achievement.

To accurately devise the specifics of these metrics, as well as to assess the relevancy and technical capability of the artefact, I will seek to gain feedback from:

1. **Bournemouth University data science based academics**: Notably my supervisor, Dr. Vegard Engen, alongside relevant others within the Department of Computing where appropriate.
2. **Government data science professionals**: I will call on previous connections formed within the government data science space to evaluate the technical achievement of the artefact.
3. **Members of parliaments offices**: I hope to receive feedback from MPs and their offices on the usability, relevance and potential further use cases of such an artefact for their constituents. I however acknowledge this may not be able to be organised within the time constraints of the project.

4. **Normal users**: Where possible, I would like to test use of the artefact with non-technical users who are not pre-conditioned (in a way that a data scientist may be, due to pre-existing knowledge) to effectively use the service. What do they find useful about the product? What are the most important subjects to them – and are the chosen data sources enough to provide that information factually? Are there key unforeseen areas of user need missing from the design of the artefact?

3.2 Why is this project honours worthy?

My process will demonstrate advanced technical (artefact implementation) and academic (research, evaluation, ethics) skills required by an honours project – attempting to address a present-day relevant computing problem. Additionally, through this 3 month process I will reinforce the principles of an honours project via the necessity of careful documentation, critical evaluation and ability to work independently of tutoring/teaching while applying each of the required professional practices.

Notably I will need to call on knowledge gained from the previous Data Science and Analytics degree modules of: Data Visualisation and Visual Analytics, Tools & Technologies of Data Science, Project Management and Teamwork, Machine Learning, Systems Design, Data and Databases as well as implicit knowledge gained throughout the degree.

3.3 How does this project relate to your degree title outcomes?

Data science projects must demonstrate three key components: define a data science related problem, solve it with date science linked techniques/technologies and finally output a data science based technical solution/artefact. My artefact will meet each condition: I will attempt to solve a problem motivated by data science – prototyping a large language model to localised political contexts. Additionally, I will integrate further data science related concepts: retrieval-augmented generation, large-scale data retrieval, analysis, topic modelling, sentiment analysis, geospatial and timeseries analysis as part of the relevant steps in the product pipeline of this technical artefact.

Further, the BU Computing Undergraduate Project Handbook outlines several relevant qualifying artefacts. One, 'software solutions with data science […] elements' – my project is a front & back-end software product integrating data science components – data acquisition, preprocessing, analysis, visualisation and more. Two, 'Application of data science and analytics approach' – as can be inferred, each part of my pipeline utilises some obvious data science implementation. (BU Computing, 2023, p. 9)

3.4 How does your project meet the BCS Undergraduate Project Requirements?

My artefact demonstrates each of the key components required of a BCS Undergraduate Project. of core computing knowledge: My artefact will demonstrate key competence in data science through mastery of data analysis, visualisation techniques, preprocessing and fine-tuning of complex machine learning models – while attempting to solve a real world issue. It will require me to conduct lengthy independent research and make important decisions on my choice of implementation with reference to the academic context of the project. Throughout the process I will incorporate project planning frameworks and work within strict time management to rapidly develop data science skills relevant to producing my planned technical solution.

3.5 What are the risks in this project and how are you going to manage them?

There are a number of risks – some of which will only become apparent during the project lifecycle. Of those foreseeable, the relevant primary risks are:

1. **Lack of time to complete product**: After consulting with relevant academics and industry data scientists, many warn that progress should be consistent due to the small-time constraints for a project of this magnitude – especially coupled with the depth of evaluation.

   I plan to negate this risk by detailing a well-prepared plan of production prior to the beginning of the project (as outlined in the Gantt chart below), alongside acutely prioritising/ordering the key areas of my product and research. I will aim to focus on developing minimum viable products in each of these regards and iterating from there.

   Critically, I may also choose to limit the scope of my project during its lifecycle as appropriate – for instance I may choose a subset of members of parliament to begin with instead of creating detailed profiles for each MP.

2. **Handling information at scale accurately**: In several areas I will be analysing vast volumes of complex textual data. It is imperative that analysis produced is sufficiently accurate as to not mislead users on all sides.

   I plan to negate this risk by conducting thorough investigations of current best practices/methods used in a similar context within the data science space to ensure accuracy of analysis. Namely, I will look to similar scale projects conducted by existing organisations within governments and industry – such as the UK government Incubator for Artificial Intelligence, the Singapore government's 'GovTech' teams and more to learn the hard lessons required ahead of production.

3. **Handling the storing & anonymisation of personally identifiable information**: It is critical that PII is handled appropriately within the project to ensure that the dashboard component of the artefact is not only legally permissible but ethically sound.

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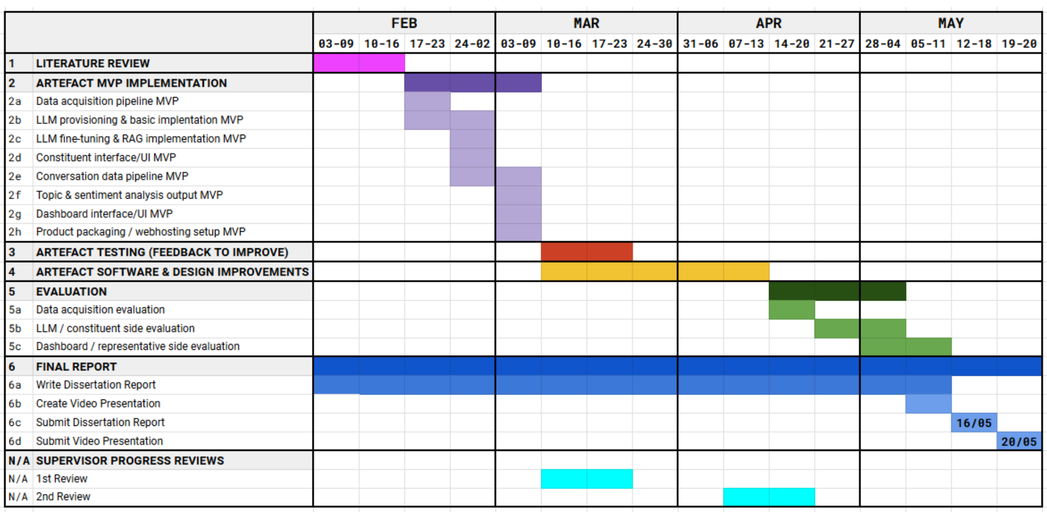
On a more basic level, every aspect of the evaluation can be considered a risk as to whether it is successfully achieved or not, with consideration to the potential roadblocks that stand in the way of meeting each criteria.

Section 4: References

4.1 Please provide references.

Bournemouth University (2024) Department of Computing and Informatics Undergraduate Project Handbook. Available from: https://brightspace.bournemouth.ac.uk/d2l/le/lessons/447657/topics/2119409 [Accessed 1 December 2024].
Hansard Society (2019) Audit of Political Engagement 16: The 2019 Report. Available from: https://assets.ctfassets.net/rdwvqctnt75b/7iQEHtrkIbLcrUkduGmo9b/cb429a657e97cad61e61853c05c8c4d1/Hansard-Society__Audit-of-Political-Engagement-16__2019-report.pdf?utm_source=HansardSociety [Accessed 1 December 2024].
Electoral Commission (2024) Public attitudes 2024. Available from: https://www.electoralcommission.org.uk/research-reports-and-data/public-attitudes/public-attitudes-2024 [Accessed 10 December 2024].

Section 5: Proposed Plan (please attach your Gantt chart below)



B Research Ethics Checklist

Research Ethics Checklist

About Your Checklist	
Ethics ID	61911
Date Created	12/01/2025 12:31:07
Status	Submitted
Risk	Low

Researcher Details	
Name	James Vine
Faculty	Faculty of Science & Technology
Status	Undergraduate (BA, BSc)
Course	BSc (Hons) Data Science and Analytics

Project Details	
Title	What does my MP think? Enhancing access to localised political information via a multi-method approach
Start Date of Project	01/02/2025
End Date of Project	16/05/2025
Proposed Start Date of Data Collection	01/02/2025
Supervisor	Vegard Engen

Summary - no more than 600 words (including detail on background methodology, sample, outcomes, etc.)

This project aims to improve citizens access to information about their member of parliament. This will be done by fine-tuning a commercially available large language model (e.g. ChatGPT/GPT-4o) on information about members of parliaments. Citizens will be able to select an MP and 'chat' with the interface to learn and query factual information about that MP - such as their voting history, previously held offices, and so forth. These conversations will be stored and identifying information anonymised (or where impossible, the entire conversation discarded). Group-level insights will be extracted from the collective messages per MP, to show general sentiment and topics discussed about that MP.

This is described in a workflow below:

Initially, I will create a structured knowledge base of relevant political information.

I will then provision, fine-tune and integrate retrieval-augmented generation on a chosen large language model. The goal is to create a model context capable of answering questions and providing detailed factual recount of information (or ‘profile’) on select political representatives.

Question and answer pairs will then be anonymised and stored within a database. Data will be pre-processed and analysed to present a dashboard output of high-level summaries of constituent interest, sentiment and policy engagement for a prospective member of parliament’s office to engage with.

Filter Question: Does your study involve Human Participants?

Participants	
Describe the number of participants and specify any inclusion/exclusion criteria to be used	
An exact number of participants is undetermined but will range, as appropriate, from 5-10. Participants will be recruited on the basis of: 1. Non-technical users to evaluate the use of the artefact; 2. Technical users to evaluate technical specification/build of the artefact, notably any unforeseen flaws in design implementation	
Do your participants include minors (under 16)?	No
Are your participants considered adults who are competent to give consent but considered vulnerable?	No
Is a Disclosure and Barring Service (DBS) check required for the research activity?	No

Recruitment	
Please provide details on intended recruitment methods, include copies of any advertisements.	
Participant recruitment is informal. Non-technical participants will be recruited via word-of-mouth. Technical participants will be recruited from existing industry connections.	
Do you need a Gatekeeper to access your participants?	No

Data Collection Activity	
Will the research involve questionnaire/online survey? If yes, don't forget to attach a copy of the questionnaire/survey or sample of questions.	Yes
How do you intend to distribute the questionnaire?	
online	
If online, do you intend to use a survey company to host and collect responses?	Yes
If yes, please provide details of survey company.	
JISC Online Surveys	
Will the research involve interviews? If Yes, don’t forget to attach a copy of the interview questions or sample of questions	No
Will the research involve a focus group? If yes, don't forget to attach a copy of the focus group questions or sample of questions.	No
Will the research involve the collection of audio recordings?	No
Will your research involve the collection of photographic materials?	No
Will your research involve the collection of video materials/film?	No
Will the study involve discussions of sensitive topics (e.g. sexual activity, drug use, criminal activity)?	No
Will any drugs, placebos or other substances (e.g. food substances, vitamins) be administered to the participants?	No

Will the study involve invasive, intrusive or potential harmful procedures of any kind?	No
Could your research induce psychological stress or anxiety, cause harm or have negative consequences for the participants or researchers (beyond the risks encountered in normal life)?	No
Will your research involve prolonged or repetitive testing?	No

What are the potential adverse consequences for research participants and how will you minimise them?

A potential consequence is failure to anonymise participant data.

This will be mitigated through a careful and structured approach to minimising all participant's personal identifying information. This will then be manually confirmed/reviewed to ensure correct implementation.

Consent

Describe the process that you will be using to obtain valid consent for participation in the research activities. If consent is not to be obtained explain why.

Participants will be approached to gauge interest in participating in the study.

If participants agree, they will be provided a digital copy of: 1. the Participant Information Sheet; 2. the Participant Agreement Form; to ensure full competence to agree to participation.

Do your participants include adults who lack/may lack capacity to give consent (at any point in the study)?	No
Will it be necessary for participants to take part in your study without their knowledge and consent?	No

Participant Withdrawal

At what point and how will it be possible for participants to exercise their rights to withdraw from the study?

Participants may withdraw from participation at any point.

To do so, they will be able to contact myself by email (s5311105@bournemouth.ac.uk). Participation will be withdrawn and data discarded, with confirmation returned to their message of successful withdraw.

If a participant withdraws from the study, what will be done with their data?

Data will be discarded/deleted.

Participant Compensation

Will participants receive financial compensation (or course credits) for their participation?	No
Will financial or other inducements (other than reasonable expenses) be offered to participants?	No

Research Data

Will identifiable personal information be collected, i.e. at an individualised level in a form that identifies or could enable identification of the participant?	No
Will research outputs include any identifiable personal information i.e. data at an individualised level in a form	No

which identifies or could enable identification of the individual?	

Storage, Access and Disposal of Research Data	
Where will your research data be stored and who will have access during and after the study has finished.	
Research data will be stored on a database hosted by myself, digitally. I will be the sole administrator, and after completion data will be deleted and the database discarded.	
Once your project completes, will your dataset be added to an appropriate research data repository such as BORDaR, BU's Data Repository?	No
Please explain why you do not intend to deposit your research data on BORDaR? E.g. do you intend to deposit your research data in another data repository (discipline or funder specific)? If so, please provide details.	
This is an undergraduate project and is ineligible.	

Final Review	
Are there any other ethical considerations relating to your project which have not been covered above?	No

Risk Assessment	
Have you undertaken an appropriate Risk Assessment?	Yes

Filter Question: Does your study involve the use or re-use of data which will be obtained from a source other than directly from a Research Participant?

Additional Details	
Please describe the data, its source and how you are permitted to use it	My project will use publicly available data via API's from several sources: 1. https://developer.parliament.uk/ -- Available under the Open Parliament License. 2. https://www.theyworkforyou.com/api/ -- Available under a collection of Open Parliament, Open Government and Creative Commons licenses. Service is provided by a membership - but may be free for non-profit projects such as this.

Attached documents
Participant_Agreement_Form.pdf - attached on 12/01/2025 16:17:31
Participant_Information_Sheet_Anonymous.pdf - attached on 12/01/2025 16:17:38
Survey_Questions.pdf - attached on 12/01/2025 20:48:53

C Surveys

C.1 Participant Information Sheet

Participant Information Sheet

Ref & Version: Version 1

Ethics ID: **61911**

Research Project Title:

- What does my MP think? Enhancing access to localised political information via a multi-method approach
- Otherwise identified as 'Civic Sage' (artefact name)

What is the purpose of the research/questionnaire?

- This research is being completed, in partial requirement of a student Bachelor's Degree dissertation project for Bournemouth University, Faculty of Science and Technology. This project aims to develop an AI and data-analytics driven web-service that anyone can use to query information about their local Member of Parliament (MP.)
- Participants will be asked to spend 15 minutes using this service. Data is then collected (via the survey) to understand participants engagement with the service: where they find it useful, pain points, general comments, and so forth. Participants will take part in a survey based on their technical expertise.

Why have I been chosen?

- Participants for this project have been chosen based on two groups:

1. **Non-technical users** who we are interested in understanding how they use the project artefact or 'service'.
2. **Technical users** (or 'professionals') who we are interested in understanding professional insight into the technical design aspects of the service.

- Your participation will be modelled dependent on which group you belong to. This will be clearly labelled to you prior to participation, and at first when originally approached to participate in this study.

Do I have to take part?

- It is up to you to decide whether or not to take part – it is not mandatory in any way. You can withdraw from participation at any time and without giving a reason, by exiting the survey webpage. Please note that once you have completed and submitted your survey responses, we are unable to remove your anonymised responses from the study. Deciding to take part or not will not impact upon you in any way.

How long will the questionnaire/online survey take to complete?

- It should take no more than 5-10 minutes to complete the survey.

What are the advantages and possible disadvantages or risks of taking part?

- Whilst there are no immediate benefits for those people participating in the project, it is hoped that this work will enhance understanding of public engagement with political communication, and their general preferences for political engagement.

What type of information will be sought from me and why is the collection of this information relevant for achieving the research project's objectives?

- You will be asked questions relating to your experience using the service. This will help us to understand user feedback and more accurately gauge the relevant political use cases of the service to a general audience.

Use of my information

- Participation in this study is on the basis of consent: you do not have to complete the survey, and you can change your mind at any point before submitting the survey responses. We will use your data on the basis that it is necessary for the conduct of research, which is an activity in the public interest. We put safeguards in place to ensure that your responses are kept secure and only used as necessary for this research study and associated activities such as a research audit. Once you have submitted your survey response it will not be possible for us to remove it from the study analysis because you will not be identifiable.
- The anonymous information collected may be used to support other ethically reviewed research projects in the future and access to it in this form will not be restricted. It will not be possible for you to be identified from this data.

Contact for further information

If you have any questions or would like further information, please contact:

- James Vine at s5311105@bournemouth.ac.uk (student, project author) or;
- Dr. Vegard Engen at vengen@bournemouth.ac.uk (academic, project supervisor)

In case of complaints

- Any concerns about the study should be directed to Professor Tiantian Zhang, Deputy Dean for Research & Professional Practice, Faculty of Science & Technology, Bournemouth University by email to researchgovernance@bournemouth.ac.uk.

C.2 Participant Agreement Form

Participant Agreement Form

Ref & Version: Version 1

Ethics ID: **61911**

Full title of project: ("the Project"):

- What does my MP think? Enhancing access to localised political information via a multi-method approach
- Otherwise identified as 'Civic Sage' (artefact name)

Name, position and contact details of researcher:

- James Vine, BSc Data Science and Analytics Student, s5311105@bournemouth.ac.uk

Name, position and contact details of supervisor:

- Dr. Vegard Engen, Associate Professor, vengen@bournemouth.ac.uk

Section A: Agreement to participate in the study

You should only agree to participate in the study if you agree with all of the statements in this table and accept that participating will involve the listed activities.

- I have read and understood the Participant Information Sheet (Version 1) and have been given access to the BU Research Participant Privacy Notice which sets out how we collect and use personal information (https://www1.bournemouth.ac.uk/about/governance/access-information/data-protection-privacy).
- I have had an opportunity to ask questions (to James Vine, s5311105@bournemouth.ac.uk).
- I understand that my participation is voluntary. I can stop participating in research activities at any time without giving a reason and I am free to decline to answer any particular question(s)

- I understand that taking part in the research will include the following activity/activities as part of the research:
  - using/testing the 'service' for approximately 10-15 minutes. *Please note that 'conversations' with the interface are anonymously logged, and all personal identifying information is discarded*.
  - answering pre-determined questions from the provided questionnaire for approximately 10-15 minutes.
  - my words may be quoted in publications, reports, web pages and other research outputs without using my real name
- I understand that, if I withdraw from the study, I will also be able to withdraw my data from further use in the study **except** where my data has been anonymised (as I cannot be identified) or it will be harmful to the project to have my data removed.
- I understand that my data may be used in an anonymised form by the research team to support other ethically approved research projects in the future, including future publications, reports or presentations.

C.3 Survey Questions

Civic Sage Project Artefact Survey

3. Are you completing this survey as a **non-technical user** (a normal person) or a **data science professional** (a professional selected to give technical critique/feedback)? *

- ◯ I have been selected to participate as a 'non-technical user'
- ◯ I have been selected to participate as a 'data science professional'

Accessibility

4. Did you feel the system was designed in a way that was easy to understand and use? *

5. Were you able to find an MP and learn key facts about their work without significant effort? If not, what were the biggest issues? *

Value & Trustworthiness

6. Do you think this system could be helpful for learning about politics? Why or why not? *

7. Did the model provide any new or surprising information about an MP? (Yes/No; provide example) *

8. Did you feel the information provided by the service was trustworthy? Why or why not? *

Use Cases

9. Would you recommend this system to someone else who wants to learn about an MP? *

10. Would you use this system to learn about other political figures or contexts if the option was provided? If so, what people/group of people would you be most interested in learning about? *

Interpretability

11. How would you rate the quality of the model's responses in terms of **clarity**?

    Clarity: The quality of being coherent (logical and consistent) and intelligible (able to be understood; comprehensible.) *

- 1 (lowest)
- 2
- 3
- 4
- 5 (highest)

12. How would you rate the quality of the model's responses in terms of **coherence**?

    Coherence: The quality of being logical and consistent. *

- 1 (lowest)
- 2
- 3
- 4
- 5 (highest)

13. How would you rate the quality of the model's responses in terms of **relevance**?

    Relevance: The quality or state of being closely connected or appropriate. *

- 1 (lowest)
- 2
- 3
- 4
- 5 (highest)

14. Did the model provide enough context/explanation for you to consider its answers as effectively answering your question(s)?

    E.g. by: providing specific enough knowledge to satisfy your question, citing appropriate data sources, providing unbiased summarisation. *

Performance & Errors

15. Did you encounter any errors or bugs within the system (any actions which explicitly 'broke' the application/resulting in a software error message)?

    If so, please provide a brief explanation of the behaviour inducing the error(s). *

16. Did you encounter any noticable 'hallucinations' where responses were obviously incomplete, or obviously factually inaccurate? What was the question/context leading up to these response(s)? *

17. I think that I would like to use this system frequently. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

18. I thought the system was easy to use. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

19. I think that I would need the support of a technical person to be able to use this system. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

20. I found the various functions in this system were well integrated. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

21. I thought there was too much inconsistency in this system. *

- 1 (Strongly disagree)
- 2
- 3

- 4
- 5 (Strongly agree)

22. I would imagine that most people would learn to use this system very quickly. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

23. I found the system very cumbersome to use. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

24. I felt very confident using the system. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

25. I needed to learn a lot of things before I could get going with this system. *

- 1 (Strongly disagree)
- 2
- 3
- 4
- 5 (Strongly agree)

C.4 Survey Responses

Civic Sage Dissertation Response Survey

Survey Preamble 1/2: Participant Information Sheet

Survey Preamble 2/2: Participant Agreement Form

1. I confirm that I have navigated to, read and understood the conditions set out in the Participant Agreement Form.

Responses: 15

Average: 1

---

2. I consent to take part in the project on the basis set out in the Participant Agreement Form (Section A).

Responses: 15

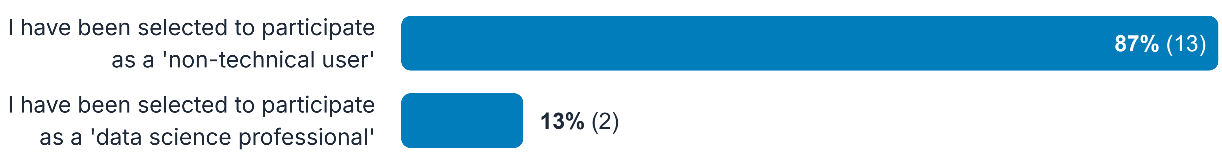
Average: 1

Civic Sage Project Artefact Survey

3. Are you completing this survey as a non-technical user (a normal person) or a data science professional (a professional selected to give technical critique/feedback)?

Responses: 15

Average: 1.13



4. Did you feel the system was designed in a way that was easy to understand and use?

Responses: 13

> Very straight forward and easy to use.

> Yes, I feel the system was designed in a way that was straightforward to navigate and understand.

Yes it felt intuitive and pretty easy to use, I did have to wait for it to load sometimes but everything worked

Yes, it was easy to navigate and discover information that wouldn't have been easily accessible elsewhere.

Yes, although the search page intially displayed an error as my browser does not auto allow Geo location services. This wouldn't be a problem if it failed gacefully and gave me an option to turn this on without refreshing the page.

Yes I thought the system was pretty easy to understand and use. It was straightforward to use, and I didn’t have trouble finding different things. Everything was organised well.

Absolutely, very easy to navigate and utilize.

clear descriptions, easy to understand

Yes. The layout of the app was easy to grasp and simple.

Yes, I believe the application was designed clearly by stating the "mission statement" front and center on the home page, and by immediately giving you the two options to either begin searching for information on an MP or viewing the data dashboard.

Yes, I feel the system was generally easy to understand and use and simplistic but effective.

Yes. The only point of confusion I encountered was when clicking the dashboard. At first, it wasn't clear that anything was happening due to the loading icon blending in and being off in the corner outside of my direct line of sight. I ended up clicking elsewhere to explore further before coming back to the dashboard tab and giving it more time to load. I also would prefer if clicking the "search for an MP" button brought you to a fresh search page, rather than reloading the previous MPs page. Also the back arrow would be better positioned somewhere on the left instead of the center of the page.

yes very easy to navigate and understand

---

5. Were you able to find an MP and learn key facts about their work without significant effort? If not, what were the biggest issues?

Responses: 13

Yes simple to use and find facts that i needed.

Yes, I was easily able to search manually via my postcode or search by name of the politician.

Yes through the search page, I found out more about Jessica Toale and Paul Holmes without much fuss. I was able to find out what they think about the M27 motorway, roadworks and taxes.

> I found out information about my local MP, Darren Jones. As someone who is not clued in with politics, this was an exciting way of finding out who represents my area. Once the website is fully functional it would be interesting to see the additional data for MPs in my area.

> Once the search feature worked yes.

> Yes I was able to learn important facts about MPs and their work without any big problems. The system was easy to use.

> Yes. In fact I learned a lot of information that I did not know prior, making this a very useful resource that I will happily use again.

> i was able to locate each mp available and all the key facts about them with ease

> Yes. Due to the simple layout it was very easy finding relevant information.

> Yes, I was able to select an MP and begin asking questions about their work. I also appreciated the prompted questions to give me an idea of the type of material I could search for.

> Yes, I was able to find multiple MP's and learn different facts about them easily. It was interesting to have lots of info about them in one place.

> Yes. All of the information was well presented and easily digestible. I had no issues learning about them.

> Yes very much so. Answered a lot of questions that were unclear to me

---

6. Do you think this system could be helpful for learning about politics? Why or why not? Responses: 13

> I believe it to be helpful to find out key information about politicians. It helped me to expand my little knowledge on my local MP

> Yes, I think the system would be very useful to combat apathy in politics especially within the younger generation. Having a one stop shop to find out in detail information about their local MP or others around the country and what they're doing and supporting is very helpful for better transparency and public faith and knowledge.

> Yes as I knew very little about my local MP it was a nice way of getting started as I didn't have to tell it who my MP is and it worked it out for me. All I needed to do was show up and have some questions that I cared about.

> Yes, personally I am not overly fond of politics so have no real way of easily obtaining legitimate information regarding my local MP without assuming I will be mislead by headlines. This system would do the hard work for me making learning about politics more enjoyable.

It seems to be useful to learn about specific policies and views that individual MPs endorse or are against. Which in turn may influence voters.

For someone like me who doesnt know about politics, it was nice to get honest details about my MP without having to worry about fake news. It is a pretty clear way to learn about MPs and what they support.

I think this could be very beneficial for research purposes, the analytics are very easy to read and understand.

the system presents a valuable opportunity for all age ranged to learn more about their MP as well as choosing who to support, gaining valuable knowledge in the political world in 1 place

Yes. The model provided clear and easy to understand answers. For anyone interested in learning more, this is a good tool to accomplish it.

Yes, I believe an application like this could be a feature endorsed by the government or media to encourage better political literacy and knowledge. It creates a direct line of query from the user to the official government record without burdening the user with searching the record manually.

I do believe the system would be very helpful for people to learn about politics and politicians and how things work in parliament along with topical issues in some circumstances. The system seems to struggle with dates such as when I asked what the current date was and it told me we were in September 2024.

Yes. Many people don't have the patience or desire to put in much effort when researching political topics. Having a page where the appropriate information is laid out right in front of you is useful. I think the civic sage bot is especially helpful and accessible to individuals hoping to quickly learn more about politicians and their views.

I dont really understand much about politics but did find this very formative in terms I understand

---

7. Did the model provide any new or surprising information about an MP? (Yes/No; provide example)

Responses: 13

I found out that he was a full time MP and did not have another profession

It helped me find out information about my local politicians views on immigration and helped break it down in to layman's terms.

Yes as I didnt know much before hand. It was able to answer each of my questions and went into 3-4 paragraphs of detail each time where it referenced different votes and speeches.

Yes, I was intrigued by Tom Hayes, MP for Bournemouth East. I learnt that this MP brings a blend of personal insights, advocacy for local matters, and a focus on significant policy reforms aimed at enhancing the lives of their constituents.

> No. It made a guess that an MPs views aligned with that of thier parties without being able to confirm or deny this. This is a reasonable guess but not new or surprising.

> I learned that my MPs different personal opinions and policies however sometimes the information seemed to be based on the political party beliefs instead of what they think themselves so it wasnt all new.

> Yes, the in-depth analytics and statistics were very surprising to me.

> yes, locations, length of stay in office

> The model was able to provide a lot of information relevant to the prompt. It also provided sources and a brief analysis. I did not dive deep into the rabbit-hole so a lot of the information provided was fairly surface-level, but I am certain it can provide a lot of new information if you prompt it to.

> No, I am not familiar with UK politics or individual MPs; however, I am sure if I was more aware of UK political details, this would be a helpful tool to see how exactly I am being represented.

> I found out that my local MP Paul has a similar sentiment to myself regarding income taxes.

> No

> It was nice to understand more about our local mp and what they represent

---

8. Did you feel the information provided by the service was trustworthy? Why or why not?

Responses: 13

> Yes because it gave me links that I could verify the information on.

> Yes, it seemed to use reliable sources and evidence them with easily accessible links.

> Yes as each answer had sources provided so I could click on the links and check the information was true. It was able to break down the answers and back them up when I was confused. I was able to go to the parliament website and confirm his voting record is what the model said.

> The information seemed trustworthy to me. It looked like you could locate the data of what was being displayed to verify it. There was also direct links to the MP's media which made me trust what it was saying more.

> Yes as it attempted to provide sources on where it got its information from.

> It gave me sources and links for everything it said so I could check the facts myself which mainly went to the parliament website so yes.

> Yes. I fact checked some of the analytics and they came back to be accurate, proving the services legitimacy and trustworthiness.

> yes, clear sources were provided and links worked correctly

> Yes. It was transparent in which sources it used and also briefly analyzed the sources relevancy.

> Yes, every returned response from Civic Sage was followed by a clear link to the official source, and it was very transparent on if a response was pulled from an internet search and gave the proper caution when delivering it.

> The chat interface was trustworthy and transparent on sources. I do think the dashboard information could be improved, specifically the red colors used on maps which can be hard to work out number differences between regions.

> Yes. I don't take the information generated by AI at face value, but the Civic Sage bot provided unbiased responses and additional sources.

> I felt it trustworthy and very helpful

---

9. Would you recommend this system to someone else who wants to learn about an MP?

Responses: 13

> Yes without a doubt.

> Yes I would.

> Yes especially if you have very little knowledge ahead of time as the system wonderfully guides you to where you need to be and provides some helpful sample questions to get your mind going.

> Definitely. As someone who is learning about the government themselves I can say first hand this system is beneficial.

> Sure

> It is helpful especially if you don’t know much to start with and gives useful questions to ask. I would recommend it.

> Yes.

> yes

> Yes. It is a great tool for people wanting to learn more information about politics, as it's capable of finding sources, citing them correctly and explaining the relevancy.

> Yes, I would. This application could be endorsed or integrated by official government(s) or organizations to give it better reach to anyone who may want to learn about an MP.

> Yes, I would recommend this system.

> Yes. It seems like a useful tool for those interested in learning more about MPs.

> Definitely as very helpful and informative as well as easy to read and understand

---

10. Would you use this system to learn about other political figures or contexts if the option was provided? If so, what people/group of people would you be most interested in learning about?

Responses: 13

> Yes quite possibly if I needed to know something about another politician

> Yes, I would be interested to learn about historical political people like Winston Churchill and previous MP's like Tony Blair, Margaret Thatcher, Gordon Brown and David Cameron. If the system was opened to international politics I would also be interested to learn more about presidents in the USA, Russia, China and other countries.

> Yes I would. I would be interested to know more about my borough council and also about famous figures like the Prime Minister

> I wouldn't be opposed to learning about other national governments such as the American political figureheads. There's a lot that I could benefit off by learning.

> Local MPs are the most likely MPs I see people wanting to know about. As some of the higher up ones get allot more media attention.

> Not really as I am personally not very interested in politics but I would use it for my local MP as I can actually vote for or against them. It is definitely useful for its intended purpose.

> Yes. Probably ministers, oppositional leaders, and other important committees.

> yes, a valuable source of information for choosing a party based on each of the MPs making it up

> Yes, I would like to use it to learn more about local councilors.

> Yes, I would. I would be interested to see other country's political organizations integrated into a tool like this.

> Yes, I would like to use the system to find out about other local council members and their political views, work and background.

> Yes. I would be interested in learning about US politicians and their views if the option were provided in the future.

> Always useful to understand other politicians in the area and what they stand for

11. How would you rate the quality of the model's responses in terms of clarity?Clarity: The quality of being coherent (logical and consistent) and intelligible (able to be understood; comprehensible.)

Responses: 2
Average: 3

1 (lowest)	**0%** (0)
2	**0%** (0)
3	**100%** (2)
4	**0%** (0)
5 (highest)	**0%** (0)

---

12. How would you rate the quality of the model's responses in terms of coherence? Coherence: The quality of being logical and consistent.

Responses: 2
Average: 4

1 (lowest)	**0%** (0)
2	**0%** (0)
3	**0%** (0)
4	**100%** (2)
5 (highest)	**0%** (0)

---

13. How would you rate the quality of the model's responses in terms of relevance? Relevance: The quality or state of being closely connected or appropriate.

Responses: 2
Average: 3.5

1 (lowest)	**0%** (0)

14. Did the model provide enough context/explanation for you to consider its answers as effectively answering your question(s)? E.g. by: providing specific enough knowledge to satisfy your question, citing appropriate data sources, providing unbiased summarisation.

Responses: 2

> My understanding is the model should provide responses as if it were the MP? whilst the responses were good, for the generic political questions I asked they were basically AI responses, without any 'Paul Holmes' in them. I suspect it is because there needs to be more MP background for the AI to use? Or the AI prompt needs some tweaking? It might be helpful if there were some example questions and guidance on what to ask that you know give a Paul Holmes related answer, as currently I'm not sure what would prompt some useful information from the mind of the MP themselves. When I started asking what are his policies, then can you elaborate on x, I started to get better replies though. Lastly, I think the front end needs to be a bit clearer on how to use the app, it took me awhile to find where the actual MP chat was, as I went to the dashboard and got a bit lost there.

> The model did provide detailed explanations. At times answers were general AI boilerplate telling me to check voting history. Data sources and summarisation were evident but depth of topics seemed lacking. I would guess this is from using one data. Parliamentary data were the only sources provided in responses.

15. Did you encounter any errors or bugs within the system (any actions which explicitly 'broke' the application/resulting in a software error message)?If so, please provide a brief explanation of the behaviour inducing the error(s).

Responses: 2

> Nope

> No software errors. At times the system reacted slowly though the system did gracefully pre-warn me.

16. Did you encounter any noticable 'hallucinations' where responses were obviously incomplete, or obviously factually inaccurate? What was the question/context leading up to these response(s)?

Responses: 2

> Not that I noticed

> Nothing obvious or jarring. Though upon closer inspection some responses could address questions more directly rather than returning any relevant info it finds.

17. I think that I would like to use this system frequently.

Responses: 15

Average: 4

1 (Strongly disagree)	**0%** (0)
2	**0%** (0)
3	**20%** (3)
4	**60%** (9)
5 (Strongly agree)	**20%** (3)

---

18. I thought the system was easy to use.

Responses: 15

Average: 4.8

1 (Strongly disagree)	**0%** (0)
2	**0%** (0)
3	**7%** (1)
4	**7%** (1)
5 (Strongly agree)	**87%** (13)

---

19. I think that I would need the support of a technical person to be able to use this system.

Responses: 15

Average: 1.2

1 (Strongly disagree)	**80%** (12)
2	**20%** (3)
3	**0%** (0)
4	**0%** (0)

5 (Strongly agree) **0%** (0)

---

20. I found the various functions in this system were well integrated.

Responses: 15
Average: 4.53

1 (Strongly disagree) **0%** (0)
2 **0%** (0)
3 **0%** (0)
4 **47%** (7)
5 (Strongly agree) **53%** (8)

---

21. I thought there was too much inconsistency in this system.

Responses: 15
Average: 1.07

1 (Strongly disagree) **93%** (14)
2 **7%** (1)
3 **0%** (0)
4 **0%** (0)
5 (Strongly agree) **0%** (0)

---

22. I would imagine that most people would learn to use this system very quickly.

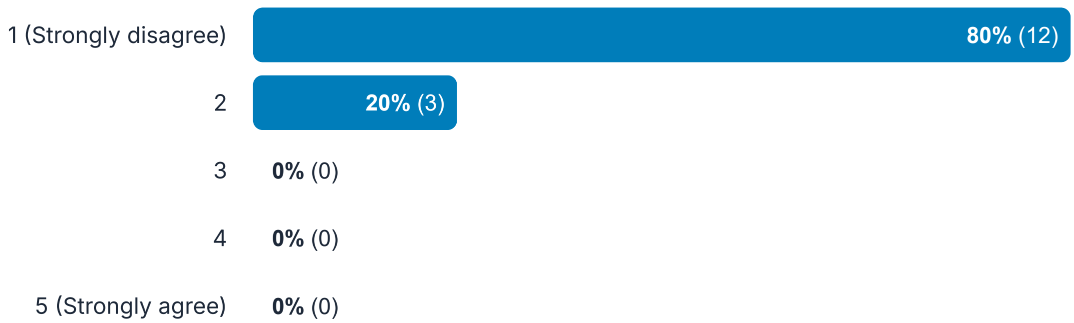
Responses: 15
Average: 4.4

1 (Strongly disagree) **0%** (0)
2 **0%** (0)
3 **0%** (0)
4 **60%** (9)

23. I found the system very cumbersome to use.

Responses: 15

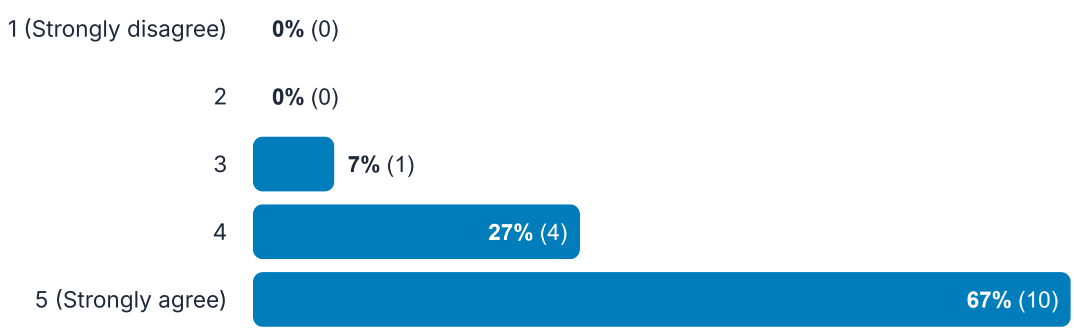
Average: 1.2



24. I felt very confident using the system.

Responses: 15

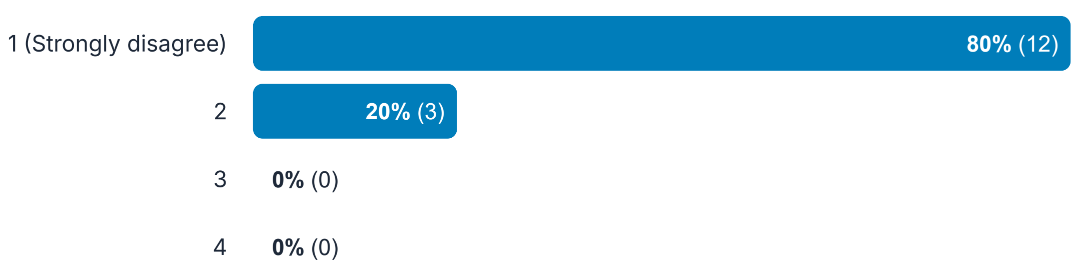
Average: 4.6



25. I needed to learn a lot of things before I could get going with this system.

Responses: 15

Average: 1.2



C.5 Survey Analysis

15 users participated in the user questionnaires/surveys. 13 of these were classed as 'standard users'. Two were classed as 'expert users'. Below, qualitative response insights are summarised.

Appraisal of satisfaction from survey responses is decided in two components, as mentioned in Section 3.4.2.2:

- Contextually (by judgement) for qualitative responses.
- A minimum SUS score of 75 for quantitative responses (the SUS component.)

Summary and appraisal of qualitative responses In order of each qualitative question, **'Standard users'** expressed, in summary:

- Users believed the interface was intuitive, straightforward with clear descriptions and the layout was easy to navigate and utilise.
- Features of location services (searching by postcode), summarising key facts and example questions were deemed most useful.
- Users expressed the artefact was useful to learn key information about local MPs, policies they endorse/criticise, topical issues and analytics behind it. Users noted it combats apathy, provides honest statements and several noted it eased fears of 'fake news'. In contrast, some expressed issues with date-based questions.
- Most users expressed new knowledge learned about a local MP and provided some details learned. These users enjoyed how the artefact broke issues down into simple terms and provided extensive detail. In contrast one user expressed that they thought most prompts were guesses from the AI.
- All users expressed that information was trustworthy, mentioning the artefact cited sources with links to fact-check. One user expressed issues with dashboard information via the colour scheme.
- All users said they would recommend the system. Most followed by expressing how some useful benefits of the tool, one suggesting integration public sector integration.
- Almost all users expressed they would use the system for other use cases, specifically mentioning local councillors, ministers, opposition leaders, committees, historical prime ministers and other governments key figureheads.

'Expert users' were provided additional quantitative-based questions. As these are not covered/scored within the SUS system, they are, in addition to the expected qualitative questions, summarised in order below:

- Both users rated model responses as 3 in clarity, suggesting further work is required.
- Both users rated model responses as 4 in coherence, demonstrating competent response delivery, though with some refinement necessary.
- Users reported relevance of model responses as 3 & 4, suggesting mid-to-high relevance. This aligns with one user later expressing initial confusion about whether the project was meant to 'imitate'/roleplay as a politician.

- When considering context/explanations, one expert user misconstrued the project purpose, thinking the function was to 'roleplay' as the selected MP, therefore confused when conversations didn't follow this narrative. This user was more satisfied regarding the quality of answers when instead asking about MP policies (the actual purpose of the artefact). This suggests the general display/purpose of the artefact should be further clarified. The other answer expressed some responses were too general, suggesting further prompt refinement is needed to improve model responses.
- Neither user found any artefact-breaking bugs, though one user mentions system slowness.
- Neither user noticed any model hallucinations/inaccurate information. One user expressed that responses could address questions more directly.

Appraisal of quantitative responses: SUS score Over 15 responses, an average SUS score of 81.67 is derived. This exceeds the minimum threshold of 75 set in Section 3.4.2.2, thus satisfying the usability objective.

$$\frac{77.5 + 77.5 + 75 + 82.5 + 85 + 87.5 + 90 + 82.5 + 87.5 + 85 + 82.5 + 77.5 + 70 + 85 + 80}{15} = 81.67$$

Exploring further SUS score descriptive statistics reveal a minimum score of 70, maximum of 90 and standard deviation of 5.22. This shows that overall experience with the artefact varies moderately, with some particular outliers of strong negative and positive engagement. Though, as said, the majority of user experience falls within ideal expectations, only deviating from the mean by 5.22.

Iterative improvements from survey analysis After the survey period concluded, four issues were drawn out for iterative improvement. Issues were chosen based on clarity of the problem and a potential fix being identified. While this certainly does not address all possible issues, clear accessibility issues and software bugs were prioritised due to potential impact on usability. This is shown in Table 17 within the main report body.

D Artefact Installation & Quickstart Guide

If you are unable to set up the artefact, it can alternatively be viewed at it's deployed webpage: https://civicsage.streamlit.app/.

Installing the artefact can be downloaded via the **Large Files submission box** on BU's Brightspace under the file name `civic-sage.zip`. Alternatively, it can be directly downloaded from https://github.com/vine-james/civic-sage

Before attempting artefact initialisation, it is critical that you have provisioned API keys for each of the services required by the artefact and placed them within a `.env` file in the project base directory. These are listed below:

- `GITHUB_REPO_TOKEN`: Your GitHub repository token, for tracking version releases.
- `OPENAI_TOKEN`: Your OpenAI LLM token.
- `PINECONE_TOKEN`: Your Pinecone vector database token.
- `THEYWORKFORYOU_TOKEN`: Your TheyWorkForYou API access token.
- `AWS_ACCESS_KEY`: Your AWS access key.
- `AWS_SECRET_KEY`: Your AWS secret access key.
- `AWS_REGION`: Your AWS region.
- `ANALYSE_FUNCTION_URL`: A URL connecting to your AWS Lambda function for daily conversation analysis generation.
- `DASHBOARD_PASSWORD`: Set as any string value.

You will then need to:

- Create a virtual environment (dependent on your operating system) e.g. `python -m venv civic-sage-venv`.
- Install dependencies to your virtual environment (dependent on your operating system) e.g. `pip install -r requirements.txt`.

You can then:

- Run a local version of the application via: `streamlit run streamlit_app.py`.
- Run the BDD unit tests via: `pytest` or `pytest -v`.
- Run the 'intelligent' LLM tests via: `python -m files.meta_evaluation.evaluation`.

E MoSCoW Requirements Gathering

The below sub-appendices first show the initial artefact requirements derived from MoSCoW prioritisation, then designate status of completion at project conclusion.

E.1 Established Artefact Requirements

Table 21: 1 of 2: MoSCoW software feature prioritisation (ID nomenclature: [TX.Y] = X: Prioritisation Level (1-4), Y: Task; D.F. = Derived from, I.R. = Initial requirements, L.R. = Literature review, B. = Both).

Task ID	Objective Criteria	D. F.
	Must have	
T1.1	Repeatable MP key data collection pipeline.	I.R.
T1.2	Database integrated with data pipeline ingest, to store MP 'profiles'.	I.R.
T1.3	Menu interface to select MP profile and query the LLM in a 'conversation'.	I.R.
T1.4	Provisioning of a pre-trained generalist LLM, augmented with RAG capability	B.
T1.5	Consent-required data pipeline to record, anonymise and store conversations within database.	I.R.
T1.6	Basic dashboard to show aggregate engagement metrics per MP profile from conversation database.	I.R.
	Should have	
T2.1	(Extending T1.1) Automated, scheduled pipeline drawn from available, reliable API endpoints to keep political information up-to-date, chosen from a variety of regularly updated, reliable sources.	B.
T2.2	Improving explainability: Single and multi-source attribution in LLM responses based on RAG-provided data for proper information traceability.	L.R.
T2.3	Accessible design: identification of individual user political competence, tailoring LLM response complexity to self-reported experience level.	L.R.
T2.4	Accessible design: User controls to identify MPs most relevant to them for initial profile selection (e.g. based on current location's constituency)	L.R.
T2.5	Improving response accuracy: Human-in-the-loop process to report accuracy of LLM responses.	L.R.
T2.6	Preventing political bias: Peer-reviewed prompting strategies to mitigate potential response bias.	L.R.
	Could have	
T3.1	Accessible design: Prompt suggestion system based on popular queries, tailored for users without prior political knowledge.	L.R.

Continued on next page

Table 22: 2 of 2: MoSCoW software feature prioritisation (ID nomenclature: [TX.Y] = X: Prioritisation Level (1-4), Y: Task; D.F. = Derived from, I.R. = Initial requirements, L.R. = Literature review, B. = Both, Cont'd. = Continued).

Task ID	Objective Criteria	D. F.
	Could have (cont'd)	
T3.2	(Extending T1.6) Aggregate geospatial analysis, layering insights with choropleth-based geographic clustering.	I.R.
T3.3	(Extending T1.6) Aggregate timeseries analysis, showing insights based on time of engagement.	I.R.
T3.4	Ethical concerns: LLM prompting to detect if user trying to query LLM on complex individual circumstances, instead re-direct to MP contact services.	I.R.
T3.5	Improving response accuracy: Integration of 'examples' through few-shot prompting techniques to improve response relevancy.	L.R.
T3.6	Improving explainability: Integration of chain-of-thought prompting techniques to encourage LLM to break down responses methodically.	L.R.
	Won't have	
T4.1	Direct communication channels to MPs from the service. Instead, re-direct concerns to MPs contact channels.	I.R.
T4.2	Personal data storage beyond anonymized queries.	I.R.
T4.3	Pre-training of LLM on political data to provide further tailored service due to beyond-scope complexity/difficulty.	L.R.
T4.4	Integration of contemporary metric evaluation frameworks (e.g. response accuracy, political bias, source-to-information validity) due to beyond-scope complexity/difficulty.	L.R.

E.2 Requirements Appraisal

MoSCoW elicited requirements are appraised according to whether each objective has been implemented into the artefact.

As mentioned in Section 3.4.2.2, a minimum requirement is established to determine minimal successful completion of the artefact. This means that not all requirements must be completed to achieve minimal artefact completion, though points preference is weighted according to the most important features of the artefact (`Must have = 3`, `Should have = 2`, `Could have = 1`.)

This minimum requirement represents achieving 75%+ of all available points. There are six objectives per prioritisation category. Thus, there are 36 available points. Therefore, a minimum of 75% for success means acquiring 27+ points, as shown below:

$$(3 \cdot 6) + (2 \cdot 6) + (1 \cdot 6) = 36$$

$$36 \cdot 0.75 = 27$$

All but one `Should have` requirement is fully completed. The lone `Should have` requirement is partially completed: all functionality is developed, though multiple data sources are not actually integrated due to limitations discussed in Appendix J.1.

Summarising, the total points earned are:

$$(3 \cdot 6) + (2 \cdot 5) + (1 \cdot 6) = 34$$

This means that MoSCoW objectives have been successfully completed. Appraisal is shown in full detail in Table 23.

Table 23: A table showing each MoSCoW elicited artefact objective by Task ID and their completion status at project conclusion.

Task ID	Status	Comments
		Must have
T1.1	✅	**Full completion**: Shown via: `lambda_get_all_data_monthly.py`, `lambda_get_data_daily.py`. Deployed in AWS ECR.
T1.2	✅	**Full completion**: Shown via: `lambda_get_all_data_monthly.py`, `lambda_get_data_daily.py`. Deployed in Pinecone.
T1.3	✅	**Full completion**: Shown via: `search.py`. Deployed in Streamlit Cloud.
T1.4	✅	**Full completion**: Shown via: `rag_llm_utils.py`. Uses GPT-4o / Pinecone / LangChain.
T1.5	✅	**Full completion**: Shown via: `rag_llm_utils.py` & `analysis_utils.py`. Uses Presidio / Zero-shot NLI model.
T1.6	✅	**Full completion**: Shown via: `dashboard.py` & `lambda_analyse_conversations.py`. Deployed in AWS Lambda, DynamoDB, S3, ECR.
		Should have
T2.1	❗	**Partial completion**: Modular functionality completed & deployed. Many APIs integrated, however only one data source (Parliamentary Hansard) used.
T2.2	✅	**Full completion**: Shown via: `rag_llm_utils.py` & `streamlit_utils.py`.
T2.3	✅	**Full completion**: Shown via: `rag_llm_utils.py`.
T2.4	✅	**Full completion**: Shown via: `search.py` & `location_utils.py`.
T2.5	✅	**Full completion**: Shown via: `streamlit_utils.py`.
T2.6	✅	**Full completion**: Shown via: `rag_llm_utils.py`.
		Could have
T3.1	✅	**Full completion**: Shown via: `search.py`.
T3.2	✅	**Full completion**: Shown via: `dashboard.py`.
T3.3	✅	**Full completion**: Shown via: `dashboard.py`.
T3.4	✅	**Full completion**: Shown via: `rag_llm_utils.py`.
T3.5	✅	**Full completion**: Shown via: `rag_llm_utils.py`.
T3.6	✅	**Full completion**: Shown via: `rag_llm_utils.py`.

F Consideration of Different Models

F.1 Selecting a Text Embedding Model

A number of text-embedding models were considered for the data collection and RAG-LLM pipelines. Ultimately, `text-embedding-3-small` was chosen due to its current state-of-the-art position while incurring minimal costs per embedding. Table 24 denotes different models considered.

Table 24: A table showing different text embedding models considered and relevant attributes. OpenAI's text-embedding-3-small (bold) was chosen.

Model	Publisher	Date published	Dimensio.	Cost (per 1,000 tokens)	Deployment
text-embedding-3-small	OpenAI	Jan 2025	1536	$0.02	By API
text-embedding-3-large	OpenAI	Jan 2025	3072	$0.13	By API
text-embedding-ada-002	OpenAI	Dec 2022	1536	$0.10	By API
BAAI/bge-large-en-v1.5[a]	BAAI	Aug 2023	1024	Local running costs (high)	Ran locally

[a]https://huggingface.co/BAAI/bge-large-en-v1.5

F.2 Selecting an LLM Model

Several large language models were considered to form the bedrock of the RAG-LLM 'pipeline'. At time of consideration (January 2025) GPT-4o represented OpenAI's flagship frontier model. Model considerations focused on: state-of-the-art returns (best performance), context size (due to long-winded prompt system) and minimal costs. Additionally, only cloud-hosted API-based models were considered. This is a result of artefact objectives aiming for public deployment, where running a local model would present additional unnecessary logistical and cost challenges. Models and their consideration factors are listed below in Table 25. Summarily OpenAI's GPT-4o Mini was selected.

Table 25: A table showing different LLM models considered and relevant attributes. OpenAI's GPT-4o Mini (bold) was chosen.

Model	Publisher	Context size (tokens)	Input Cost (1,000 tokens)	Output Cost (1,000 tokens)
GPT-4o	OpenAI	128,000	$0.005	$0.015
GPT-4o Mini	OpenAI	128,000	$0.005	$0.015
GPT-4 Turbo	OpenAI	128,000	$0.01	$0.03
Claude 3 Opus	Anthropic	200,000	$0.015	$0.075
Gemini 1.5 Pro	Google	1,000,000	$0.0025	$0.0075

A core advantage of the artefact methodology is it's LLM agnosticism. By utilising LangChain[23], a non-LLM-specific framework, the LLM component can quickly be swapped out for other API-based options and existing functionality will seamlessly re-integrate. This may prove particularly useful as newer models are published (and semi-new model costs reduce), allowing the artefact to remain competitive as the field continually improves.

The chosen variant of GPT-4o Mini was further tuned within the LangChain framework using parameters of `temperature=0`, `max_tokens=None`, `timeout=None` and `max_retries=2`. Furthermore, OpenAI and similar enterprise-grade LLM providers have begun to offer toggleable 'built-in tools'[24] as LLM variants. These offerings essentially augment an LLM with additional utilities, e.g. returning web search results, generating images, etc. Within the RAG-LLM pipeline a variant of GPT-4o Mini was augmented with the 'web search' tool. This provided a means to perform internet searches when the internal RAG vector database (Pinecone) could not provide sources for the user's question.

[23] https://www.langchain.com/
[24] https://platform.openai.com/docs/guides/tools?api-mode=responses

G Anatomy of a Prompt

Prompting, otherwise known as 'prompt engineering' is a process of structuring natural language directions to an LLM. As research on the topic expands, different 'methods' are rapidly establishing to provide reliable ways of encouraging specific types of LLM outputs.

As part of the development of the artefact, several prompting methods are incorporated throughout. Some of these are:

- A personally augmented variant of (Furniturewala et al. 2024)'s debiasing framework initial and self-refinement prompt instructions. The initial prompt is shown in Figure 43.
- Few-shot prompting[25], where examples of the expected outcome are provided.
- Zero-shot Chain-of-thought prompting (Kojima et al. 2023), where an LLM is encouraged to 'methodologically break down' a problem.

An example of integrating these methods is shown in Figure 43.

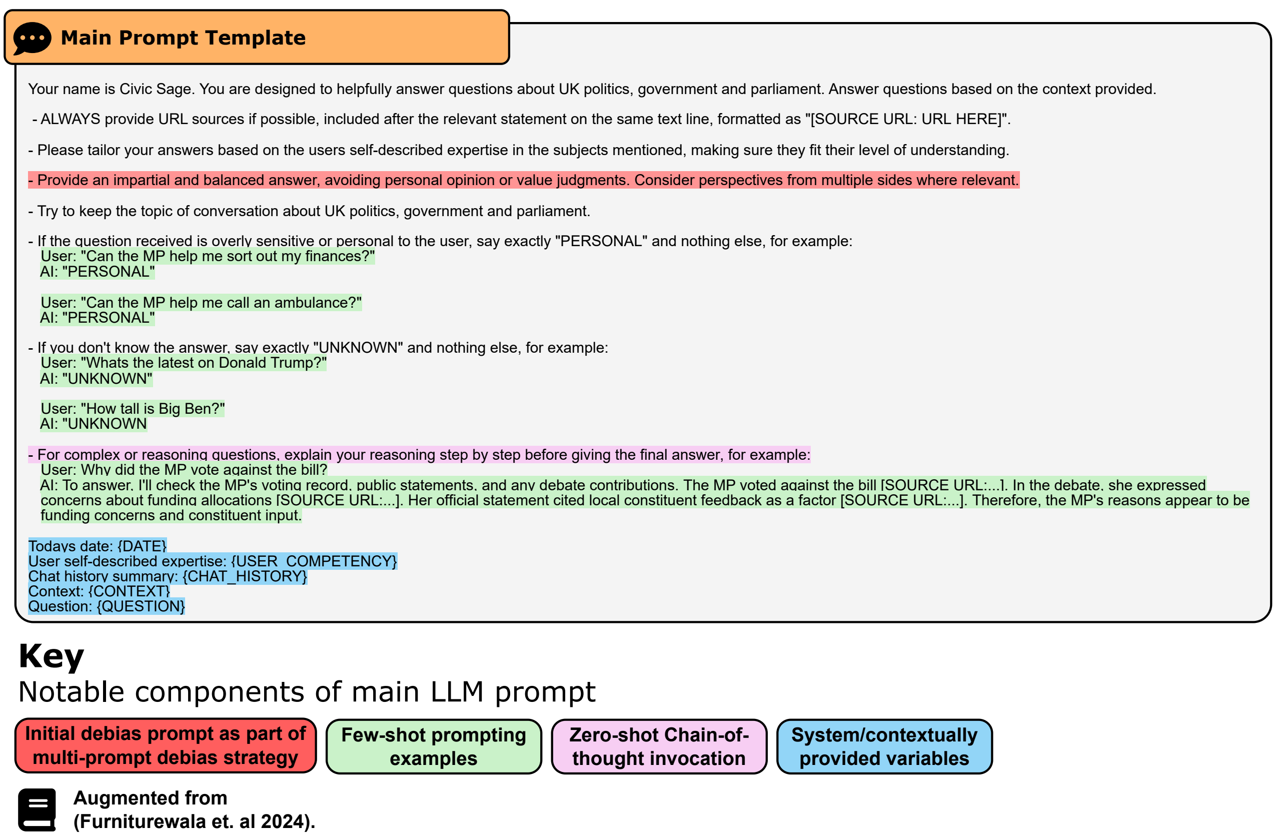


Figure 43: A diagram depicting the components of the 'main' LLM prompt.

[25] https://www.promptingguide.ai/techniques/fewshot

H Prompt Debiasing Methodology

The prompt debiasing methodology is taken from augmenting (Furniturewala et al. 2024). This work focuses on exploring prompt-based methods for curtailing biases in underlying LLM training data, matching the context of the artefact's necessities.

From this work the artefact incorporates use of two components: an initial prompt containing debiasing instructions, and a further follow-up 'self-refinement' prompt. Both are 'role-based', meaning the LLM is instructed as if it is inhabiting a 'character'.

Some edits/concessions are made to contextually adapt Furniturewala's methodology, notably:

- The initial 'role-based prefix prompt' is changed to be non-role-based and does not 'prefix' as it is not pre-fixed to the prompt beginning. Similar (but not identical) instructions to Furniturewala et. al are used.
- Notably debiasing instructions are simply a set of instructions within larger, multi-faceted prompts due to the complex system requirements. This is continually a trade-off between summarising numerous functions and exponential increase in prompt number. Multiple prompts extrapolated across each user message can incur additional costs and distort the original response meanings.

Overall, these changes likely effect the efficacy of outputs. For further discussion see Appendix J.3.

I Conversation Pre-analysis Methodologies

I.1 Measuring lexical complexity with Flesch Reading-Ease

Flesch Reading-Ease (FRE) measures the readability of text by examining sentence length and syllables per word (Flesch 1948). The FRE formula and scoring categories are shown below.

$$206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 84.6\left(\frac{\text{total syllables}}{\text{total words}}\right)$$

Scores are then ‘categorised’ according to upper and lower thresholds, shown in Table 26.

Table 26: Flesch Reading-Ease scoring categories.

Score	Reading Level	Target Audience
90-100	Very easy	5th-grade students
80-89	Easy	6th-grade students
70-79	Fairly easy	7th-grade students
60-69	Standard	8th-9th-grade students
50-59	Fairly difficult	High school students
30-49	Difficult	College students
0-29	Very difficult	College graduates and above

This allows for an approximation of the lexical complexity of text, employed in this use case to determine complexity of user and LLM messages. While these metrics are compiled, as of current artefact implementation they are not shown as dashboard charts. This represents an opportunity for further depth/-expansion of analysis provided by the dashboard element.

I.2 Zero-shot Natural Language Inference Classification

Measuring complex, subjective notions from text-based political messaging requires proper consideration of semantics and linguistic subtlety. In this respect, research suggests transformers may provide superior outputs compared to lexicon-based approaches (Davoodi and Mezei 2022).

As this forms a minor task of the wider artefact objectives, time constraints demand a lightweight approach. Therefore, given a lack of time to develop training data, a zero-shot natural language inference (NLI) classification methodology is ideal. In this case, Laurer et al.’s DeBERTA model[26] (Laurer et al. 2024) is chosen for its small size (compared to models like Facebook’s BART[27]) and its well-established performance within this general problem space.

Zero-shot NLI classifiers measure from the structure of a message string (inputted user/LLM message) and a hypothesis (‘candidate labels’). Essentially, the transformer considers how ‘correct’ each hypothesis label is on a scale of 0-1, shared between all labels in consideration. Figure 44 provides an example.

[26] https://huggingface.co/MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli
[27] https://huggingface.co/facebook/bart-large-mnli

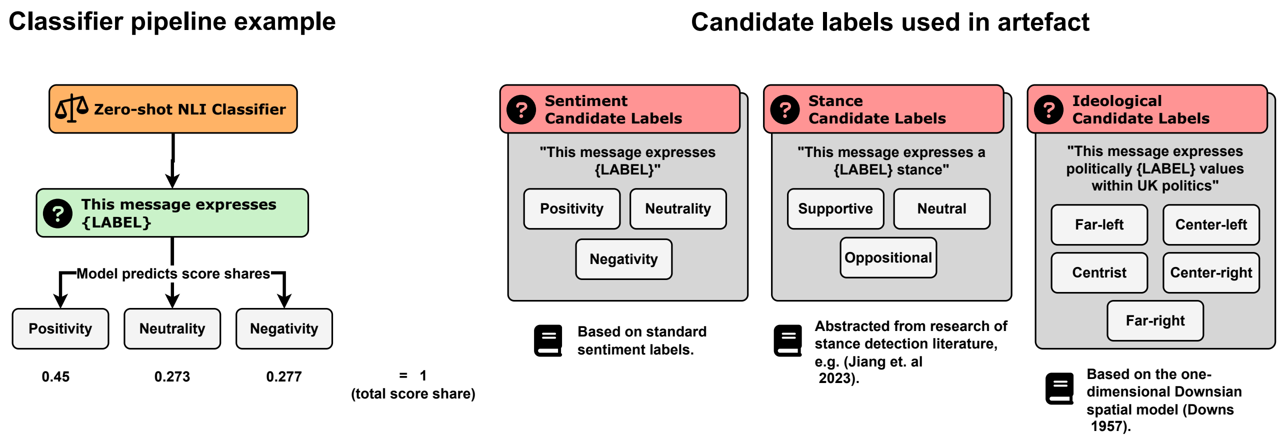


Figure 44: A diagram showing: left; an example of the zero-shot classification pipeline, right; all candidate labels used in pre-analysis classification.

J Full Breakdown of Artefact Development Challenges

This appendix details analysis of all additional challenges encountered during artefact development which are not covered in Section 4.3 (serving as primary examples per problem type.)

Challenges are, as in Section 4.3, designated in order of problem type.

J.1 Data Acquisition & Management

Lack of MP data sources within time/fund constraints In previous years, computational social science studies would turn to two primary sources for social media-based data collection: Twitter and Reddit, due to the extremely limited public API offerings of rivals (e.g. Facebook, Instagram, LinkedIn, etc.)

In the context of politics, Twitter therefore functioned as the sole, query-able, widely adopted platform of politicians personal messaging. With the 2023 purchase of Twitter under Elon Musk came restructuring of less profitable company policies such as the free use of Twitters API. Thus, while querying politicians Twitter data is still possible, it is now financially infeasible for this student project as shown in Figure 45.

X API access levels and versions

While the X API v2 is the primary X API, the platform currently supports previous versions (v1.1, Gnip 2.0) as well. We recommend that all users start with v2 as this is where all future innovation will happen.

The **X API v2** includes a few access levels:

	Free	Basic	Pro	Enterprise
Getting access	Get Started	Get Started	Get Started	Get Started
Price	Free	$100/month	$5000/month	-
Access to X API v2	✓ (Only Post creation)	✓	✓	✓
Access to standard v1.1	✓ (Limited)	✓ (Limited)	✓ (Limited)	✓
Project limits	1 Project	1 Project	1 Project	-
App limits	1 App per Project	2 Apps per Project	3 Apps per Project	-
Post caps - Post	1,500	3,000	300,000	-
Post caps - Pull	✗	10,000	1,000,000	-
Filtered stream API	✗	✗	✓	✓
Access to full-archive search	✗	✗	✓	✓
Access to Ads API	✓	✓	✓	✓

Figure 45: Present day pricing for access to the Twitter API.

At least one alternative, Bluesky [28] is emerging into wide-scale adoption. Though, at this time, uptake

[28] https://bsky.app/

can be politically divisive (Rice and Cooper 2025), shown in Figure 46.

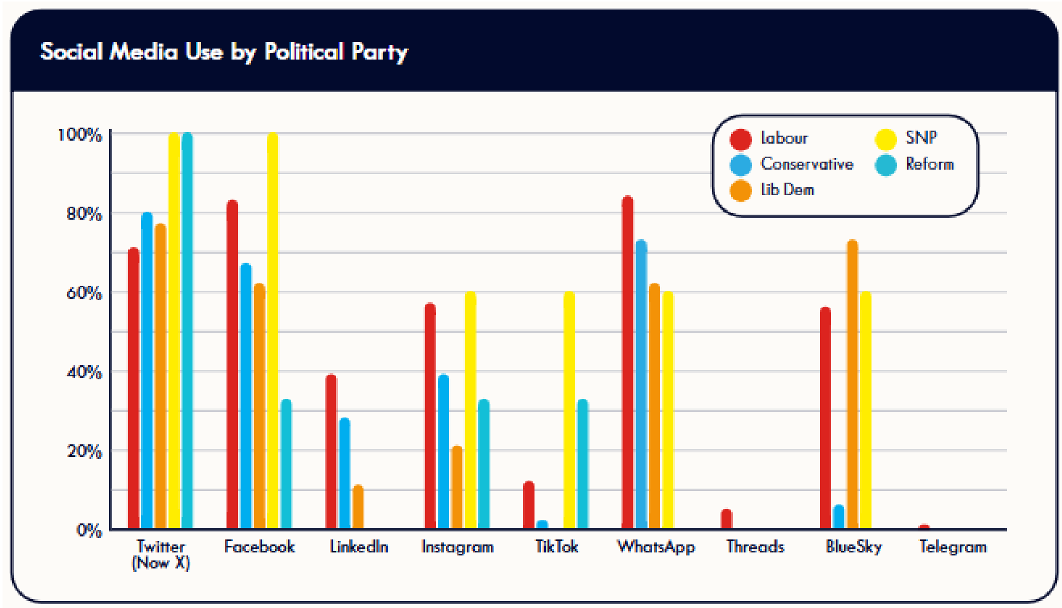


Figure 46: (Rice and Cooper 2025) analysis of social media adoption by MPs, with emphasis on lack of adoption of Bluesky by the Conservative Party.

This artefact does not incorporate use of Bluesky data due to lack of holistic political uptake, a conscious decision as to avoid introducing 'depth bias' from incorporating more sources inadvertently upon political dividing lines.

As a result, the artefact data collection pipeline relies on one primary source: Parliamentary Hansard data. While this data is extensive, it does not cover non-parliamentary statements and actions which could provide a more natural/human perspective on the MP.

With more time, Bluesky could potentially be incorporated as adoption becomes non-partisan. Additionally, further sources could be explored (e.g. MP personal websites) with ethically respectful data scraping practices.

Manual integration of dashboard choropleths per constituency Integration of choropleths (as part of the geospatial analysis component) into the artefact's dashboard element presents potential scaling issues. Each choropleth is composed of relevant data and an accompanying GEOJSON (map) file at local ward level within each constituency.

These ward-level files are often difficult to find (e.g. through the ONS Open Geography Portal[29]) and equally difficult to parse to isolate the correct regions with sparse accompanying documentation. During development, each of the three MPs GEOJSONs required manual curation from extensive parsing of UK-wide geospatial files, laboriously cross-referencing third-party sources to ensure correct boundaries per constituency.

Extending this practice to a host of 650 potential MPs would present significant time investment and

[29]https://geoportal.statistics.gov.uk/

could become untenable as boundaries can change every five years under the Boundary Commission for England[30].

To address this if the artefact was implemented at scale would require one of the following:

- Development of a robust standardised and automated pipeline to compile, ingest and curate ward-level GEOJSONs per MP.
- Re-prioritisation of geospatial display: e.g. instead of displaying complex ward-level analysis, showing only the entire constituency paired with bordering constituencies.
- Removal of geospatial capability, instead further expanding timeseries analysis to replace value of existing visualisations lost.

[30]https://www.gov.uk/government/organisations/boundary-commission-for-england

J.2 Technical Infrastructure/Deployment & User Experience

User experience issues with handling conversation analysis within Streamlit Section 4.3.1 references the use of a 'exit button' to establish a conversation analysis/storage mechanism. Expanding on this, the underlying analysis performed through this trigger presents its own challenges.

Upon button trigger, some conversation 'pre-analysis' is conducted and then stored within the AWS DynamoDB database. This 'pre-analysis' is lengthy and conducted prior to upload to: (a) anonymise information before upload & (b) avoid expensive AWS computation.

Streamlit facilitates element interaction linearly, meaning the website is 'frozen' while the execution of an element (e.g. a button) is handled. Therefore, when users trigger the exit page/conversation pipeline, their engagement is suspended until pipeline completion. This can on average require a 10-20 second delay.

As mentioned in Section 4.3.1, while this is clarified (and shown with loading bars) to testers, in production would present an undesirable, uncomfortable user experience. In future work, asynchronous task queues should be explored to mitigate this, i.e. the Celery package[31], allowing the user to progress unencumbered.

Manual setup of Lambda 'data-getters' at scale The current initial data collection pipeline presents scaling challenges when deployed within AWS Lambda. AWS requires Lambda functions to have a maximum 'time-out' threshold, where script execution terminates prematurely, having exceeded it's time constraints. The upper time limit is presently 15 minutes. The objectives of this research necessitate the querying of numerous data sources by API. To do this ethically, and in compliance with vendors, requires implementing responsible delays (e.g. `time.sleep()` in Python) between calls.

Therefore, even with 3 MPs, this pipeline can exceed the maximum Lambda threshold of 15 minutes.

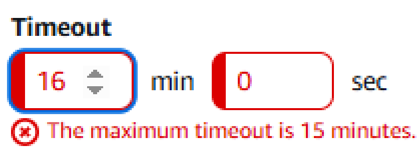


Figure 47: AWS Lambda 'timeout' thresholds.

To address this the data pipeline code is duplicated, set to collect data for one specific MP, and then packaged within a Docker Image. Each Docker Image is uploaded as a separate Lambda function, meaning one Lambda instance per MP.

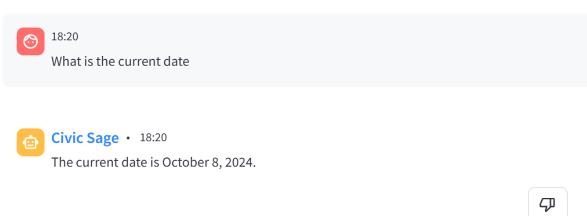
When this practice is applied to 650 MPs, this may be time-consuming to update if needing to add/remove API sources. Likewise, this could incur additional costs at execution time (650 functions executing once a day at the same time.) While this is unremarkable for an enterprise, maintaining such costs is not possible for a student-funded project.

[31]https://docs.celeryq.dev/en/stable/

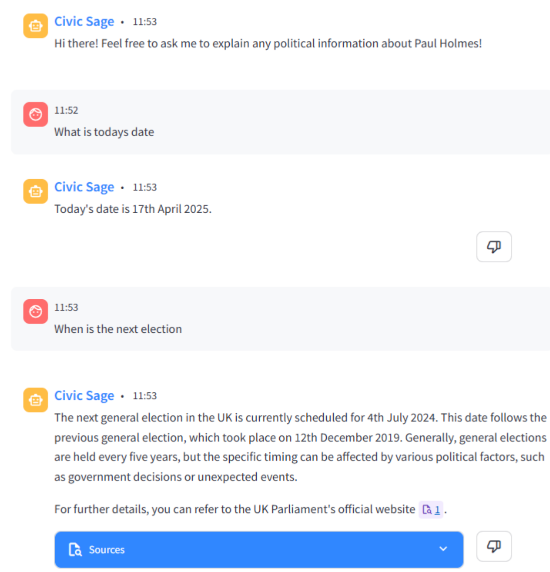
J.3 LLM & Conversation Analysis Implementation

Time-based LLM prompt problems Through user testing surveys it was discovered that LLMs can suffer in date-based reasoning. This presents a unique challenge within the context of a political information system, where time is a fundamental aspect of politics (e.g. attempting to query when the next election is.)

This is often a result of LLMs abstracting the date of their training data (their 'knowledge cut-off') with the present day, operating as if time is 'paused' in that moment. To counteract this, the artefact implements a mechanism to include the current date in each starting prompt. This is however an admittedly crude solution, which the model may occasionally ignore and refer to its own training context, shown in Figure 48.



(a) An example prompt prior to the inclusion of date in prompt templates.



(b) An example of the LLM ignoring date inclusion and referring to its training context when attempting a reasoning task.

Figure 48: Two sub-plots showing date-based issues with LLM reasoning tasks.

This is an ongoing issue, where the most definitive solution is simply to use the newest models. Though, as discussed throughout this report, is a constant challenge between incurring higher costs, versus real-terms value of newer models.

Necessary deviations from prompt political debiasing implementation Given the unique context of the artefact, deviation from a strict implementation of (Furniturewala et al. 2024) LLM debiasing techniques were necessary. Ergo it is possible that these changes, which have not been empirically evaluated, could have negatively impacted debiasing efficacy. With the time constraints and scope of this project, it is not possible to sufficiently assess whether these changes have significantly affected debiasing efficacy. In future works, this should be investigated further, replicating (Furniturewala et al. 2024)'s evaluation methodology.

K LLM Meta-evaluation Topics

As detailed in Section 3.4.2.1, the artefact develops a form of meta-evaluation, or ‘intelligent tests’, using an agent-driven testing approach, integrating concepts of agentic systems paired-evaluation and multi-turn iterative feedback from the Agent-as-a-judge-methodology (Zhuge et al. 2024).

An implementation explanation is available in Section 3.4.2.1. Tests can be viewed from the artefact `evaluation.py` file within the `meta_evaluation` directory.

Six ‘intelligent tests’ archetypes are created. These cover topics of, per MP:

- Election temporal and spatial history.
- Current political party details.
- History of official roles held while a member of the HM Official Opposition[32] party.
- Latest election statistics.
- Voting behaviour on a recent vote.
- History of all registered financial interests and amounts.

Topics are chosen to analyse a wide census of potential simple and complex MP factual information. These are extrapolated out for each of the three MPs featured in the artefact proof-of-concept, giving a total of 18 customised test cases. Each test case, except the recent vote subject, asks the same ‘archetype question’, though the criterion to satisfy test completion is tailored to the individual MP.

The initial statements/questions per test archetype are shown in Table 27:

Table 27: A table showing meta-evaluation subjects and questions.

Subject	**Archetype question**
All elections	”Where and when have they won an election?”
Current party	”What party do they belong to?”
Opposition roles	”What are all the roles they have held while part of the opposition?”
Latest election	”What was their latest election statistics and who came 2nd?”
Recent vote	”What did they vote for on Tobacco and Vapes legislation?”[a]
Registered interests	”What are all of their registered interests, and how much were they paid?”

[a]Example, as recent votes are contextual.

An example of a specific test case for the MP Paul Holmes is shown in Table 28 for clarity:

It is important to clarify these tests are based on retrieving factual information from the artefact and rely on statically inputted information inputted into the test cases (correct as of April 2025).

It is possible that some MPs information (e.g. roles) may have changed since then, potentially causing tests to fail. In the future, test cases should focus on static, unchanging facts, or a more sophisticated methodology should be incorporated to rectify this. For the purpose of testing a variety of MP information

[32]Commonly referred to by terms of ‘the opposition’ or ‘shadow cabinet’, the largest political party by majority within the House of Commons not in Government. This is currently the Conservative Party. https://members.parliament.uk/Opposition/Cabinet

Table 28: A table showing an example of a meta-evaluation test case for Paul Holmes, showing subject, initial question and criteria needed to pass the test case.

Subject	Question	Criteria
Latest election	”What was their latest election statistics and who came 2nd?”	”Paul Holmes (Conservatives) came 1st in the election, achieving 19,671 votes (36.4% share) Prad Bains (Liberal Democrats) came 2nd in the election, achieving 14,869 votes (27.5% share).”

and project time constraints, a contextual decision is made to acknowledge and disregard this outcome as a controlled limitation of the research.

Finally, **all established tests successfully pass** within two conversational turns (a maximum of three before test failure), shown in Figure 41. This indicates that the artefact satisfactorily passes all intelligent tests created.

L Large Files Submission Content

The contents submitted to the 'Large Files' submission box on BU's Brightspace are as follows:

- `civic-sage.zip`